Polynomial Upper Bounds on Characteristic-Data Length for Multiple Context-Free Languages under Fixed Finite-Semigroup Typing

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Abstract

We establish polynomial upper bounds on the total encoded length $\|K\|_+ = \sum_{w\in K}(|w|+1)$ of characteristic positive data for multiple context-free languages (MCFLs) that are tuple-substitutable under a fixed finite-semigroup typing. Qualitative reconstruction and positive-data identification follow Yoshinaka's distributional framework; the new results control the lengths of the required witnesses relative to an MCFG presentation. The alphabet, fan-out, rule-rank cap, and typing are fixed parameters. For reduced good MCFGs G, with rule thickness $\bar{\theta}_G$, we prove $O(\|G\|^2\bar{\theta}_G)$ bounds for prefix–suffix-window typing, including combinations with arbitrary finite, possibly noncommutative, groups. The proof simultaneously shortens interleaved tuple components while preserving the ordered group products in every output component.

Using the classical local-group/delay-theory representation, we extend the upper bound to finite typings whose positive image lies in the pseudovariety of local groups, obtaining $O(\|G\|^3\bar{\theta}_G)$. For arbitrary standard input MCFGs, including deletion and empty components, we obtain corresponding $O(n^3t)$ and $O(n^4t)$ bounds, where n is input grammar size and t is component-pattern thickness. The latter parameter may grow exponentially in n, so these are parameterized bounds, not grammar-size-only polynomial guarantees. Rank-one input grammars have a cubic size bound for every fixed typing. A separate conservative learner provides Gold identification. We exhibit a schematic cross-serial language with finite feature licensing that is window–group typed but not untyped multidimensionally substitutable, and a second example of intrinsic fan-out and rank two. Two further fan-out-two targets separate local-group from window–group typings, and nonabelian from window–abelian typings, respectively. At fan-out one the local-group upper bound strictly extends the companion locally-trivial–group bound; together with the companion semilattice lower bound it makes **LG** the exact pseudovariety boundary there, relative to presentation size and rule thickness. Whether this boundary persists at fan-out two or more remains open.

Keywords: grammatical inference; multiple context-free grammars; positive data; finite-semigroup typing; multidimensional substitutability; Gold identification; rule rank.

1 Introduction

Positive-data grammatical inference asks when a language can be recovered from an enumeration of positive examples alone [1,2]. Distributional learning builds grammatical constituents from recurring word or tuple contexts, beginning with substitutable CFGs [3, 12] and extending to MCFGs with discontinuous and cross-serial dependencies [13, 14, 30, 31].

The resource is encoded data length, not sample cardinality. Yoshinaka's finite-signature generalization already provides positive-data identification and polynomially many characteristic witnesses in the corresponding bounded-rank specialization [15]. But a sample of polynomial *cardinality* may contain exponentially long words. We therefore bound $\|K\|_+ = \sum_{w\in K}(|w|+1)$, not merely $|K|$. This follows the polynomial *thick-data* perspective reviewed by Eyraud, Heinz and Yoshinaka [10, Section 2.5.2]: the size of a grammar and the length needed to realize its productive rules are distinct resources. We study presentation-sensitive characteristic data for a set-driven reconstruction operator and a separate conservative Gold learner.

The original thickness, not the cost of a typed refinement. There is a fixed two-element typing $s_{\mathtt{c}}$ and a family of rank-two CFG presentations G_n with $L(G_n) = \{\mathtt{a},\mathtt{c}\}^+$, $\|G_n\| = O(n)$ and ordinary rule thickness $\theta_{G_n} \le 2$, but the trimmed full yield-type refinement has $\theta_{\widetilde{G}_{n,0}} \ge 2^n$ (Proposition 4.5). The example isolates an exponential cost of typed refinement relative to original thickness, motivating an analysis directly on the original presentation. The central question is when the algebra of a fixed typing permits simultaneous shortening *before* paying that cost.

Why supply a finite typing in advance? Finite features such as morphological compatibility or case licensing can constrain substitutions between discontinuous constituents, while an MCFG handles the unbounded dependencies. For example, let $\mathcal{R}_{\mathrm{feat}} = \{(0,0),(0,1),(1,0)\}$ and

$$L_{\mathrm{feat}} = \{\mathtt{a}^m\mathtt{p}_i\mathtt{b}^n\mathtt{c}^m\mathtt{q}_j\mathtt{d}^n : m,n \ge 1,\ (i,j) \in \mathcal{R}_{\mathrm{feat}}\}.$$

The a/c and b/d dependencies cross; the internal markers $\mathtt{p}_i, \mathtt{q}_j$ carry a finite, nonrectangular compatibility relation. A fixed product of a prefix–suffix window and a finite abelian-group feature value admits this language as an (f,s)-typed fan-out-two target, although it is neither context-free nor in Yoshinaka's untyped $\mathsf{YSub}(f)$. Even the boundary window alone is insufficient for its substitutability (Proposition 5.3). The example is therefore covered by the window–group upper bound, not just by the qualitative reconstruction theorem. The learner is *given* the finite feature bias but must still infer the unbounded crossing dependencies from positive data. This gives a schematic example of finite case licensing.

Our principal quantitative results give $O(\|G\|^2\bar{\theta}_G)$ characteristic-data bounds for window and window–group typings and $O(\|G\|^3\bar{\theta}_G)$ for positive-image local groups, with alphabet, fan-out, rule rank and typing fixed. Classical sliding-factor coordinates are used only as input to the new owner-wise simultaneous shortening of noncommutative, interleaved tuple components. Arbitrary input-grammar normalization and the rank-one case are treated separately.

This paper fixes, as part of the learning bias, a surjective finite-semigroup homomorphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ and compares observed *nonempty* tuple components only when their s-values agree. The canonical fresh-identity completion $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$ is used only to evaluate grammar templates, including empty terminal gaps. The typing and the fan-out bound are fixed before any data are seen, separating the choice of a finite bias from learning under that bias [5]. Proposition 5.3 uses a fixed window–group product: the group component records a finite feature relation, while neither a target grammar nor the unbounded crossing counts are supplied. Prefix–suffix windows and DFA transition morphisms are other instances. Other biases restrict derivations by regular control sets [6, 7] or restrict the evidence used to compare factors [9, 11]; positive-data learning of other mildly context-sensitive formalisms is studied in [19], and MCFG learning with membership queries or dual context sets in [16, 18, 20]. A recent substitution lemma for MCFLs [23] is a necessary condition for membership, not a learning criterion.

Prior work and the exact novelty boundary. The following comparison concerns the guarantees used in this paper; it does not attempt an exhaustive classification of prior learning models.

Source	Established scope	Present additional question
Yoshinaka (2008) [12]	(k,ℓ)-substitutable CFGs; characteristic-data size with a grammar-thickness parameter	Higher-fan-out, fixed-group, and LG-typed shortening
Yoshinaka (2009, 2011) [13, 14]	Multidimensionally substitutable MCFGs and positive-data learning; polynomial identification in the corresponding grammar regime	Original-thickness bounds under supplied finite-semigroup typing
Yoshinaka (2015) [15]	Finite-signature and disjoint-sort learning principles, including finite witness cardinality under fixed bounded-rank signatures	Encoded-length upper bounds that survive the typing refinement
Kuriyama, MCFG preprint [26]	Supplied/latent finite observations and tuple reconstruction; restricted polynomial-data regimes including single-spine presentations	General good-grammar bounds via noncommutative owner-wise shortening and local groups
Companion CFG revision [25]	Fixed-semigroup-typed CFG quantitative results: an **LI** $\vee$ **Grp** upper bound and a two-element-semilattice lower bound	Multiple-component shortening, the **LG** upper bound (new already at fan-out one), support-pattern normalization

The Const/Link/Comp construction specializes Yoshinaka's qualitative learning principle rather than claiming it as new. In particular, the companion CFG revision gives a sufficient regime through **LI** $\vee$ **Grp** together with a semilattice lower bound, whereas the positive-image **LG** upper bound is established in this paper. The fan-out-one boundary statement (Corollary 4.20) combines our new local-group upper bound with the companion semilattice lower bound, reproduced in Appendix G for self-containment. The independent contribution is quantitative: noncommutative owner-wise shortening (Proposition 4.10), the resulting window–group and local-group bounds (Theorems 4.14 and 4.18), and normalization controlled by component-pattern thickness (Proposition 4.23). The underlying finite-semigroup identities, local-group/delay coordinates, and finite characteristic-data cardinality results are attributed to their original sources; the rank-one obstruction uses [21,22]. The distinction between this paper and the earlier MCFG preprint, including its restricted polynomial-data cases, is summarized in the preceding table.

Contributions. Our object is the *total encoded length* of characteristic positive data relative to an MCFG presentation, for typed *tuples* at higher rule rank, where several component boundaries must be preserved simultaneously. An untyped pruning argument cannot simply shorten child derivations while retaining their types: in noncommutative finite groups, blocks from one subtree may appear in different orders across different output components. The new quantitative step is an *owner-wise simultaneous* selection of replacements preserving all ordered root products, rather than preserving each omitted subtree's group value separately. The bounds below hold with $f, r_{\max}, |\Sigma|$ and the typing fixed; their hidden constants can grow rapidly with these fixed parameters. For a reduced good MCFG G put $\bar{\theta}_G = \max\{1, \theta_G\}$, where θ_G is rule thickness (Section 4). With the alphabet, fan-out, rule-rank bound, and typing fixed, the principal length bounds are as follows. For a reduced good grammar G, write $\bar{\theta} = \bar{\theta}_G$. For any standard input grammar H, put $n = \max\{1, \|H\|\}$ and $t = \bar{t}_{\mathrm{pat}}(H_{\mathrm{red}})$.

Fixed typing / rank	Reduced good $(\|G\|, \bar{\theta})$	Any standard input (n,t)	Source
Rule rank ≤ 1; any typing	—	$O(n^3)$	Thm. 4.6
Prefix–suffix window	$O(\|G\|^2\bar{\theta})$	$O(n^3 t)$	Thm. 4.8
Window with finite group	$O(\|G\|^2\bar{\theta})$	$O(n^3 t)$	Thm. 4.14
Positive image in **LG**	$O(\|G\|^3\bar{\theta})$	$O(n^4 t)$	Thm. 4.18

The arbitrary-input columns follow from Corollary 4.24; the rank-one row is a separate normalization argument. Bounds measure the total encoded length, not an intrinsic minimum grammar size. Hidden constants depend on fixed f, rule rank, alphabet and typing; for noncommutative-group shortening they contain $M_{A,f} = U_f \max\{3, |A|^f\}^{2^{f^2}}$, which may grow doubly exponentially with f. The abelian case has sharper constants (Theorem 4.15).

The proofs establish a local-group sufficient condition strictly beyond $\mathbf{LI} \vee \mathbf{Grp}$ (Proposition B.6), a rule-only computable budget (Proposition 4.26), an abelian-filtered bound (Theorem 4.28), and an intrinsically rank-two example (Theorem 5.11).

Scope. The typing, alphabet and arity bounds are fixed before data arrive. The characteristic-data guarantees apply to the specified Const/Link/Comp reconstruction operator and the stated grammar presentations; the operator-independent fan-out-one lower bound is attributed to the companion and reproduced in Appendix G. No presentation-independent stage of Gold convergence is claimed. The local-group condition is sufficient at every fan-out; whether it is also necessary at fan-out two or more is open (Open Problem 7.1). The polynomial exponents are not established as optimal. The distinct qualitative results of Yoshinaka and the companion CFG construction are cited for attribution. Sections 2–5 develop the setting, operator, upper bounds and examples. The conclusion isolates the higher-arity question; the appendices supply the requisite proofs, including a self-contained version of the companion lower-bound argument.

2 Preliminaries

Throughout, Σ is a finite alphabet, ε the empty word, and $\Sigma^+ := \Sigma^* \setminus \{\varepsilon\}$. If $\Sigma = \emptyset$, the only languages are $\emptyset$ and $\{\varepsilon\}$ and are handled directly.

Notation. The table lists the default meanings of frequently used symbols; locally rebound symbols are specified where they occur.

Symbol	Meaning
K	a finite sample; $\|K\|_+ = \sum_{w \in K}(\|w\| + 1)$
$G,\ H$	a (reduced good) MCFG; an arbitrary standard input MCFG
B, C, X, Y	nonterminals; X, Y also denote typed nonterminals where specified
$\|G\|$, θ_G, t_G	encoding size, rule thickness, nonterminal thickness
s, $\widehat{s}$, S	positive typing, its fresh-identity completion, its positive image
$\preceq$	kernel refinement (Definition 2.3), also between monoid morphisms on Σ^*
A	a finite group in group-theoretic lemmas (otherwise locally defined); A an abelian group in Section 4.8
g	a monoid morphism $\Sigma^* \to A$ into a finite group
ν	a sliding-factor labelling $\Sigma^{k+1} \to A$; $\lambda_{k,\nu}$, $\gamma_{k,\nu}$ as in Definition B.1
λ	an auxiliary finite-monoid morphism used to type derivation nodes
N	a finite monoid, codomain of λ
K	the minimal ideal of a finite semigroup (Appendix B)
$\hat{k}$	an auxiliary padding length at least $\max\{k, \ell, 1\}$, chosen even when required
T	a derivation tree
$\ell_{\mathrm{in}},\ \ell_{\mathrm{out}}$	minimum inside yield and outside context lengths (Section 4.7)

A rule or grammar is λ-*free* if no template component is empty, following Yoshinaka's terminology [14]. This condition concerns template components and is distinct from excluding the empty sentence from the generated language.

2.1 Positive-Data Learning

Definition 2.1 (Text, TxtEx identification, conservativeness)**.** Fix a pause symbol $\perp \notin \Sigma$. A *text* for $L \subseteq \Sigma^*$ is an infinite sequence over $L \cup \{\perp\}$ in which every element of L occurs; thus $\perp, \perp, \ldots$ is the text for $\emptyset$. The *content* of a finite prefix is the set of elements of Σ^* occurring in it. A *learner* is a computable map from finite prefixes to grammars. It *TxtEx-identifies* L [1] if on every text for L its outputs are, from some stage on, one fixed grammar H with $L(H) = L$, and it TxtEx-identifies a class if it does so for each member. A learner is *conservative* if it changes its hypothesis only on a datum $w \in \Sigma^*$ not generated by the current hypothesis. We fix a hypothesis H_0 with $L(H_0) = \emptyset$.

Definition 2.2 (Set-driven operator and characteristic sample)**.** A *set-driven reconstruction operator* $\mathcal{B}$ maps each finite $K \subseteq \Sigma^*$ to a grammar $\mathcal{B}(K)$ depending only on the set K; it is *sample-consistent* if $K \subseteq L(\mathcal{B}(K))$. For a language L, a finite $\mathcal{F} \subseteq L$ is a *characteristic sample for* $\mathcal{B}$ if $\mathcal{F} \subseteq K \subseteq L$ implies $L(\mathcal{B}(K)) = L$ for every finite K (cf. [4]). Unlike TxtEx identification this is a statement about generated languages: $\mathcal{B}(K)$ need not stabilize syntactically as K grows.

For finite $K \subseteq \Sigma^*$ put $\|K\|_+ := \sum_{w \in K}(|w| + 1)$, and for a tuple $\vec{u} = (u_1, \ldots, u_d)$ put $\|\vec{u}\|_1 := \sum_i |u_i|$. Every polynomial bound below states whether it concerns cardinality or $\|\cdot\|_+$.

2.2 Fixed Finite-Semigroup Typing

Following the companion CFG paper [25], the learning bias is a surjective semigroup morphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ onto a finite semigroup, given by its multiplication table and the letter images (so $s(w)$ is computable in linear time). A finite codomain can always be replaced by the positive image, so surjectivity loses no generality.

Definition 2.3 (Fresh completion, tuple types, refinement)**.** Let $\mathsf{S}^\sharp = \mathsf{S} \uplus \{\mathbf{1}_\sharp\}$ be obtained by adjoining a *fresh* identity (even if S already has one). The canonical monoid extension $\widehat{s} : \Sigma^* \twoheadrightarrow \mathsf{S}^\sharp$ satisfies $\widehat{s}(\varepsilon) = \mathbf{1}_\sharp$ and $\widehat{s}(w) = s(w)$ for $w \in \Sigma^+$. For $\vec{w} = (w_1, \ldots, w_d)$ write $\widehat{s}^{(d)}(\vec{w}) := (\widehat{s}(w_1), \ldots, \widehat{s}(w_d))$; on $(\Sigma^+)^d$ it agrees with $s^{(d)}$. The fresh identity evaluates empty template factors but does not create an additional positive type. For typings $s, t : \Sigma^+ \to$ finite semigroups write $s \preceq t$ if $\ker t \subseteq \ker s$ on Σ^+, and (s, t) for their product restricted to its positive image.

Positive typing and auxiliary morphisms. Empty terminal gaps in rule templates are evaluated by $\widehat{s}$ with values in $\mathsf{S}^\sharp$. An auxiliary finite-monoid morphism $q : \Sigma^* \to N$ represents the positive typing $q|_{\Sigma^+}$, and Lemma E.1 proves equivalence with its fresh-identity completion on all nonempty components and all successful trimmed refinements. In particular, algebraic conditions such as **LG** concern the positive image S, not $\mathsf{S}^\sharp$. Write s_{triv} for the one-element positive typing.

Definition 2.4 (Prefix–suffix typing)**.** For $k, \ell \geq 0$ put $m_0 := \max\{1, k + \ell\}$, and for $|w| \geq m_0$ let $\mathrm{pref}_k(w)$, $\mathrm{suf}_\ell(w)$ be the prefix and suffix of the indicated lengths (length-zero windows are ε). Define

$$\widehat{s}_{k,\ell}(w) = \begin{cases} w, & |w| < m_0, \\ \langle \mathrm{pref}_k(w), \mathrm{suf}_\ell(w) \rangle, & |w| \geq m_0, \end{cases}$$

with disjoint tags for short words and pairs, and put $M_{k,\ell} := \widehat{s}_{k,\ell}(\Sigma^*)$. For short x, y and pairs $\langle u, v \rangle, \langle u', v' \rangle$ define

$$\begin{aligned} x \cdot y &:= \widehat{s}_{k,\ell}(xy), & x \cdot \langle u, v \rangle &:= \langle \mathrm{pref}_k(xu), v \rangle, \\ \langle u, v \rangle \cdot y &:= \langle u, \mathrm{suf}_\ell(vy) \rangle, & \langle u, v \rangle \cdot \langle u', v' \rangle &:= \langle u, v' \rangle. \end{aligned}$$

A case check gives $\widehat{s}_{k,\ell}(xy) = \widehat{s}_{k,\ell}(x) \cdot \widehat{s}_{k,\ell}(y)$; since $\widehat{s}_{k,\ell}$ is onto $M_{k,\ell}$, associativity is inherited from concatenation and $\widehat{s}_{k,\ell}(\varepsilon)$ is the identity. Thus $\widehat{s}_{k,\ell}$ is a finite-monoid morphism; its positive restriction $s_{k,\ell}$ is the boundary typing, and since the short-word tag separates ε from Σ^+, $\widehat{s}_{k,\ell}$ is isomorphic to $\widehat{s_{k,\ell}}$. A positive typing s with $s \preceq s_{k,\ell}$ is called (k,ℓ)-*boundary-determined*, or briefly a *boundary typing*.

2.3 Multiple Context-Free Grammars

Definition 2.5 (Standard MCFG presentation)**.** An MCFG [30, 31] is $G = (V, \Sigma, P, S, \mathrm{fan})$ with finite V, $S \in V$, $\mathrm{fan}(S) = 1$ and $\mathrm{fan}(A) \geq 1$. A rule of rank $r \geq 0$ is $A \to \rho(B_1, \ldots, B_r)$ with template $\rho = (\alpha_1, \ldots, \alpha_{d_0})$, $d_0 = \mathrm{fan}(A)$, $d_j = \mathrm{fan}(B_j)$, where each α_i is a word over $\Gamma_\rho := \Sigma \uplus \Xi_1 \uplus \cdots \uplus \Xi_r$, $\Xi_j := \{\xi_j^1, \ldots, \xi_j^{d_j}\}$, and every variable occurs at most once in $\alpha_1 \cdots \alpha_{d_0}$ (*linearity*). For $\vec{u}_j \in (\Sigma^*)^{d_j}$, $\rho(\vec{u}_1, \ldots, \vec{u}_r)$ substitutes $u_{j,\kappa}$ for ξ_j^κ; for $r = 0$ the rule is a constant rule. The tuple languages $L_A(G)$ form the least family closed under the rules, and $L(G) := \{w : (w) \in L_S(G)\}$. We write $A \Rightarrow_G^* \vec{u}$ for a *terminal derivation* (a finite derivation tree rooted at A evaluated bottom-up); one from S is *successful*. G is an f-MCFG if $\mathrm{fan}(A) \leq f$ for all A, and has rule-rank cap $r_{\max}$ if all ranks are at most $r_{\max}$.

We write $\mathsf{MCFL}(f, r)$ for the languages of MCFGs of fan-out $\leq f$ and rule rank $\leq r$, so that f-MCFL $= \bigcup_{r<\infty} \mathsf{MCFL}(f, r)$. Minimum rule rank of a language refers to all presentations, not to a given grammar. $\|G\|$ is the length of a fixed symbol-level encoding counting nonterminal and rule records and every terminal or variable occurrence in templates. For nonterminals of equal fan-out d, $A \to B$ abbreviates the identity-template rule $A \to (\xi_1^1, \ldots, \xi_1^d)(B)$, and $A \to \vec{c}$ a constant rule. Rule variables are always ξ_j^κ; terminals in examples are in `typewriter` font.

Fan-out and rule rank are distinct resources: there is no rank reduction preserving an arbitrary fixed fan-out [24, 33]. The reconstruction below therefore imposes neither a binary normal form nor a known rank cap.

2.4 Tuple Contexts and Substitutability

A *sentence context of arity* d is a word $E = v_0 \Box_1 v_1 \cdots v_{d-1} \Box_d v_d$ with each hole occurring once, $v_0, v_d \in \Sigma^*$ and $v_1, \ldots, v_{d-1} \in \Sigma^+$; such contexts are *admissible*, and $\mathcal{E}_d$ denotes their set ([14, Section 3.1]). $E[\vec{w}]$ fills $\Box_i$ by w_i, $|E|_{\mathrm{fix}} := \sum_{i=0}^{d} |v_i|$, and E *accepts* $\vec{w}$ in L if $E[\vec{w}] \in L$.

Definition 2.6 (Tuple occurrence)**.** For $w = a_1 \cdots a_n$ number the cut positions $\textcircled{0} a_1 \textcircled{1} \cdots a_n \textcircled{n}$ and write $w[i : i'] := a_{i+1} \cdots a_{i'}$. An *arity-$d$ tuple occurrence* in w is a sequence $0 \leq i_1 < i_1' < i_2 < i_2' < \cdots < i_d < i_d' \leq n$; it determines $\vec{x} = (w[i_1 : i_1'], \ldots, w[i_d : i_d']) \in (\Sigma^+)^d$ and the context $E \in \mathcal{E}_d$ obtained by replacing these intervals by $\Box_1, \ldots, \Box_d$, so that $E[\vec{x}] = w$. For a finite K, $\mathrm{Obs}_d(K)$ is the set of tuples $\vec{x}$ arising from occurrences in words of K, i.e. $\vec{x} \in (\Sigma^+)^d$ with $E[\vec{x}] \in K$ for some $E \in \mathcal{E}_d$, and $\mathrm{Obs}(K) := \bigcup_{d \leq f} \mathrm{Obs}_d(K)$.

For $L \subseteq \Sigma^*$ and $\vec{x} \in (\Sigma^*)^d$ the *d-tuple distribution* is $\mathcal{D}_L^{(d)}(\vec{x}) := \{E \in \mathcal{E}_d : E[\vec{x}] \in L\}$.

Definition 2.7 ((f,s)-tuple substitutability)**.** Let $f \geq 1$ and $s : \Sigma^+ \to \mathsf{S}$ a finite-semigroup morphism. A language $L \subseteq \Sigma^*$ is (f,s)-*tuple-substitutable* if for every $1 \leq d \leq f$ and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$,

$$s^{(d)}(\vec{x}) = s^{(d)}(\vec{y}) \ \text{ and } \ \mathcal{D}_L^{(d)}(\vec{x}) \cap \mathcal{D}_L^{(d)}(\vec{y}) \neq \emptyset \implies \mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y}).$$

Empty components are never compared (Yoshinaka's convention [14, Section 3.1 and footnote 3]; see [8] for CFGs); $\varepsilon \in L$ is handled at the start symbol. The restriction matters already at fan-out

one: $\mathtt{a}$ and ε share a context in $\mathtt{a}^+$ but have different distributions. At fan-out one, $\mathcal{D}_L^{(1)}(x)$ is the two-sided distribution $\{(u,v) : uxv \in L\}$. For $\vec{x}, \vec{y} \in (\Sigma^+)^d$ write $\vec{x} \equiv_{L,s}^d \vec{y}$ (or $\vec{x} \equiv_{L,s} \vec{y}$) if $\widehat{s}^{(d)}(\vec{x}) = \widehat{s}^{(d)}(\vec{y})$ and $\mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y})$; this is an equivalence relation.

Definition 2.8 (The target class)**.** $\mathcal{C}_{f,s}^{\mathrm{mcf}}$ is the class of (f,s)-tuple-substitutable f-MCFLs. A target in $\mathcal{C}_{f,s}^{\mathrm{mcf}}$ for a boundary typing $\widehat{s}$ is called *boundary-typed*.

If $s \preceq t$, then $\mathcal{C}_{f,s}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,t}^{\mathrm{mcf}}$, since equal finer types are equal coarser types; in particular $\mathcal{C}_{f,s_{\mathrm{triv}}}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$. The class is not monotone in f: only $\mathcal{C}_{f+1,s}^{\mathrm{mcf}} \cap f\text{-MCFL} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$ holds, and already for trivial typing $\{w\#w^R\}$ is 1- but not 2-dimensionally substitutable [14, Section 3.1, Example 5]. Thus f is part of the bias. For every $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$, $\widehat{s}^{-1}(\mathsf{Acc}) \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$: it is regular, and equal componentwise types give equal total types in every context.

Proposition 2.9 (Intersection and abelian filtering)**.** *Fix f and $\widehat{s}$.*

(i) If L_1, L_2 are (f,s)-tuple-substitutable, so is $L_1 \cap L_2$.

(ii) If L is (f,s)-tuple-substitutable, A is an abelian group, $\varphi : \Sigma^* \to \mathsf{A}$ a monoid morphism, $\mathsf{H} \leq \mathsf{A}$ and $\gamma \in \mathsf{A}$, then $L \cap \varphi^{-1}(\gamma + \mathsf{H})$ is (f,s)-tuple-substitutable.

Whenever the intersection is an f-MCFL, it belongs to $\mathcal{C}_{f,s}^{\mathrm{mcf}}$.

Proof. (i) A context accepting equal-type $\vec{x}, \vec{y}$ in $L_1 \cap L_2$ accepts both in each L_i, so their distributions agree in each L_i, hence in the intersection. (ii) If a context accepts $\vec{x}, \vec{y}$ in $\varphi^{-1}(\gamma + \mathsf{H})$, then $\delta := \sum_i (\varphi(y_i) - \varphi(x_i)) \in \mathsf{H}$; by commutativity $\varphi(E'[\vec{y}]) - \varphi(E'[\vec{x}]) = \delta$ for every E', so coset membership transfers in both directions. Thus $\varphi^{-1}(\gamma + \mathsf{H})$ is (f, s_{triv})-, hence (f,s)-tuple-substitutable; apply (i). □

Consequently every f-MCFL of the form $\widehat{s}^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathsf{H})$ lies in $\mathcal{C}_{f,s}^{\mathrm{mcf}}$; such factorizations are not supplied to the learner.

3 Yoshinaka's Learning Principle and Quantitative Witnesses

Realizable type vectors. For a nonterminal B of arity d and a fixed monoid morphism $\mu : \Sigma^* \to M$, we call $\mathbf{t} \in M^d$ *realizable at B* if some $\vec{u} \in L_B(G)$ satisfies $\mu^{(d)}(\vec{u}) = \mathbf{t}$. For a positive-word typing we use its canonical fresh-identity completion. The word *realizable* below always refers to this property, not to an algorithm for computing the tuple.

Fixed-rank learning is a specialization of Yoshinaka [15]. We first identify the relevant signature and its concrete implementation, retaining below only those invariants and finite witnesses required by the *total-length* analysis in Section 4 and the separate unbounded-rank extension.

3.1 Set-Driven Reconstruction

Auxiliary good form for the proofs. A rule is *nondeleting* if every variable occurs exactly once; this alone does not forbid empty output components. A nondeleting rule is *nonpermuting* if, for each j, $\xi_j^1, \ldots, \xi_j^{d_j}$ occur in this order in $\alpha_1 \cdots \alpha_{d_0}$; following [14, Section 2.3] it is *λ-free* if no α_i is empty, and a nonpermuting rule is *nonmerging* if no α_i contains a factor $\xi_j^\kappa \xi_j^{\kappa+1}$. A rule, template or presentation is *good* if it is λ-free, nondeleting, nonpermuting and nonmerging. By induction on derivations, every component of every tuple derived from a nonterminal of a λ-free grammar is nonempty.

All interaction between templates and contexts below rests on one observation.

Lemma 3.1 (Induced child contexts)**.** *Let $\rho = (\alpha_1, \dots, \alpha_e)$ be a good template of rank r with child arities $d_1, \dots, d_r$, let $E \in \mathcal{E}_e$, let $1 \le j \le r$, and fix $\vec{u}_i \in (\Sigma^+)^{d_i}$ for $i \neq j$. Let E_j be the word obtained from $E[(\alpha_1, \dots, \alpha_e)]$ by replacing ξ_j^κ by $\square_\kappa$ and every other variable ξ_i^ι by $u_{i,\iota}$. Then $E_j \in \mathcal{E}_{d_j}$ and $E_j[\vec{x}] = E[\rho(\vec{u}_1, \dots, \vec{u}_{j-1}, \vec{x}, \vec{u}_{j+1}, \dots, \vec{u}_r)]$ for every $\vec{x} \in (\Sigma^*)^{d_j}$.*

Proof. The identity holds because substitutions commute. By linearity and nondeletion each $\square_\kappa$ occurs exactly once in E_j, and by nonpermutation in the order $\kappa = 1, \dots, d_j$, since E places $\alpha_1, \dots, \alpha_e$ from left to right. Consider ξ_j^κ and $\xi_j^{\kappa+1}$. If both lie in one α_i, nonmerging puts a terminal or a variable ξ_i^ι with $i \neq j$ between them, which becomes a nonempty word. Otherwise they lie in $\alpha_i, \alpha_{i'}$ with $i < i'$, and the material between them contains the separator $v_i \in \Sigma^+$ of E. So every internal separator of E_j is nonempty. □

Definition 3.2 (Composition witness)**.** Let K be finite and $r \ge 1$. A *composition witness of rank r* in K is a triple $(\vec{z}, \rho, (\vec{u}_1, \dots, \vec{u}_r))$ with $\vec{z} \in \mathrm{Obs}(K)$, ρ a good template of rank r whose output and child arities are at most f, $\vec{u}_j \in (\Sigma^+)^{d_j}$, and $\vec{z} = \rho(\vec{u}_1, \dots, \vec{u}_r)$.

Since each $\alpha_i(\vec{u}_1, \dots, \vec{u}_r) = z_i$ fixes the positions of all child components inside $\vec{z}$, a composition witness is the same as a segmentation of an observed occurrence of $\vec{z}$ into terminal gaps and labelled child intervals. If $E[\vec{z}] \in K$ with $E \in \mathcal{E}_e$, Lemma 3.1 shows that every $\vec{u}_j \in \mathrm{Obs}(K)$. Linearity and nondeletion make the children contribute disjointly to $\vec{z}$, and each contributes at least one letter, so

$$r \le \textstyle\sum_j \|\vec{u}_j\|_1 \le \|\vec{z}\|_1 \le \max_{w \in K} |w| \le \|K\|_+. \tag{3.1}$$

Definition 3.3 (Capped and uncapped reconstruction operators)**.** Fix f and a finite-semigroup typing s with canonical completion $\widehat{s}$. For $r_{\max} \ge 1$ let $\mathcal{B}_{f,s}^{\le r_{\max}}(K) = \widehat{G}_{\le r_{\max}}(K)$ be the grammar with start symbol $\widehat{S}$, a nonterminal $[\vec{x}]$ of arity d for each $\vec{x} \in \mathrm{Obs}_d(K)$, $d \le f$, and the rules

(Start) $\widehat{S} \to [(w)]$ for $w \in K \cap \Sigma^+$, and $\widehat{S} \to \varepsilon$ if $\varepsilon \in K$;

(Const) $[\vec{x}] \to \vec{x}$ for every observed $\vec{x}$;

(Comp) $[\vec{z}] \to \rho([\vec{u}_1], \dots, [\vec{u}_r])$ for every composition witness of rank $1 \le r \le r_{\max}$;

(Link) $[\vec{x}] \to [\vec{y}]$ for observed $\vec{x}, \vec{y}$ of the same arity d with $\widehat{s}^{(d)}(\vec{x}) = \widehat{s}^{(d)}(\vec{y})$ and $E[\vec{x}], E[\vec{y}] \in K$ for some $E \in \mathcal{E}_d$.

For $K \neq \emptyset$ put $\mathcal{B}_{f,s}(K) := \widehat{G}(K) := \widehat{G}_{\le \|K\|_+}(K)$; $\mathcal{B}_{f,s}(\emptyset)$ has no rules. Fixed encoding orders make both operators deterministic and set-driven.

By (3.1), $\mathcal{B}_{f,s}(K)$ contains a Comp rule for every composition witness in K. Both operators are sample-consistent, by $\widehat{S} \to [(w)] \to (w)$ and $\widehat{S} \to \varepsilon$. Every $[\vec{x}]$ derives only tuples with nonempty components, since Comp templates are λ-free; so $\widehat{S} \Rightarrow^* \varepsilon$ only by the rule $\widehat{S} \to \varepsilon$. For trivial typing and $\varepsilon \notin K$, Comp/Const, Link and Start are exactly the rules of Types I, II and III of [14, Section 4.1], and $\mathcal{B}_{f,s}$ is the operator of Yoshinaka's unbounded-rank learner $\mathcal{A}(p, *)$ [14, Corollary 2].

Inherited qualitative learning principle. For fixed f, r, s, finite-sort specialization of Yoshinaka's signature theorem [15, Theorem 1] provides qualitative positive-data identification and polynomial hypothesis updates in this fixed-signature specialization. The relation to Yoshinaka's signature formulation is described in Appendix D; soundness and completeness for the actual Const/Link/Comp operator are established by direct arguments, without appealing to equivalence with SUBST$_{\mathrm{P}}$. No total-length bound is inherited from characteristic-set *cardinality*: this is the new quantitative question.

Soundness interface. The standard substitution invariant for the witnessed reconstruction is Proposition D.3; its typed Link/Comp verification is in Appendix D.

3.2 Yield-Typed Refinement and Witnesses

Definition 3.4 (Yield-type refinement)**.** Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a finite nondeleting MCFG and $s : \Sigma^+ \to \mathsf{S}$ a finite-semigroup typing, with canonical extension $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$. For a template ρ and $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{d_j}$, let $\mathrm{out}_\rho(\mathbf{t}_1, \ldots, \mathbf{t}_r) \in (\mathsf{S}^\sharp)^{d_0}$ have ith component the image of α_i under the monoid morphism $\Gamma_\rho^* \to \mathsf{S}^\sharp$ extending $a \mapsto \widehat{s}(a)$ and $\xi_j^\kappa \mapsto (\mathbf{t}_j)_\kappa$. The *full yield-type refinement* $G^{\widehat{s}}$ has a fresh start symbol $\widetilde{S}$, nonterminals $A_{\mathbf{t}}$ for $A \in V$ and $\mathbf{t} \in (\mathsf{S}^\sharp)^{\mathrm{fan}(A)}$, start rules $\widetilde{S} \to S_{(\eta)}$ for $\eta \in \mathsf{S}^\sharp$, and for every rule $A \to \rho(B_1, \ldots, B_r)$ (including $r = 0$) and all $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{\mathrm{fan}(B_j)}$ the rule $A_{\mathrm{out}_\rho(\mathbf{t}_1,\ldots,\mathbf{t}_r)} \to \rho(B_{1,\mathbf{t}_1}, \ldots, B_{r,\mathbf{t}_r})$. The *trimmed refinement* $\widetilde{G}_0$ keeps exactly the typed nonterminals and rules occurring in some successful derivation from $\widetilde{S}$. Unless $G^{\widehat{s}}$ is named, *typed nonterminal* and *typed rule* refer to the non-start nonterminals and rules of $\widetilde{G}_0$, and $L_X := \{\vec{u} : X \Rightarrow^*_{\widetilde{G}_0} \vec{u}\}$.

Typed rules have the templates of their sources, so $\widetilde{G}_0$ is good whenever G is; each L_X is nonempty, with nonempty components if G is λ-free.

Proposition 3.5 (Yield-type invariants)**.** *(i) If $A_{\mathbf{t}} \Rightarrow^* \vec{u}$ in $G^{\widehat{s}}$, then $\widehat{s}^{(\mathrm{fan}(A))}(\vec{u}) = \mathbf{t}$, and erasing indices gives a G-derivation of $\vec{u}$.*

(ii) Every terminal derivation $A \Rightarrow^*_G \vec{u}$ lifts uniquely to $A_{\widehat{s}^{(\mathrm{fan}(A))}(\vec{u})} \Rightarrow^*_{G^{\widehat{s}}} \vec{u}$; a successful derivation of w lifts uniquely through $\widetilde{S} \to S_{(\widehat{s}(w))}$.

(iii) $L(\widetilde{G}_0) = L(G^{\widehat{s}}) = L(G)$.

The proof is given in Appendix E.

The typed homogeneity of Lemma E.3 (Appendix E) is weaker than the homogeneity of congruential MCFGs [17], whose contexts may have empty separators.

Definition 3.6 (Canonical tuples, contexts and the witness set)**.** Let G be good. For every typed nonterminal X fix a *canonical tuple* $\omega(X) \in L_X$ and a *canonical context* $\chi(X)$ as in Lemma E.2, with $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Let $\mathsf{W}(\widetilde{G}_0)$ consist of the words

$$\chi(X)[\omega(X)] \ \ (X \text{ typed}), \qquad \chi(X)[\rho(\omega(Y_1), \ldots, \omega(Y_q))] \ \ (X \to \rho(Y_1, \ldots, Y_q) \text{ typed}, \ q \geq 0).$$

Since $\omega(X)$ and $\rho(\omega(Y_1), \ldots, \omega(Y_q))$ belong to L_X, $\mathsf{W}(\widetilde{G}_0)$ is a finite subset of $L(G)$, nonempty if $L(G) \neq \emptyset$. Components of canonical tuples are nonempty, so $\|\omega(X)\|_1 \geq \mathrm{fan}(X)$. Canonical objects are target witnesses, not inputs of the operator; qualitatively any choice works, while Section 4 chooses short ones.

Proposition 3.7 (Cardinality of the witness set)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be good of fan-out $\leq f$ and rule rank $\leq r_{\max}$, and let n_{typ} and n_{rule} be the numbers of typed nonterminals and typed rules of $\widetilde{G}_0$. Then*

$$|\mathsf{W}(\widetilde{G}_0)| \leq n_{\mathrm{typ}} + n_{\mathrm{rule}} \leq |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}}.$$

Proof. One word is selected per typed nonterminal and per typed rule. A typed nonterminal is some $A_{\mathbf{t}}$ with $\mathbf{t} \in (\mathsf{S}^\sharp)^{\mathrm{fan}(A)}$, and a typed rule is determined by its source rule and the child indices $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{\mathrm{fan}(B_j)}$. □

3.3 Completeness and Exact Reconstruction

The completeness argument [14, Lemma 5] needs only that the sample exposes, for every typed rule, one same-type link and one decomposition. We state it for arbitrary typed representatives, so that it also applies to the relaxed witnesses of Section 4. No assumption that $\iota(X) \in L_X$ is made: the two required witnessed sentence words are instead explicit hypotheses.

Proposition 3.8 (Completeness from a witness assignment)**.** *Let G be a finite good MCFG of fan-out $\le f$, let r_G be the maximum rank of a typed rule of $\widetilde{G}_0$ ($r_G := 0$ if none), and assign to each typed nonterminal $X = A_{\mathbf{t}}$ a tuple $\iota(X) \in (\Sigma^+)^{\mathrm{fan}(A)}$ of type $\mathbf{t}$ and a context $\chi(X) \in \mathcal{E}_{\mathrm{fan}(A)}$, with $\chi(X) = \square_1$ for $\widetilde{S} \to X$. Suppose K contains $\chi(X)[\iota(X)]$ for each X and $\chi(X)[\rho(\iota(Y_1), \dots, \iota(Y_q))]$ for each typed rule $X \to \rho(Y_1, \dots, Y_q)$, $q \ge 0$. Then*

$$L(G) \subseteq L(\mathcal{B}_{f,s}^{\le r_{\max}}(K)) \quad (r_{\max} \ge \max\{1, r_G\}), \qquad L(G) \subseteq L(\mathcal{B}_{f,s}(K)).$$

Neither substitutability nor $\iota(X) \in L_X$ is assumed; the canonical choice $\iota = \omega$ gives $\mathsf{W}(\widetilde{G}_0) \subseteq K$.

The verification is given in Appendix D.

Good normalization needed for exact reconstruction. A nonterminal is *reachable* if it is reachable from S in the graph with an edge $A \to B$ whenever B is a child in a rule for A, and *productive* if $L_A(G) \neq \emptyset$; G is *reduced* if all nonterminals are both.

Proposition 3.9 (Good normalization with rank preservation)**.** *Let G_{in} be a finite f-MCFG of rule rank at most $r_{\max}$. There is a finite good f-MCFG G_{good} of rule rank at most $r_{\max}$ with $L(G_{\mathrm{good}}) = L(G_{\mathrm{in}}) \setminus \{\varepsilon\}$ and, for fixed $f, r_{\max}$, $\|G_{\mathrm{good}}\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|)$.*

The proof is given in Appendix F. The proposition controls size, fan-out and rank, not thickness; thickness transfer for reduced λ-free inputs is Proposition F.2.

Theorem 3.10 (Exact reconstruction for a fixed typing)**.** *Fix $f \ge 1$ and $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$, and put the completion $\widehat{s}$. Let $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$. If $L \cap \Sigma^+ = \emptyset$ put $\mathcal{F}_L := L$ and $r_G := 0$. Otherwise let G be a good presentation of $L \cap \Sigma^+$ (Proposition 3.9), and put $\mathcal{F}_L := \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$. Then $\mathcal{F}_L \subseteq L$ is finite, and every finite K with $\mathcal{F}_L \subseteq K \subseteq L$ satisfies*

$$L(\mathcal{B}_{f,s}(K)) = L, \qquad L(\mathcal{B}_{f,s}^{\le r_{\max}}(K)) = L \quad \text{for every } r_{\max} \ge \max\{1, r_G\}.$$

If $L \in \mathsf{MCFL}(f, r)$, G can be chosen of rank $\le r$, so $r_{\max} = \max\{1, r\}$ suffices (also when $r = 0$).

The verification is given in Appendix D.

3.4 Qualitative learning and effectivity

Fixed-rank polynomial hypothesis construction (Theorem D.4) and Gold/TxtEx convergence (Corollary 3.12) use Yoshinaka's conservative update policy. The proofs, the exact update-time interpretation and the unbounded-rank supplement are collected in Appendix D. These statements do not imply short characteristic data; that is the subject of Section 4.

Definition 3.11 (Conservative sequential learner)**.** The learner $\mathcal{A}_{f,s}$ (respectively $\mathcal{A}_{f,s}^{\le r_{\max}}$) starts with H_0, keeps its hypothesis on $\perp$ and on every datum the hypothesis generates, and otherwise replaces it by $\mathcal{B}_{f,s}(K)$ (respectively $\mathcal{B}_{f,s}^{\le r_{\max}}(K)$), where K is the content read so far. This is the policy of [14, Algorithm 1].

Corollary 3.12 (Gold identification)**.** *Fix $f \geq 1$ and s.*

(i) *$\mathcal{A}_{f,s}$ TxtEx-identifies $\mathcal{C}^{\mathrm{mcf}}_{f,s}$.*

(ii) *For fixed $r_{\max}$, $\mathcal{A}^{\leq r_{\max}}_{f,s}$ TxtEx-identifies $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r_{\max})$, in time per datum polynomial in the size of the content read so far.*

4 Characteristic-Data Bounds

Under a rule-rank cap, Proposition 3.7 already gives polynomially many characteristic words; this section controls their total length.

The bounds by typing regime are summarized in the table in the Introduction. For arbitrary fixed typings, the refined-thickness baseline is given by Lemma 4.4.

Dependency map of the quantitative proof. Order homogenization (Lemma A.7) and the noncommutative boundary-state identity (Lemma A.5) produce simultaneous block replacement (Proposition 4.10). This yields the window–group shortening theorem (Theorem 4.14); combining the same replacement with classical Rees/sliding-factor coordinates and Lemma B.3 yields Theorem 4.18. The standard-presentation bounds additionally require the support-pattern normalization of Proposition 4.23.

The first row is the unconditional baseline. Full yield-type refinement can make its thickness exponentially larger than the original presentation parameters (Proposition 4.5); rank one removes the thickness dependence, and the remaining rows transfer the *original* thickness. For window-only typings the window-skeleton lemma (Lemma 4.7) gives the bound; for window–group typings the marked heavy-node lemma (Lemma 4.12) combines boundary marks and simultaneous group-product preservation in one heavy-node count, invoking the block-replacement lemma (Appendix A). For general fixed typings the refined-thickness baseline remains available, but it need not give an upper bound polynomial in the original thickness. The present results isolate sufficient regimes where that loss can be avoided. Polynomial update time in the observed sample is independent of these data-length refinements (Theorem D.4 and Corollary 3.12).

4.1 Thickness and the witness-length template

For the quantitative results, presentations are reduced and *good*, as in Yoshinaka's fixed-dimension, fixed-rank good form [14, Section 2.3, after Example 2]. For a rule R of G, define its *rule thickness* by

$$\theta_G(R) := \min\{\, |w| : w \in L(G) \text{ has a successful derivation using } R \,\}.$$

Thus length means the number of terminal letters in the *derived sentence*, not the number of steps of its derivation. Put $\theta_G := \max_{R \in P} \theta_G(R)$, $\bar{\theta}_G := \max\{1, \theta_G\}$, and $t_G := \max_A \min\{\|\vec{w}\|_1 : \vec{w} \in L_A(G)\}$. Thus θ_G is rule thickness and t_G nonterminal thickness; Yoshinaka writes them τ_G and t_G [14, Section 4.3]. If a target contains ε, every statement below is applied to its λ-free core and the single datum ε is added, exactly as in Theorem 3.10; this adds one to the encoded length.

Lemma 4.1 (Thickness lemma)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG.*

(i) *Every rule writes at most θ_G terminal symbols explicitly.*

(ii) *$t_G \leq \theta_G$; that is, every $A \in V$ has a tuple $\vec{u} \in L_A(G)$ with $\|\vec{u}\|_1 \leq \theta_G$.*

(iii) $\theta_G \le 2\|G\| \max\{1, t_G\}$.

(iv) *Every $A \in V$ of fan-out d has a context $E_A \in \mathcal{E}_d$ with $|E_A|_{\mathrm{fix}} \le \theta_G$ and $E_A[\vec{u}] \in L(G)$ for every $\vec{u} \in L_A(G)$, and $E_S = \square_1$. For every finite typing $\widehat{s}$ and every non-start typed nonterminal $X = A_{\mathbf{t}}$ of $\widetilde{G}_0$, the choice $\chi(X) := E_A$ meets Definition 3.6, and E_A arises from a successful typed derivation by deleting an occurrence of an X-subtree.*

Proof. Fix a rule R with left-hand side A and a successful derivation D_R using R of length $\theta_G(R)$; it exists because G is reduced. By nondeletion, every terminal written by R and every component of the tuple produced at that occurrence of A survives exactly once in the final sentence. This gives (i), and also (ii) because every productive A has a rule. For (iv) with $A \neq S$, delete the subtree rooted at that occurrence of A and put ordered holes in place of its components; iterating Lemma 3.1 along the path to the root shows that the result E_A lies in $\mathcal{E}_d$, and its fixed material is part of the selected sentence. Substituting any $\vec{u} \in L_A(G)$ yields another successful derivation. Since $L_X \subseteq L_A(G)$ by Proposition 3.5(i), E_A accepts L_X; a typed nonterminal with a start link has the form $S_{\mathbf{t}}$, for which $E_S = \square_1$. Filling E_A's derivation with an X-derivation and lifting by Proposition 3.5(ii) places X at the hole, so E_A is derivational for X.

For (iii), follow a shortest dependency path from S to A; it repeats no nonterminal, so its rules are distinct and differ from R. Keep the path child open, complete every sibling and every child of R by a shortest terminal tuple (of length at most t_G), and apply R. The distinct rules used write at most $\|G\|$ terminals and have at most $\|G\|$ child occurrences, so the resulting sentence has length at most $\|G\| + \|G\| t_G \le 2\|G\| \max\{1, t_G\}$. □

Hence the two thickness parameters are polynomially equivalent on reduced good presentations. Yoshinaka's efficiency criterion asks for polynomial hypothesis construction, characteristic-set cardinality, and total positive data in target size and rule thickness [14, Definition 1]; see also [10]. Proposition F.2 transfers the size–thickness scale to reduced λ-free inputs; genuine empty components may instead cause the exponential growth of Proposition F.3.

All higher-rank bounds below are assembled by the following two lemmas.

Lemma 4.2 (Witness-length template)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \ge 1$, let $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$ be a finite typing, and let $\widetilde{G}_0$ be the trimmed $\widehat{s}$-refinement. If every non-start typed nonterminal has a canonical tuple with $\|\omega(X)\|_1 \le b_{\mathrm{can}}$ and $\chi(X) := E_A$ is taken from Lemma 4.1(iv), then every word of $\mathsf{W}(\widetilde{G}_0)$ has length at most $2\bar{\theta}_G + r_{\max} b_{\mathrm{can}}$, and*

$$|\mathsf{W}(\widetilde{G}_0)| \le |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}}, \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \le (|V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}})(1 + 2\bar{\theta}_G + r_{\max} b_{\mathrm{can}}).$$

If moreover $L(G) = L \cap \Sigma^+$ for some $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for $\mathcal{B}^{\le r_{\max}}_{f,s}$.

Proof. A witness is $\chi(X)[\omega(X)]$, $\chi(X)[\vec{c}]$ or $\chi(X)[\rho(\omega(Y_1), \dots, \omega(Y_q))]$. The context contributes at most $\bar{\theta}_G$ terminals, the source template at most $\bar{\theta}_G$ by Lemma 4.1(i), and there are at most $r_{\max}$ child tuples. The count is Proposition 3.7; multiply. The characteristic property is Proposition 3.8 (the cap $r_{\max}$ bounds every typed rank) together with Proposition D.3 and the Start rule for ε. □

Lemma 4.3 (Type-preserving replacement)**.** *Let G be reduced and good, let $s : \Sigma^+ \to \mathsf{S}$ be a finite-semigroup typing and $\lambda : \Sigma^* \to N$ an auxiliary morphism with $s \preceq \lambda|_{\Sigma^+}$, and let $X = B_{\mathbf{t}}$ be a non-start typed nonterminal of the trimmed $\widehat{s}$-refinement $\widetilde{G}_0$. If $\vec{u} \in L_X$, $\vec{u}' \in L_B(G)$ and $\lambda^{(d)}(\vec{u}') = \lambda^{(d)}(\vec{u})$, then $\vec{u}' \in L_X$; in particular $\vec{u}'$ may be chosen as $\omega(X)$.*

Proof. Goodness makes all components nonempty, so $\widehat{s} \preceq \lambda$ gives $\widehat{s}^{(d)}(\vec{u}') = \mathbf{t}$. Proposition 3.5(ii) lifts a G-derivation of $\vec{u}'$ to a $G^{\widehat{s}}$-derivation rooted at X. Inserting it at an occurrence of X in a successful derivation of $\widetilde{G}_0$ gives a successful derivation, because typed rules depend only on typed symbols; hence all its rules survive trimming. □

4.2 Baselines and Rule Rank One

For a reduced good presentation G of a λ-free target, let $\widetilde{G}_0$ be its trimmed full yield-type refinement and put $\bar{\theta}_{\mathrm{typ}} := \max\{1, \theta_{\widetilde{G}_0}\}$, where $\theta_{\widetilde{G}_0}$ is rule thickness computed in the refined grammar. No comparison with the original untyped thickness is assumed.

Lemma 4.4 (Refined-thickness baseline)**.** *Fix $f \geq 1$, $r_{\max} \geq 1$, a finite alphabet Σ, and a finite typing $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$. Let G be a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of a λ-free language in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, and let $\widetilde{G}_0$ be its trimmed yield-type refinement. Then the target witness set has cardinality polynomial in $\|G\|$ and total positive length polynomial in $\|G\|$ and $\bar{\theta}_{\mathrm{typ}} := \max\{1, \theta_{\widetilde{G}_0}\}$.*

Proof. $\widetilde{G}_0$ is reduced and good, so Lemma 4.1(i),(ii),(iv) applies to it with $\bar{\theta}_{\mathrm{typ}}$ in place of $\bar{\theta}_G$: canonical tuples and (derivational) canonical contexts of length at most $\bar{\theta}_{\mathrm{typ}}$ exist, with $\chi(X) = \square_1$ for start links. Each typed rule writes at most $\bar{\theta}_{\mathrm{typ}}$ terminals and has at most $r_{\max}$ children, so every witness has length at most $(r_{\max}+2)\bar{\theta}_{\mathrm{typ}}$, and

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq (|V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}})(1 + (r_{\max}+2)\bar{\theta}_{\mathrm{typ}})$$

by Proposition 3.7. □

For trivial typing no productive nonterminal has more than one yield type, so the original thickness scale is retained; this recovers Yoshinaka's untyped estimates $|K_G| \leq |P|$ and $\|K_G\| \leq |P|\tau_G$ [14, Lemma 7 and Theorem 1]. The quantitative claims below therefore concern typed, multi-component settings. Without uniqueness the refinement may be much thicker:

Proposition 4.5 (Exponential blow-up of full yield-type refinement)**.** *There are a fixed two-element typing $s_{\mathtt{c}}$, reduced good fan-out-one rank-two grammars G_n of size $O(n)$, and a constant bound $\theta_{G_n} \leq 2$, such that $L(G_n) = \{\mathtt{a}, \mathtt{c}\}^+$ and the trimmed full yield-type refinement $\widetilde{G}_{n,0}$ satisfies $\theta_{\widetilde{G}_{n,0}} \geq 2^n$.*

Proof. Let $M_{\mathtt{c}} = \{1, \zeta\}$ with $\zeta^2 = \zeta$, $s_{\mathtt{c}}(\mathtt{a}) = 1$, and $s_{\mathtt{c}}(\mathtt{c}) = \zeta$. For $n \geq 1$, view

$$S \to SS \mid \mathtt{a} \mid \mathtt{c} \mid A_0 \mid \cdots \mid A_n, \qquad A_0 \to \mathtt{a}, \qquad A_i \to A_{i-1}A_{i-1} \mid \mathtt{c} \quad (1 \leq i \leq n)$$

as a fan-out-one MCFG. It is reduced and good, has size $O(n)$ and rule rank two, generates $\{\mathtt{a}, \mathtt{c}\}^+$, and every rule has a successful use of length at most two, so $\theta_{G_n} \leq 2$. In the full refinement, the retained copy $(A_n)_1$ can derive only words without $\mathtt{c}$; recursively the unique such yield is $\mathtt{a}^{2^n}$. Hence the typed rule $S_{(1)} \to (A_n)_1$ has successful thickness at least 2^n. □

This is a limitation of the full-refinement route, not of the language $\{\mathtt{a}, \mathtt{c}\}^+$. In particular, an exponential increase in the thickness of the *chosen refined presentation* is not an operator-independent lower bound on characteristic-data length. This is why the subsequent proofs work directly with the original presentation.

Theorem 4.6 (Rank-one characteristic-data bound)**.** *Fix f, Σ, and a finite-semigroup typing s. For any standard rank-≤ 1, fan-out-$\leq f$ grammar G_{in} presenting a target in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, the operator $\mathcal{B}^{\leq 1}_{f,s}$ admits a characteristic sample K with*

$$\|K\|_+ = O_{f,s,\Sigma}(\|G_{\mathrm{in}}\|^3).$$

Proof. Normalize without increasing rank (Proposition 3.9) and trim, obtaining a reduced good rank-$\leq$ 1 grammar G with $\|G\| = O_f(\|G_{\mathrm{in}}\|)$, and trim its typed refinement. Let $n_{\mathrm{typ}} \leq |V||\mathsf{S}^\sharp|^f$ be the number of non-start typed nonterminals and $b_G := \max\{1, \text{largest explicit terminal count of a rule}\} \leq \max\{1, \|G\|\}$. A terminal derivation from X is a chain of unary rules ending in a constant rule. Choose one minimizing first the yield length and then the number of rules. No typed nonterminal repeats: splicing in the later suffix would not increase the yield (nondeletion) and would shorten the chain. Hence $\|\omega(X)\|_1 \leq n_{\mathrm{typ}} b_G$. For the context, take $\Box_1$ if $\widetilde{S} \to X$ is present; otherwise follow a shortest path $\widetilde{S} \to X_1 \to \cdots \to X_p = X$ in the typed dependency graph, so $p \leq n_{\mathrm{typ}}$. Composing its unary templates and putting ordered holes for the variables of X gives, by Lemma 3.1 iterated with no siblings, a derivational context $\chi(X) \in \mathcal{E}_{\mathrm{fan}(X)}$ with $|\chi(X)|_{\mathrm{fix}} \leq n_{\mathrm{typ}} b_G$ accepting all of L_X. Every witness word therefore has length at most $n_{\mathrm{typ}} b_G + b_G + n_{\mathrm{typ}} b_G$, and Proposition 3.7 with $r_{\max} = 1$ gives

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq (n_{\mathrm{typ}} + |P||\mathsf{S}^\sharp|^f)(1 + (2n_{\mathrm{typ}} + 1)b_G) = O_{f,s}(\|G_{\mathrm{in}}\|^3).$$

This set, with ε when $\varepsilon \in L$, is characteristic for $\mathcal{B}^{\leq 1}_{f,s}$ by Propositions 3.8 and D.3. □

Polynomial-time construction and Gold convergence follow separately from Theorem D.4 and Corollary 3.12.

This extends the rank-one untyped scale [14, Lemma 7] and the fan-out-one argument of [25] to arbitrary fixed finite typing at every fixed fan-out. Rank two cannot generally yield a grammar-size-only bound: the size-$O(n)$ grammar $A_0 \to \mathsf{c}$, $A_i \to A_{i-1}A_{i-1}$, $S \to A_n$ generates $\{\mathsf{c}^{2^n}\}$, which is (f, s)-tuple-substitutable for every fixed f, s (over a one-letter alphabet, sharing a sentence context forces equal total component lengths), and every characteristic sample for it contains a word of length at least 2^n.

4.3 Boundary Windows and Auxiliary Invariants

The refined thickness need not be polynomially controlled by the original good presentation (Proposition 4.5). A transfer is nevertheless possible whenever the typing is determined by a fixed prefix–suffix window together with an auxiliary finite invariant that can itself be preserved by short replacements. We use the thickness-sensitive representation model of Yoshinaka [14, Section 4.3].

Lemma 4.7 (Window skeletons)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$. Fix $k, \ell \geq 0$, a morphism $\kappa : \Sigma^* \to N$ into a finite monoid, and a finite typing $\widehat{s} \preceq (\widehat{s}_{k,\ell}, \kappa)$. Let $\widetilde{G}_0$ be the trimmed $\widehat{s}$-refinement and n_{typ} its number of non-start typed nonterminals. Suppose that $\beta \geq 0$ is such that for every $C \in V$ and $\vec{x} \in L_C(G)$ there is $\vec{x}' \in L_C(G)$ with $\kappa^{(\mathrm{fan}(C))}(\vec{x}') = \kappa^{(\mathrm{fan}(C))}(\vec{x})$ and $\|\vec{x}'\|_1 \leq \beta$. Then every non-start typed nonterminal X has a canonical tuple with*

$$\|\omega(X)\|_1 \leq f(k + \ell + 1)\, n_{\mathrm{typ}}\, (\bar{\theta}_G + r_{\max}\beta).$$

For trivial κ, $\beta = \bar{\theta}_G$ is admissible by Lemma 4.1(ii), so $\|\omega(X)\|_1 \leq f(k + \ell + 1) n_{\mathrm{typ}}(r_{\max} + 1)\bar{\theta}_G$ whenever $\widehat{s} \preceq \widehat{s}_{k,\ell}$.

Proof. Choose a terminal derivation D from X in $\widetilde{G}_0$ with the minimum number of grammar-rule nodes, and let $\vec{u}$ be its root tuple. No typed nonterminal occurs twice on a root-to-leaf path: replacing the upper subtree by the lower one would give a derivation from X with fewer nodes. Hence every path has at most n_{typ} grammar nodes.

Mark in each component of $\vec{u}$ the terminals that determine its $\widehat{s}_{k,\ell}$-type (Definition 2.4): all terminals of a component shorter than $m_0 = \max\{1, k+\ell\}$, and otherwise its first k and last ℓ terminals (one arbitrary terminal when $k = \ell = 0$). There are at most $f(k+\ell+1)$ marks. Each terminal is written explicitly by exactly one node, so the union $\mathcal{K}$ of the root paths to the marked terminals has at most $f(k+\ell+1)n_{\mathrm{typ}}$ nodes. Call a child of a node of $\mathcal{K}$ that is not itself in $\mathcal{K}$ *omitted*. By linearity and nondeletion, each component of an omitted child is one contiguous block of exactly one root component, and this block contains no mark.

Replace each omitted subtree, with underlying nonterminal C and yield $\vec{x}$, by a G-derivation of a tuple $\vec{x}'$ as in the hypothesis, and forget all typed labels; this gives a G-derivation from the underlying nonterminal B of X with root tuple $\vec{u}'$. A short root component consists only of marked terminals, so it receives no replaced block and is unchanged. In a long component every replaced block lies strictly between the marked first k and last ℓ terminals, and goodness keeps it nonempty; the marked prefix and suffix remain literal, and the component stays on the long side of the threshold. Hence $\widehat{s}_{k,\ell}$-vectors agree. Each root component is the ordered product of explicit terminals and child components, and every replaced block keeps its κ-value; so κ-vectors agree, with no commutativity used. Thus $(\widehat{s}_{k,\ell}, \kappa)^{(d)}(\vec{u}') = (\widehat{s}_{k,\ell}, \kappa)^{(d)}(\vec{u})$, and Lemma 4.3 gives $\vec{u}' \in L_X$. Each node of $\mathcal{K}$ writes at most $\bar{\theta}_G$ terminals (Lemma 4.1(i)) and has at most $r_{\max}$ omitted children, each of length at most β; the bound follows. □

Theorem 4.8 (Polynomial data for prefix–suffix-determined typings)**.** *Fix $f \geq 1$, $r_{\max} \geq 1$, k, ℓ, a finite alphabet Σ, and a finite-semigroup typing $s : \Sigma^+ \to \mathsf{S}$ satisfying $s \preceq s_{k,\ell}$; write $\widehat{s}$ for its canonical completion. Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and let G be a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar with $L(G) = L \cap \Sigma^+$. Then $\mathcal{B}^{\leq r_{\max}}_{f,s}$ is polynomial-time in the positive sample. A characteristic sample for L is $K = \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$, with*

$$|\mathsf{W}(\widetilde{G}_0)| = O_{f,r_{\max},s}(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,s,\Sigma}(\|G\|^2 \bar{\theta}_G),$$

where $\bar{\theta}_G = \max\{1, \theta_G\}$. For a reduced λ-free input presentation G_{in} (deleting rules allowed), the structural good-form conversion of Proposition F.2 yields the explicit presentation-relative bound

$$O_{f,r_{\max},k,\ell,s,\Sigma}\Big(\|G_{\mathrm{in}}\|^3 \max\{1, t_{G_{\mathrm{in}}}\}\Big).$$

Proof. By Lemma E.1 we may use $\widehat{s}$. Lemma 4.7 with trivial κ gives $b_{\mathrm{can}} = f(k+\ell+1)n_{\mathrm{typ}}(r_{\max}+1)\bar{\theta}_G$, and $n_{\mathrm{typ}} \leq |V||\mathsf{S}^\sharp|^f = O(\|G\|)$. Lemma 4.2 bounds every witness by $O(\|G\|\bar{\theta}_G)$ and their number by $O(\|G\|)$, and gives the characteristic property; the datum ε adds at most one. Theorem D.4 gives the construction time. For the input clause, Proposition F.2 gives $\|G\| = O(\|G_{\mathrm{in}}\|)$ and $\bar{\theta}_G = O(\|G_{\mathrm{in}}\| \max\{1, t_{G_{\mathrm{in}}}\})$; nonterminal thickness is used because deleting source rules may erase material before good-form conversion. □

Corollary 3.12 gives polynomial time per datum for the capped conservative learner. At fan-out one, polynomial time-and-thick-data learning for fixed windows is already due to Yoshinaka [12, Theorem 1]. As explained in the Introduction, our quadratic total-encoded-length bound is measured relative to a reduced good presentation; the degree in that presentation size is uniform for fixed window parameters, without claiming an improvement over Yoshinaka's bounds after normalization. The contribution of Lemma 4.7 is to preserve boundary information simultaneously across *multiple*

tuple components at every fixed $f \geq 2$. For fan-out-one window–group sufficient conditions, we credit the companion CFG manuscript [25]. Theorem 5.11 shows that this higher-rank regime is needed at the language level: it gives a non-context-free $(1,1)$-typed target of minimum fan-out two and minimum rule rank two over all finite fan-outs, whereas the rank-one examples are covered by Theorem 4.6.

4.4 Finite-Group Invariants

Window–group typings require short tuples with *prescribed* componentwise group values. The length bound is obtained from a single count of heavy nodes, which we state for an arbitrary finite-monoid invariant because Appendix B reuses it.

Lemma 4.9 (Heavy-node count)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$, let $\lambda : \Sigma^* \to N$ be a morphism into a finite monoid, and let $b \geq \bar{\theta}_G$ and $c \geq 0$. Let T be a terminal derivation from $B \in V$ that minimizes first the total yield length and then the number of nodes among derivations from B with the same root λ-vector. Call a node* heavy *if its subtree yield has total length greater than b. If every set of pairwise incomparable heavy nodes of T has at most c elements, then*

$$\|\textit{root tuple of } T\|_1 \leq (r_{\max}+1)\, b\, \max\{1, c\,|V|\,|N|^f\}.$$

Proof. Label each node by its nonterminal and the λ-vector of its yield; there are at most $|V|\,|N|^f$ labels. If an ancestor and a proper descendant had the same label, replacing the former subtree by the latter would keep the root λ-vector (templates evaluate homomorphically, as in Proposition 3.5), would not increase the yield (nondeletion) and would remove nodes. So every root-to-leaf path has at most $|V|\,|N|^f$ nodes. If the root is light, its yield has length at most b. Otherwise the heavy nodes form an ancestor-closed subtree, whose leaves are pairwise incomparable; hence there are at most $c|V|\,|N|^f$ heavy nodes. Every terminal of the root yield is written either by a heavy node (at most $\bar{\theta}_G \leq b$ per node, Lemma 4.1(i)) or inside a light child of a heavy node (at most $r_{\max}$ children of length at most b each). □

For a finite abelian group H write $\mathsf{d}(H)$ for the maximal length of a zero-sum-free sequence over H, so that $\mathsf{D}(H) = \mathsf{d}(H) + 1$ is its Davenport constant. For a finite group A and $f \geq 1$ put

$$N_{A,f} = \max\{3, |A|^f\}, \quad U_f = f^{f+1} f!, \quad M_{A,f} = U_f\, N_{A,f}^{2^{f^2}}, \qquad c_{A,f} := \begin{cases} \mathsf{d}(A^f), & A \text{ abelian}, \\ M_{A,f} - 1, & \text{otherwise}. \end{cases}$$

The lower floor $N_{A,f} \geq 3$ is required for the mixed-direction cases in the block-order decomposition; the $|A|^f$ term supplies repeated group boundary states.

Proposition 4.10 (Simultaneous block replacement: the exact interface)**.** *Let A be a finite group and $f \geq 1$. There are d ordered output rows ($1 \leq d \leq f$). Row c is a fixed-order product of q_c blocks, interspersed with unaltered constants $a_{c,0}, \ldots, a_{c,q_c} \in A$. A block $\beta_{c,j}$ has an owner $o(c,j) \in [m]$ and an original and a replacement value $x_{c,j}, x'_{c,j} \in A$. Every owner has at most f blocks over all rows. Different owners' blocks may interleave arbitrarily; all material not replaced is absorbed into the intervening constants. For $J \subseteq [m]$ put*

$$\mathrm{val}_c(J) = a_{c,0} y_{c,1} a_{c,1} \cdots y_{c,q_c} a_{c,q_c}, \qquad y_{c,j} = \begin{cases} x'_{c,j}, & o(c,j) \in J, \\ x_{c,j}, & o(c,j) \notin J. \end{cases}$$

With $c_{A,f}$ as defined above, if $m > c_{A,f}$ there exists a nonempty $J \subseteq [m]$ such that $\mathrm{val}_c(J) = \mathrm{val}_c(\varnothing) \quad (c = 1, \ldots, d)$. *For abelian A, one may take $c_{A,f} = \mathsf{d}(A^f)$; for an arbitrary finite group, $c_{A,f} = M_{A,f} - 1$ suffices. Individual block or owner values need not be preserved.*

Proof. This is Lemma A.2 of Appendix A, written with root-component notation. For abelian A, associate with each owner its difference vector in A^d and select a nonempty zero-sum subsequence. In a nonabelian group one must not commute factors. Instead fix a common owner pattern and homogenize pairwise order comparisons by Lemma A.7; Lemma A.4 then decomposes the output rows into consecutive rank chunks, and Lemma A.5 selects a nonempty interval of owners preserving all ordered row products. The threshold is at most $M_{A,f}$. The detailed extraction and constants appear in Appendix A. □

Theorem 4.11 (Short tuples with prescribed finite-group values)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and maximum rule rank $r_{\max} \geq 1$. Let $g : \Sigma^* \to A$ be a homomorphism into* any *finite group A. For every $B \in V$ and every realizable $\mathbf{s} \in A^{\mathrm{fan}(B)}$ there is $\vec{u} \in L_B(G)$ with $g^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{s}$ and*

$$\|\vec{u}\|_1 \leq \beta_g := (r_{\max} + 1)\bar{\theta}_G \max\{1, c_{A,f}|V||A|^f\}.$$

In particular $\beta_g = O_{A,f,r_{\max}}(\|G\|\bar{\theta}_G)$; when A is abelian the antichain constant is the zero-sum constant $\mathsf{d}(A^f)$.

Proof. Let T be minimal as in Lemma 4.9 with $\lambda = g$ and $b = \bar{\theta}_G$, and let $v_1, \ldots, v_m$ be pairwise incomparable heavy nodes. For each v_i, labelled C_i, fix a C_i-derivation of length at most $\bar{\theta}_G$ (Lemma 4.1(ii)); for $J \subseteq [m]$ let T_J replace the subtrees at v_i, $i \in J$. By linearity and nondeletion each component of v_i is a contiguous block of one root component, and blocks of different v_i are disjoint. The g-value of a root component of T_J is the ordered product of the g-values of its fixed segments and of its blocks, each block contributing its original or its replacement value according to whether its owner lies outside or inside J. This is a block-replacement family over A in the sense of Definition A.1. If $m > c_{A,f}$, Proposition 4.10 yields a nonempty J for which T_J has the root g-vector of T; but T_J is strictly shorter, since every replaced subtree has length greater than $\bar{\theta}_G$. This contradicts minimality, so Lemma 4.9 applies with $c = c_{A,f}$ and $N = A$. □

Lemma 4.12 (Marked heavy nodes for simultaneous window–group shortening)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$. Fix $k, \ell \geq 0$, a finite group A, and a homomorphism $g : \Sigma^* \to A$. Put $\lambda = (\widehat{s}_{k,\ell}, g) : \Sigma^* \to N := M_{k,\ell} \times A$, $h := f(k + \ell + 1)$, and $n_\lambda := |V||N|^f$. For every $B \in V$ and realizable vector $\mathbf{t} \in N^{\mathrm{fan}(B)}$, there exists $\vec{u} \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$ and*

$$\|\vec{u}\|_1 \leq b_{\mathrm{mark}} := (r_{\max} + 1)\bar{\theta}_G \max\{1, (c_{A,f} + h)n_\lambda\}.$$

Hence $b_{\mathrm{mark}} = O(\|G\|\bar{\theta}_G)$ with $f, r_{\max}, k, \ell, A, g, \Sigma$ fixed. If A is abelian, $c_{A,f}$ can be taken as $\mathsf{d}(A^f)$.

Proof. Among B-derivations with root componentwise λ-value $\mathbf{t}$, choose T minimizing total yield length, then node count. As in Lemma 4.9, a root–leaf path repeats no pair of a nonterminal and its componentwise λ-value; its number of nodes is at most n_λ.

Mark the first k and last ℓ terminal positions in each root component of length at least $m_0 := \max\{1, k + \ell\}$, and all positions in a shorter component. If $k = \ell = 0$, mark one position per component to ensure that nonemptiness is explicitly preserved. There are at most $h = f(k + \ell + 1)$ marks in total.

Call a node heavy if its subtree yields more than $\bar{\theta}_G$ terminals. Fix an arbitrary antichain Q of heavy nodes. Since incomparable subtrees occupy disjoint sets of terminal positions, at most h nodes of Q contain marked terminals. Let Q_0 consist of all other nodes. For each $v \in Q_0$ with nonterminal C_v, fix a derivation producing a tuple of total length at most $\bar{\theta}_G$, which exists by Lemma 4.1(ii). Each replacement strictly shortens the original heavy subtree.

Suppose $|Q_0| > c_{A,f}$. By linearity and nondeletion, every node in Q_0 contributes at most f nonoverlapping blocks distributed among at most f ordered root components. Everything outside these subtrees remains fixed. Since g is a homomorphism, each root group value is the ordered product of the unchanged pieces and the original or replacement block values. Proposition 4.10 thus selects a nonempty set of owners whose simultaneous replacement preserves all root group values, without assuming commutativity.

No selected subtree contains a mark. Short root components are untouched, while the marked prefixes and suffixes of long components retain their letters and order. Every replacement component is nonempty by goodness. Consequently all root $\widehat{s}_{k,\ell}$-values are preserved. The root λ-vector is unchanged, while the total yield length strictly decreases, a contradiction to the choice of T. Every heavy antichain therefore has at most $c_{A,f} + h$ members. Lemma 4.9, with $b = \bar{\theta}_G$, $c = c_{A,f} + h$ and $N = M_{k,\ell} \times A$, gives the displayed bound. For abelian A use Lemma A.2(a). □

Example 4.13 (Two components with crossed owner order). Two owners with two blocks each may appear in the root tuple in the order $(\beta_{1,1}\beta_{2,1},\ \beta_{2,2}\beta_{1,2})$. Take $A = S_3$, put $\sigma = (12)$ and $\tau = (23)$, and assign value σ to the blocks of owner 1 and τ to those of owner 2. The root group vector is $(\sigma\tau, \tau\sigma)$, with $\sigma\tau \neq \tau\sigma$. Thus commuting owner contributions would be invalid. This illustrates a block-replacement arrangement, not a proof that any particular target language requires a nonabelian typing. Proposition 4.10 preserves the two ordered root products simultaneously.

Theorem 4.14 (Polynomial characteristic data for window–finite-group typing)**.** *Fix $f, r_{\max}, k, \ell, \Sigma$, a finite group A, a homomorphism $g : \Sigma^* \to A$, and a fixed positive-word typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ whose fresh-identity completion satisfies $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$. For every target $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G with $L(G) = L \cap \Sigma^+$, the capped reconstruction operator $\mathcal{B}^{\leq r_{\max}}_{f,s}$ has a characteristic sample with*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,\widehat{s},g,\Sigma}(\|G\|^2\bar{\theta}_G).$$

Proof. Apply Lemma 4.12 to $\lambda = (\widehat{s}_{k,\ell}, g)$. Since $\widehat{s} \preceq \lambda$, every realizable $\widehat{s}$-typed non-start nonterminal $X = B_{\mathbf{s}}$ has an untyped B-tuple of the same root λ-value and total length at most $b_{\mathrm{mark}} = O(\|G\|\bar{\theta}_G)$. Lemma 4.3 lifts its derivation to X, so $b_{\mathrm{can}} \leq b_{\mathrm{mark}}$. The witness template (Lemma 4.2) gives $O(\|G\|)$ witnesses, each of length $O(\|G\|\bar{\theta}_G)$; the total encoded length is $O(\|G\|^2\bar{\theta}_G)$. Handle ε by its direct start rule. □

Theorem 4.15 (Polynomial data for window–abelian typing)**.** *Fix $f, r_{\max}, k, \ell, \Sigma$, a finite abelian group A, a homomorphism $g : \Sigma^* \to A$, and a fixed positive-word typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ whose fresh-identity completion satisfies $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$. For every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G with $L(G) = L \cap \Sigma^+$, the capped operator $\mathcal{B}^{\leq r_{\max}}_{f,s}$ has a characteristic sample satisfying*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,\widehat{s},g,\Sigma}(\|G\|^2\bar{\theta}_G).$$

In particular the bound uses the marked-node constant

$$b_{\mathrm{mark}} = (r_{\max}+1)\bar{\theta}_G \max\{1, (\mathsf{d}(A^f) + f(k+\ell+1))|V|(|M_{k,\ell}||A|)^f\}.$$

Proof. Apply the proof of Theorem 4.14 using the abelian case of Lemma 4.12. Lemma A.2(a) permits $c_{A,f} = \mathsf{d}(A^f)$ rather than $M_{A,f} - 1$, with $\mathsf{d}(A^f) = O(\exp(A) f \log |A|)$ (Remark A.3). □

Commutativity is used only in the zero-sum case of Lemma A.2. For nonabelian A, the order-homogenization proof of Appendix A.1 controls blocks interleaving across multiple root components. The finite-group block-replacement lemma and its full proof, including the nonabelian order-homogenization and interval-state arguments, are in Appendix A.

4.5 An Algebraic Criterion for Polynomial Characteristic Data

Write **LI** for the pseudovariety of finite locally trivial semigroups and **Grp** for the pseudovariety of finite groups. Their join $\mathbf{LI} \vee \mathbf{Grp}$ consists of divisors of finite products $S \times A$, with $S \in \mathbf{LI}$ and A a finite group [28, 29]. The following criterion makes the window–group hypothesis intrinsic to the positive image of the fixed typing; its finite-semigroup ingredient also occurs in the companion CFG manuscript [25].

Lemma 4.16 (Locally trivial–group typings admit window–group refinements)**.** *Let Σ be a nonempty finite alphabet and let $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ be a finite-semigroup typing; write $\widehat{s}$ for its canonical completion. If $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$, then there exist $k, \ell \geq 0$, a finite group A, and a morphism $g : \Sigma^* \to A$ such that $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$ on all of Σ^*. If $\mathsf{S} \in \mathbf{LI}$, the group factor can be taken trivial, so $\widehat{s} \preceq \widehat{s}_{k,\ell}$.*

Proof. Choose a finite locally trivial semigroup S, a finite group A, a subsemigroup $T \subseteq S \times A$, and an onto semigroup morphism $\pi : T \to \mathsf{S}$. Lifting each $s(a)$ to some $(s_a, g_a) \in T$ defines a morphism $\rho : \Sigma^+ \to T$ with $\pi\rho = s$; let $\sigma : \Sigma^+ \to S$ and $g : \Sigma^* \to A$ (with $g(\varepsilon) = 1_A$) be its coordinates. The positive image $S_0 = \sigma(\Sigma^+)$ is locally trivial. Put $q = |S_0| + 1$. The finite locally trivial identities $S_0^q = S_0 E(S_0) S_0$ and $euf = ef$ for idempotents e, f and $u \in S_0$ [27] show that $\sigma(w)$ is determined by the length-q prefix and suffix of w when $|w| \geq 2q$: writing $w = pxz$ with $|p| = |z| = q$, $\sigma(p) = aeb$, $\sigma(z) = cfd$ and $\sigma^1(\varepsilon) = 1$, $\sigma(pxz) = ae\,(b\,\sigma^1(x)\,c)\,fd = aefd = \sigma(pz)$, where the parenthesized factor lies in S_0 because $b, c \in S_0$. Words shorter than $2q$ are recorded exactly by $\widehat{s}_{q,q}$. Hence equal $(\widehat{s}_{q,q}, g)$-values on Σ^+ give equal ρ- and $\widehat{s}$-values; the window typing separates ε from Σ^+. Take $k = \ell = q$. If $\mathsf{S} \in \mathbf{LI}$, use $S = \mathsf{S}$, $\pi = \mathrm{id}$ and the trivial group. □

Corollary 4.17 (Algebraic sufficient conditions for polynomial data)**.** *Fix a nonempty finite Σ, a fan-out cap $f \geq 1$, a rule-rank cap $r_{\max} \geq 1$, and a finite-semigroup typing $s : \Sigma^+ \to \mathsf{S}$; write $\widehat{s}$ for its canonical completion. Let G be any reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of a target in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, and put $\bar{\theta}_G = \max\{1, \theta_G\}$. The capped reconstruction operator has a characteristic sample of cardinality $O_{f,r_{\max},s}(\|G\|)$ with:*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},s,\Sigma}(\|G\|^2 \bar{\theta}_G) \qquad \text{if } \mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}.$$

The locally trivial case **LI** *is included, with a trivial group factor.*

Proof. Combine Lemma 4.16 with Theorem 4.8 or Theorem 4.14; the refining parameters depend only on s and Σ, not on G. □

The local-group characterization and the detailed shortening argument appear in Appendix B. A finite semigroup belongs to **LG** (the *local groups*) if eSe is a group for every idempotent $e \in S$. This condition concerns the *positive image*, never the fresh-identity completion.

Theorem 4.18 (Polynomial characteristic data for local-group typing)**.** *Fix $f, r_{\max}, \Sigma$ and a finite-semigroup typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ with $\mathsf{S} \in \mathbf{LG}$; write $\widehat{s}$ for its canonical fresh-identity completion. For every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G with $L(G) = L \cap \Sigma^+$, the capped operator $\mathcal{B}^{\leq r_{\max}}_{f,s}$ has a characteristic sample with*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},s,\Sigma}(\|G\|^3 \bar{\theta}_G).$$

Proof. By the classical finite local-group representation, implemented in Theorem B.5 of Appendix B, there are fixed k, a finite group A and a sliding-factor labelling ν such that $\widehat{s} \preceq \lambda_{k,\nu}$. Lemma B.7 constructs, for each realizable $\lambda_{k,\nu}$-tuple type, a representative of length $O(\|G\|^2 \bar{\theta}_G)$: first replace heavy subtrees by short window-compatible tuples, then select a nonempty simultaneous family whose ordered group products remain unchanged (Lemmas B.3 and A.2). Type-preserving replacement (Lemma 4.3) transfers these representatives to the typed refinement, which has $O(\|G\|)$ relevant rules. The witness-length template (Lemma 4.2) gives the cubic total-length bound. The coordinate construction and all intermediate constants are proved in Appendix B. □

The strict inclusion $\mathbf{LI} \vee \mathbf{Grp} \subsetneq \mathbf{LG}$ is proved in Proposition B.6. The Rees/sliding-factor coordinates are classical; their combination with owner-wise MCFG block shortening is the quantitative step here. Window–group typings retain the sharper quadratic bound of Theorem 4.14.

4.5.1 The fan-out-one boundary

For a pseudovariety $\mathbf{V}$ of finite semigroups and $f \geq 1$, say that $\mathbf{V}$ *admits polynomial characteristic data at fan-out f* if for every nonempty finite alphabet Σ, every $r_{\max} \geq 1$ and every surjective typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ with $\mathsf{S} \in \mathbf{V}$ there is a polynomial $p_{s,r_{\max}}$ such that every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G with $L(G) = L \cap \Sigma^+$ admit a characteristic sample K for $\mathcal{B}^{\leq r_{\max}}_{f,s}$ with $\|K\|_+ \leq p_{s,r_{\max}}(\|G\|, \bar{\theta}_G)$. The polynomial may depend on the typing; only its existence is at issue.

Theorem 4.19 (Local-group sufficiency at fan-out one)**.** *Every pseudovariety $\mathbf{V} \subseteq \mathbf{LG}$ admits polynomial characteristic data at fan-out one; the polynomial may be taken of the form $O_{r_{\max},s,\Sigma}(\|G\|^3 \bar{\theta}_G)$.*

Proof. Let $\mathsf{S} \in \mathbf{V} \subseteq \mathbf{LG}$ and apply Theorem 4.18 with $f = 1$. □

At fan-out one an MCFG is a CFG, so Theorem 4.19 is a statement about typed CFL learning. It strictly extends the companion sufficient condition $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$ [25], by Proposition B.6. Combined with the companion lower bound it yields the following boundary statement.

Corollary 4.20 (The fan-out-one boundary)**.** *For a pseudovariety $\mathbf{V}$ of finite semigroups, $\mathbf{V}$ admits polynomial characteristic data at fan-out one if and only if $\mathbf{V} \subseteq \mathbf{LG}$.*

Proof. Sufficiency follows from Theorem 4.19. For necessity, let $\mathsf{S} \in \mathbf{V} \setminus \mathbf{LG}$. By Theorem B.5(ii), S contains a subsemigroup isomorphic to the two-element semilattice U_1. Pseudovarieties are closed under subsemigroups, so $U_1 \in \mathbf{V}$. Appendix G records, with attribution to the companion [25, Proposition 6.4], a surjective typing s_{U} onto U_1 and finite targets $L_n \supsetneq L_n^-$ in $\mathcal{C}^{\mathrm{mcf}}_{1,s_{\mathrm{U}}}$. By Proposition G.3, L_n has reduced good rank-two presentations G_n with $\|G_n\| = O(n)$ and $\theta_{G_n} \leq 4n$. Both targets are finite, so they have reduced good rank-two presentations, and Theorem 3.10 gives characteristic samples for the capped set-driven operator $\mathcal{B}^{\leq 2}_{1,s_{\mathrm{U}}}$ on both targets. Lemma G.1 forces every characteristic sample for L_n to contain w_n^*, whose length is $3 \cdot 2^n - 2$. Consequently its total encoded length cannot be bounded by a polynomial in $(\|G_n\|, \bar{\theta}_{G_n})$, contradicting the fan-out-one polynomial-data property with $r_{\max} = 2$. □

Remark 4.21 (Scope of the boundary statement). Corollary 4.20 concerns pseudovarieties, not individual typings: it asserts that *some* typing with positive image in $\mathbf{V}$ forces superpolynomial data, and only relative to presentation size and rule thickness (each L_n is finite, so a listing presentation has size at least 2^n). The new ingredient here is sufficiency; the lower bound originates in the companion and is reproduced in Appendix G for verification. Proposition 4.5 shows the same semilattice obstruction on the side of the full yield-type refinement. At fan-out two or more no lower bound is claimed (Section 6 and Open Problem 7.1).

4.6 Unrestricted Standard Input MCFGs

The preceding bounds use a reduced good presentation to make the quantitative witness argument transparent. Goodness, nondeletion, nonpermutation, nonmerging, and nonemptiness of tuple components need *not* be imposed on the grammar supplied for the target language. What matters when empty components are permitted is the cost of realizing each nonempty-component pattern.

Definition 4.22 (Component-pattern thickness)**.** For a productive nonterminal A of a standard MCFG H, let $\mathrm{supp}(\vec{u}) := \{i : u_i \neq \varepsilon\}$ and $\mathsf{Pat}_H(A) := \{\mathrm{supp}(\vec{u}) : \vec{u} \in L_A(H)\}$. Put

$$t_{\mathrm{pat}}(H) := \max_{\substack{A \text{ productive} \\ I \in \mathsf{Pat}_H(A)}} \min_{\substack{\vec{u} \in L_A(H) \\ \mathrm{supp}(\vec{u}) = I}} \|\vec{u}\|_1, \qquad \bar{t}_{\mathrm{pat}}(H) := \max\{1, t_{\mathrm{pat}}(H)\},$$

with the maximum over an empty set taken as zero. In particular, the all-empty pattern costs zero.

Proposition 4.23 (Good normalization controlled by pattern thickness)**.** *Fix $f, r \geq 1$. Let H be any standard fan-out-$\leq f$, rank-$\leq r$ MCFG with $L(H) \cap \Sigma^+ \neq \emptyset$, and let H_{red} be its trimmed presentation. There exists a reduced good MCFG G for $L(H) \cap \Sigma^+$ with the same fan-out and rank bounds and*

$$\|G\| = O_{f,r}(\|H\|), \qquad t_G \leq t_{\mathrm{pat}}(H_{\mathrm{red}}), \qquad \bar{\theta}_G = O_{f,r}(\|H\| \bar{t}_{\mathrm{pat}}(H_{\mathrm{red}})).$$

Proof. Use Proposition 3.9 and the stagewise tuple-language identities of Lemma F.1. Every retained nonterminal language is an image of a union of original support-pattern fibres under component deletion, permutation and concatenation. These maps do not increase total yield length, so a shortest representative of one contributing original pattern bounds each retained nonterminal's shortest yield by $t_{\mathrm{pat}}(H_{\mathrm{red}})$. Lemma 4.1(iii) then gives the thickness estimate. The precise fibre-transfer argument is in Appendix C. □

Corollary 4.24 (Total characteristic-data length from any standard MCFG)**.** *Fix $f, r \geq 1$, a finite alphabet Σ, and a fixed finite typing s. Let H be* any *finite standard fan-out-$\leq f$, rank-$\leq r$ MCFG (deleting rules, permutations, merging and empty components permitted) for $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. Put $n := \max\{1, \|H\|\}$ and $t := \bar{t}_{\mathrm{pat}}(H_{\mathrm{red}})$. The capped reconstruction operator admits a characteristic sample $K \subseteq L$ of cardinality $O_{f,r,s}(n)$ with*

$$\|K\|_+ = \begin{cases} O(n^3 t), & \text{if } s \preceq s_{k,\ell} \text{ or } \widehat{s} \preceq (\widehat{s}_{k,\ell}, g) \text{ for a fixed finite-group morphism } g, \\ O(n^4 t), & \text{if } \mathsf{S} \in \mathbf{LG}. \end{cases}$$

The hidden constants depend on the fixed alphabet, typing, rank, fan-out, and (where applicable) window and group, not on H; an optional datum ε costs one.

Proof. For a nonempty positive core apply Proposition 4.23 and then Theorem 4.8, 4.14, or 4.18, respectively. The size is $O(n)$ and the new thickness is $O(nt)$. If $L \subseteq \{\varepsilon\}$, use $K = L$. □

4.7 Computing Short Characteristic Data

The bounds above exhibit some short tuple and some short context. Call a sentence context of a typed nonterminal X *derivational* if it arises from a successful typed derivation by deleting one occurrence of an X-subtree; such contexts are admissible. All canonical contexts used in Sections 4.1–4.5 and Appendix B are derivational (Lemma 4.1(iv) and the proofs of Lemma 4.4 and Theorem 4.6).

Proposition 4.25 (Effective short characteristic data)**.** *Fix $f, r_{\max}$ and a finite typing $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$. Given a reduced good MCFG G of fan-out at most f and rule rank at most $r_{\max}$, the following can be computed in time polynomial in $\|G\|$, with lengths written in binary:*

(i) $\theta_G(R)$ for every rule R, and hence θ_G and t_G;

(ii) the trimmed refinement $\widetilde{G}_0$ and, for each non-start typed nonterminal X, a canonical tuple $\omega(X) \in L_X$ of minimum total length and a derivational context $\chi(X)$ of minimum fixed length, with $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ belongs to $\widetilde{G}_0$;

(iii) derivation DAGs of size polynomial in $\|G\|$ representing these tuples and contexts, and hence every word of the witness set $\mathsf{W}(\widetilde{G}_0)$ of Definition 3.6 built from them.

Every word-length and total-length bound proved for $\mathsf{W}(\widetilde{G}_0)$ in Sections 4.1–4.5 and in Appendix B holds for this computed witness set. This does not apply to the relaxed samples $K^\dagger$ of Section 4.8, whose representatives need not be generated by their nonterminals. The words can be written out explicitly in time polynomial in $\|G\| + \|\mathsf{W}(\widetilde{G}_0)\|_+$.

Proof. For fixed $f, r_{\max}, s$ the full refinement $G^{\widehat{s}}$ has at most $|V||\mathsf{S}^\sharp|^f$ typed nonterminals and $|P||\mathsf{S}^\sharp|^{fr_{\max}}$ typed rules, and trimming is a linear-time productivity and reachability computation. View the typed rules as weighted deduction steps. A rule $X \to \rho(Y_1, \dots, Y_r)$ maps child weights to $(c_\rho, 1) + \sum_j (x_j, n_j)$, where c_ρ is the number of terminals written by ρ and pairs are compared lexicographically (terminal length, derivation-node count). This function is strictly superior to each argument even when no terminal is written, so the shortest-derivation algorithm in the weighted deductive form of Nederhof [36], building on Knuth's generalization of Dijkstra's method (see also Ramalingam and Reps [35]), computes $\ell_{\text{in}}(X) = \min\{\|\vec{u}\|_1 : \vec{u} \in L_X\}$ together with an acyclic choice of minimizing rules, that is, a derivation DAG for $\omega(X)$. A second Dijkstra pass computes outside lengths, breaking equal-length ties by the number of edges: it starts with $\ell_{\text{out}}(X) = 0$ when $\widetilde{S} \to X$ is present, and for every typed rule $X \to \rho(Y_1, \dots, Y_r)$ and every j adds an edge from X to Y_j of weight $c_\rho + \sum_{i \neq j} \ell_{\text{in}}(Y_i)$. Then $\ell_{\text{out}}(X)$ is the minimum fixed length of a derivational context of X, and the shortest-path tree supplies its DAG. The same two passes on G give $t_G = \max_A \ell_{\text{in}}(A)$ and $\theta_G(R) = \ell_{\text{out}}(A) + c_\rho + \sum_j \ell_{\text{in}}(B_j)$ for $R : A \to \rho(B_1, \dots, B_r)$.

Each of the cited proofs bounds a witness word through the length of its context, of its rule template and of its canonical child tuples, and does so by exhibiting *some* tuple of L_X and *some* derivational context. The computed minima are no longer, and the number of witnesses is unchanged, so every stated bound persists. Expanding a DAG takes time linear in the length of the word it represents. □

Proposition 4.26 (Rule-only witnesses and a computable data budget)**.** *In the setting of Proposition 4.25, choose all canonical tuples by the consistent acyclic minimum-weight derivations described there, and let $\mathcal{R}$ be the non-start typed rules of $\widetilde{G}_0$ (rank-zero rules included).*

(i) For each non-start typed nonterminal X, let $R_X : X \to \rho_X(Y_1, \dots, Y_q)$ be the first rule of its chosen minimizing derivation (including $q = 0$). Then $\omega(X) = \rho_X(\omega(Y_1), \dots, \omega(Y_q))$. Consequently

$$\mathsf{W}(\widetilde{G}_0) = \{\chi(\operatorname{lhs} R)[\rho_R(\omega(Y_1), \dots, \omega(Y_q))] : R \in \mathcal{R}\}, \qquad |\mathsf{W}(\widetilde{G}_0)| \le |\mathcal{R}| \le |P||\mathsf{S}^\sharp|^{fr_{\max}},$$

and this set remains characteristic for $\mathcal{B}_{f,s}^{\le r_{\max}}$ whenever the target is in $\mathcal{C}_{f,s}^{\mathrm{mcf}}$.

(ii) Let $\ell_{\mathrm{in}}(X)$ be the minimum total yield length from X and $\ell_{\mathrm{out}}(X)$ the minimum fixed length of a derivational context of X (with $\ell_{\mathrm{out}}(X) = 0$ for a direct start link). For $R : X \to \rho(Y_1, \dots, Y_q)$ put

$$b(R) := c_\rho + \sum_{j=1}^{q} \ell_{\mathrm{in}}(Y_j), \qquad \mathsf{B}_{\mathrm{rule}}(G, \widehat{s}) := \sum_{R \in \mathcal{R}} (1 + \ell_{\mathrm{out}}(\operatorname{lhs} R) + b(R)).$$

Then $\|\mathsf{W}(\widetilde{G}_0)\|_+ \le \mathsf{B}_{\mathrm{rule}}(G, \widehat{s})$, and the integer $\mathsf{B}_{\mathrm{rule}}(G, \widehat{s})$ is computable in time polynomial in $\|G\|$ for fixed $f, r_{\max}, s$, its value being represented in binary. If $L(G)$ also contains ε and its nonempty core is treated separately, the ε datum adds at most one to the budget.

Proof. (i) An optimal derivation of X starts with R_X, and each of its child derivations is optimal, since replacing a nonoptimal child by the chosen minimum would decrease the lexicographic weight. The node-count tie-breaker makes the simultaneous choice acyclic, which gives the identity, also for rank zero. Hence $\chi(X)[\omega(X)]$ is *identical* to the rule word of R_X, and every other term of Definition 3.6 is a rule word; the two sets coincide as sets of terminal words, so completeness and the characteristic-sample property are unchanged. There is at most one word per typed rule. (ii) The rule word of $R : X \to \rho(Y_1, \dots, Y_q)$ has length exactly $\ell_{\mathrm{out}}(X) + b(R)$. Summing $|w| + 1$ over the indexed family gives $\mathsf{B}_{\mathrm{rule}}(G, \widehat{s})$; passing to distinct words can only lower it. The values $\ell_{\mathrm{in}}, \ell_{\mathrm{out}}$ and the number of typed rules are computed, with polynomial binary length, by Proposition 4.25. □

This rule-only budget is presentation-sensitive and can be much smaller than the worst-case polynomial bounds; it is an efficiently computable *upper certificate*, not a claim of a minimum-size characteristic sample or an improved worst-case exponent. For the singleton $\{\mathsf{c}^{2^n}\}$ after Theorem 4.6, explicit words are exponentially longer than $\|G\|$, but the derivation DAGs remain of polynomial size.

4.8 Distributional Representatives and Abelian-Filtered Targets

For an existential characteristic-sample bound one may instead choose short representatives of the same *typed distribution classes*, even when they have no derivation from the corresponding nonterminal.

Proposition 4.27 (Relaxed distributional witnesses)**.** *Let G be a reduced good fan-out-$\le f$, rank-$\le r$ presentation of $L = L(G) \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$, and let $\widetilde{G}_0$ be its trimmed yield-type refinement. For every non-start typed nonterminal X, let $[X]$ be the $\equiv_{L,s}$-class containing L_X, and put $\kappa_G(X) = \min_{\vec{v} \in [X]} \|\vec{v}\|_1$, $\kappa_G = \max_X \kappa_G(X)$. There exists a characteristic positive sample $K^\dagger$ for both the capped operator $\mathcal{B}_{f,s}^{\le r}$ (with cap at least the rank of G) and the uncapped $\mathcal{B}_{f,s}$ satisfying*

$$|K^\dagger| \le |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{fr},$$
$$\|K^\dagger\|_+ \le (|V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{fr})(1 + \theta_G + \|G\| + (r+1)\kappa_G).$$

The bound concerns the nonempty core; if $\varepsilon \in L$, add the sample word ε at encoded cost one.

Proof. For each $X = A_{\mathbf{t}}$ choose $\omega^\dagger(X) \in [X]$ of total length $\kappa_G(X)$, and let $\chi^\dagger(X) := E_A$ from Lemma 4.1(iv), of fixed length at most θ_G ($\Box_1$ for a start link). It accepts every tuple of L_X, hence also $\omega^\dagger(X) \equiv_{L,s} \vec{u} \in L_X$. Let $K^\dagger$ consist of the words $\chi^\dagger(X)[\omega^\dagger(X)]$, the rule words $\chi^\dagger(X)[\rho(\omega^\dagger(Y_1), \dots, \omega^\dagger(Y_q))]$ and the rank-zero rule words. By Lemma D.2 the rule tuple is equivalent to a tuple of L_X, so $K^\dagger \subseteq L$; it also has the $\widehat{s}$-type of X. Apply the generalized Proposition 3.8 with $\iota(X) = \omega^\dagger(X)$ and $\chi(X) = \chi^\dagger(X)$: the first family of sample words is its representative evidence and the rule-word family is precisely its rule evidence. Componentwise types agree by the typed rules and Lemma D.2; all contexts are admissible and start-linked contexts are identities. The proposition supplies the capped and uncapped derivation simulations. Proposition D.3 gives the reverse inclusion for all larger positive samples. A nonterminal word has length at most $\theta_G + \kappa_G$ and a rule word at most $\theta_G + \|G\| + r\kappa_G$; sum the encoded costs. Unlike Proposition 4.26, the nonterminal words cannot in general be omitted. $\square$

For fixed $f, r, \widehat{s}$, a family G_n with $\|G_n\| \max\{1, \theta_{G_n}\} = n^{O(1)}$ and superpolynomial characteristic-data requirements must therefore have $\kappa_{G_n} = n^{\omega(1)}$; this is a necessary condition, not a proof that such a family exists.

Theorem 4.28 (Polynomial data for abelian-filtered targets)**.** *Fix a finite nonempty alphabet Σ, a finite semigroup typing $s : \Sigma^+ \to \mathsf{S}$ and bounds $f, r \geq 1$. Write $\widehat{s}$ for the canonical completion and $\mathsf{S}^\sharp$ for its codomain. There are constants C, e depending only on Σ, s, f, r with the following property. Let A be any abelian group, $\varphi : \Sigma^* \to \mathsf{A}$ a morphism, $\mathsf{H} \leq \mathsf{A}$, $\gamma \in \mathsf{A}$, and $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$. If $L = \widehat{s}^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathsf{H})$ has a reduced good fan-out-$\leq f$, rank-$\leq r$ MCFG presentation G of $L \cap \Sigma^+$, then $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and admits a characteristic sample for $\mathcal{B}^{\leq r}_{f,s}$ with*

$$\|K^\dagger\|_+ \leq C\big(\|G\| \max\{1, \theta_G\}\big)^e.$$

The constants do not depend on $\mathsf{A}, \varphi, \mathsf{H}, \gamma$. The datum ε is included when $\varepsilon \in L$; the degenerate cases $L \subseteq \{\varepsilon\}$ are handled directly.

Proof. Membership in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ follows from Proposition 2.9 and the remark preceding it; the positive part has the assumed presentation, and ε is admitted by Start. By Proposition 4.27 it suffices to bound κ_G by a polynomial in $\|G\|\bar{\theta}_G$ with exponent depending only on $\Sigma, \widehat{s}, f, r$.

Let $\psi : \Sigma^* \to \mathbb{N}^\Sigma$ be the Parikh map, $\Psi(\vec{u}) = \sum_i \psi(u_i)$, and let $\overline{\varphi} : \mathbb{Z}^\Sigma \to \mathsf{A}$ be the group homomorphism through which φ factors; put $\Lambda = \{z \in \mathbb{Z}^\Sigma : \overline{\varphi}(z) \in \mathsf{H}\}$. Fix for each $C \in V$ a tuple $\omega_C \in L_C(G)$ with $\|\omega_C\|_1 \leq t_G \leq \theta_G$. The context E_C accepts every C-tuple, and all accepted words lie in one coset of H, so

$$\Psi(\vec{u}) - \Psi(\omega_C) \in \Lambda \qquad (\vec{u} \in L_C(G)). \tag{$*$}$$

For $R : C \to \rho(B_1, \dots, B_q)$ let $c_R \in \mathbb{N}^\Sigma$ count its explicit letters and $\delta_R = \Psi(\omega_C) - c_R - \sum_j \Psi(\omega_{B_j})$. By ($*$) applied to $\rho(\omega_{B_1}, \dots, \omega_{B_q}) \in L_C(G)$, $\delta_R \in \Lambda$, and $\|\delta_R\|_1 \leq T := (r+1)t_G + \|G\|$. Let $\Gamma_T = \{z \in \Lambda : \|z\|_1 \leq T\}$ and $\Lambda_T = \langle \Gamma_T \rangle$. Induction on derivations, using $\Psi(\vec{u}) - \Psi(\omega_C) = \sum_j (\Psi(\vec{u}_j) - \Psi(\omega_{B_j})) - \delta_R$ for $\vec{u} = \rho(\vec{u}_1, \dots, \vec{u}_q)$, shows $\Psi(\vec{u}) - \Psi(\omega_C) \in \Lambda_T$ for every $\vec{u} \in L_C(G)$.

Now let $X = C_{\mathbf{t}}$ have arity $d \leq f$ and fix $\vec{u} \in L_X$. For each i choose, from a fixed semilinear representation of the Parikh image of the regular language $\widehat{s}^{-1}(t_i) \cap \Sigma^+$, a linear set $c_i + \sum_{p \in P_i} \mathbb{N}p$ containing $\psi(u_i)$; the number and size of all such generators depend only on Σ and $\widehat{s}$. The nonnegative integer system

$$\sum_{i=1}^{d} \Big(c_i + \sum_{p \in P_i} \eta_{i,p} p \Big) - \sum_{z \in \Gamma_T} (\mu^+_z - \mu^-_z) z = \Psi(\omega_C)$$

is feasible (use the Parikh vectors of the u_i and the Λ_T-membership of $\Psi(\vec{u}) - \Psi(\omega_C)$). It has $|\Sigma|$ equations, $O_{f,s,\Sigma}((T+1)^{|\Sigma|})$ variables, and coefficients and right-hand side of absolute value

$O_{f,s,\Sigma}(T+1)$; in particular, each coordinate of $\Psi(\omega_C)$ is at most $t_G \le \theta_G \le T$ for the nonempty core. Papadimitriou's bound for small nonnegative integral solutions of equality systems [40] (a solution with all entries at most $n(ma)^{2m+1}$ for m equations, n variables and entries bounded by a) gives a solution whose entries are bounded by a polynomial in $T+1$ with exponent depending only on $|\Sigma|$. Realize each resulting Parikh vector by a nonempty word $v_i \in \widehat{s}^{-1}(t_i)$ and put $\vec{v} = (v_1, \dots, v_d)$. Then $\|\vec{v}\|_1$ is polynomial in $T+1$, $\widehat{s}^{(d)}(\vec{v}) = \mathbf{t}$, and $\Psi(\vec{v}) - \Psi(\vec{u}) \in \Lambda_T \subseteq \Lambda$. For every sentence context E, the words $E[\vec{v}]$ and $E[\vec{u}]$ have the same $\widehat{s}$-value and $\varphi(E[\vec{v}]) - \varphi(E[\vec{u}]) = \overline{\varphi}(\Psi(\vec{v}) - \Psi(\vec{u})) \in \mathsf{H}$; hence $\vec{v} \equiv_{L,s} \vec{u}$ and $\kappa_G(X) \le \|\vec{v}\|_1$. Since $T \le (r+1)\theta_G + \|G\|$, the claim follows, with parameters independent of the auxiliary abelian group. □

Theorem 4.28 applies even when the positive image of $\widehat{s}$ lies *outside* **LG**: it is a sufficient condition on the target language, not on the finite control, and it claims no polynomial-time construction for ineffectively presented auxiliary groups.

5 Comparison with Multidimensional Substitutability

Yoshinaka's multidimensional substitutability is the direct positive-data baseline [13, 14]. Under trivial typing the present framework reproduces that semantic class; finite typing refines which observed tuples may be identified while preserving compositional rule witnesses, and $\widehat{s}_{k,\ell}$ lifts the boundary information of Yoshinaka's CFG (k,ℓ) hierarchy [12] componentwise. Since every (f,s) condition includes arity one, $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathrm{CFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{1,s}$.

5.1 Yoshinaka's Baseline and Prefix–Suffix Specialization

Definition 5.1 (Yoshinaka multidimensional substitutability)**.** Using the sentence-context sets $\mathcal{E}_d$ defined in Section 2, a language L is *fD-substitutable* if, for every $1 \le d \le f$, all $E, F \in \mathcal{E}_d$, and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$, $E[\vec{x}], \quad E[\vec{y}], \quad F[\vec{x}] \in L \Longrightarrow F[\vec{y}] \in L$. Write $\mathsf{YSub}(f)$ for this semantic class and $\mathsf{YSL}(f,r) := \mathsf{YSub}(f) \cap \mathsf{MCFL}(f,r)$, where r bounds rule rank. We refer to the displayed four-instance implication as the *substitution implication*. Yoshinaka's original definition quantifies over every arity $d \le f$, uses nonempty tuple components, and requires nonempty internal separators in d-contexts; footnote 3 separately discusses the stronger alternative that permits empty components and separators [14, Section 3.1 and footnote 3].

The trivial typing removes the componentwise type premise, so $\mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{triv}}} = \mathsf{YSub}(f) \cap f\text{-MCFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,s}$. In Yoshinaka's word-level (k,ℓ) condition [12], with $u \in \Sigma^k$, $v \in \Sigma^\ell$ and $uy_1v, uy_2v \ne \varepsilon$,

$$w_1uy_1vw_1',\ w_1uy_2vw_1' \in L \quad \Longrightarrow \quad (w_2uy_1vw_2' \in L \iff w_2uy_2vw_2' \in L).$$

The compared fragments uy_iv are nonempty although the y_i may be empty, and equality of their $\widehat{s}_{k,\ell}$-values is exactly the boundary information used here, including exact identity of words shorter than $m_0 = \max\{1, k+\ell\}$. Hence $(1, \widehat{s}_{k,\ell})$-substitutability is precisely Yoshinaka's (k,ℓ) condition, and the analogous componentwise boundary requirement at higher fan-out is a definition of this paper, not a claim that Yoshinaka (2008) formulated an MCFG (k,ℓ) notion.

A language is *strictly 2-local* if membership is determined by permitted adjacent pairs in $\vdash w \dashv$, for fixed endmarkers.

Lemma 5.2 (Strictly 2-local constraints are boundary typed)**.** *If $L_{\mathrm{sl}} \subseteq \Sigma^*$ is strictly 2-local, then for every $f \ge 1$, $L_{\mathrm{sl}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{1,1}}$.*

Proof. Let equal-$\widehat{s}_{1,1}$-type nonempty tuples $\vec{x}, \vec{y}$ share an accepting context E_0. Internal adjacent pairs of each y_i are permitted because they occur in $E_0[\vec{y}]$. If $E[\vec{x}] \in L_{\mathrm{sl}}$, every adjacent pair of $E[\vec{y}]$ is either in fixed material, internal to a y_i, or crosses a component boundary. These are permitted respectively by $E[\vec{x}]$, by $E_0[\vec{y}]$, or by equality of the first/last letters encoded by the boundary type; end-marker pairs are unchanged. Thus $E[\vec{y}] \in L_{\mathrm{sl}}$, and symmetry finishes the proof. □

Proposition 5.3 (Cross-serial dependencies with finite feature licensing)**.** *Let* $\Sigma_{\mathrm{feat}} = \{\mathtt{p}_0, \mathtt{p}_1, \mathtt{q}_0, \mathtt{q}_1, \mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}\}$, $\mathcal{R}_{\mathrm{feat}} = \{(0,0), (0,1), (1,0)\}$, *and*

$$L_{\mathrm{feat}} = \{\mathtt{a}^m \mathtt{p}_i \mathtt{b}^n \mathtt{c}^m \mathtt{q}_j \mathtt{d}^n : m, n \geq 1,\ (i,j) \in \mathcal{R}_{\mathrm{feat}}\}.$$

Let $\mathsf{F} = \mathbb{Z}_2 \times \mathbb{Z}_2$, *and define a morphism* $g : \Sigma_{\mathrm{feat}}^* \to \mathsf{F}$ *by* $g(\mathtt{p}_1) = (1,0)$, $g(\mathtt{q}_1) = (0,1)$, *and* $g(z) = (0,0)$ *for all other letters* z. *Fix the positive-image semigroup typing*

$$s = (s_{1,1}, g)|_{\Sigma_{\mathrm{feat}}^+},$$

where $s_{1,1}$ *is the first/last-letter morphism and the codomain is the finite positive image of this product. Then, for every* $f \geq 2$,

$$L_{\mathrm{feat}} \in \mathcal{C}_{f,s}^{\mathrm{mcf}} \setminus \mathsf{YSub}(f), \qquad L_{\mathrm{feat}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}.$$

Moreover L_{feat} *is not even* $(1, s_{1,1})$*-substitutable: the finite group component is needed by this example.*

Proof. Two crossed dependencies. Use fan-out-two nonterminals A, B with

$$A \to (\mathtt{a}, \mathtt{c}) \mid (\mathtt{a}\xi_1^1, \mathtt{c}\xi_1^2)(A),$$
$$B \to (\mathtt{b}, \mathtt{d}) \mid (\mathtt{b}\xi_1^1, \mathtt{d}\xi_1^2)(B).$$

For each $(i,j) \in \mathcal{R}_{\mathrm{feat}}$ add the rank-two rule

$$S \to \xi_1^1 \mathtt{p}_i \xi_2^1 \xi_1^2 \mathtt{q}_j \xi_2^2 (A, B).$$

These rules generate exactly L_{feat}, using both components of each child once and ordering them as A_1, B_1, A_2, B_2. The grammar is nondeleting, nonmerging, and has fan-out and rank two. The central crossing-count language $\{\mathtt{a}^m \mathtt{b}^n \mathtt{c}^m \mathtt{d}^n : m, n \geq 1\}$ is not context-free: for a pumping length p, a window of length p in $a^p b^p c^p d^p$ affects at most two adjacent letter blocks, and pumping cannot preserve both a/c and b/d equalities. Intersecting L_{feat} with the regular language $\mathtt{a}^+\mathtt{p}_0\mathtt{b}^+\mathtt{c}^+\mathtt{q}_0\mathtt{d}^+$ and erasing $\mathtt{p}_0, \mathtt{q}_0$ would yield this non-context-free language, so L_{feat} is not context-free.

Finite group-valued feature licensing. Put $Q = \mathtt{a}^+\{\mathtt{p}_0, \mathtt{p}_1\}\mathtt{b}^+\mathtt{c}^+\{\mathtt{q}_0, \mathtt{q}_1\}\mathtt{d}^+$. The language Q is strictly 2-local: its allowed adjacent pairs and endmarkers determine precisely these blocks and their order. Define the regular feature-compatibility filter $K_{\mathrm{feat}} = g^{-1}(\mathcal{R}_{\mathrm{feat}})$, viewing each (i,j) as the corresponding element of F. Because s includes g, this filter is the inverse image of an accepting subset of the finite s-type space. Let $\varphi : \Sigma_{\mathrm{feat}}^* \to \mathbb{Z}^2$ map $\mathtt{a}, \mathtt{c}, \mathtt{b}, \mathtt{d}$ to $(1,0), (-1,0), (0,1), (0,-1)$, respectively, and map the feature markers to $(0,0)$. Then

$$L_{\mathrm{feat}} = Q \cap K_{\mathrm{feat}} \cap \varphi^{-1}(0).$$

Lemma 5.2 applies to Q because s refines $s_{1,1}$. The inverse-image observation after Definition 2.8 applies to K_{feat}, and Proposition 2.9 to the count filter. Their intersection is (f, s)-tuple-substitutable for every f.

Why the untyped and boundary-only implications fail. Take $\vec{x} = \mathtt{ap}_0\mathtt{b}$, $\vec{y} = \mathtt{ap}_1\mathtt{b}$, $E = \Box_1\mathtt{cq}_0\mathtt{d}$, $F = \Box_1\mathtt{cq}_1\mathtt{d}$. Then $E[x], E[y], F[x] \in L_{\mathrm{feat}}$, but $F[y] \notin L_{\mathrm{feat}}$, contradicting the arity-one substitution implication. These fragments have the same first/last-letter type $(\mathtt{a}, \mathtt{b})$, but $g(x) = (0,0) \neq (1,0) = g(y)$. Thus neither trivial typing nor the boundary window alone blocks the unsound substitution, whereas the fixed product typing does. □

Corollary 5.4 (Uniform finite feature licensing)**.** *The* same *fixed typing* $s = (s_{1,1}, g)|_{\Sigma^+_{\mathrm{feat}}}$ *from Proposition 5.3 works for every relation* $R \subseteq \{0,1\}^2$*: putting*

$$L_R = \{\mathtt{a}^m\mathtt{p}_i\mathtt{b}^n\mathtt{c}^m\mathtt{q}_j\mathtt{d}^n : m, n \geq 1,\ (i,j) \in R\},$$

we have $L_R \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ *for all* $f \geq 2$*. The typing is chosen once for the entire family, not tailored to* R*.*

Proof. The grammar in Proposition 5.3 retains exactly the start rules indexed by R (or has no productive start rule if $R = \varnothing$). Its proof of typed substitutability applies with $g^{-1}(R)$ in place of $g^{-1}(\mathcal{R}_{\mathrm{feat}})$: each is an accepting union of fibres of the same fixed g. The other two filters do not depend on R. □

Remark 5.5 (What the feature example establishes). The markers represent a finite case-licensing or feature-compatibility constraint; no claim about the grammar of a specific natural language is made. Its nonrectangular three-pair relation makes untyped substitution unsound even between equal-boundary fragments. A finite group signature excludes that substitution, while two independent unbounded crossing-count dependencies remain to be learned. This illustrates a meaningful *known bias*, not an algorithm for learning the typing itself.

5.2 Targets outside window–group-determined typings

The feature-licensing example above is covered by a window and an abelian group. This subsection separates the typing classes behind Section 4 at the level of targets: one fan-out-two target is admitted by a local-group typing but by no window–group-determined typing, in particular by none in $\mathbf{LI} \vee \mathbf{Grp}$; the other is admitted by a window–S_3 typing but by no typing in $\mathbf{LI} \vee \mathbf{Ab}$, where $\mathbf{Ab}$ is the pseudovariety of finite abelian groups. Both use the same template: a flag letter $\mathtt{r}$ leaves the following word unconstrained, a flag letter $\mathtt{l}$ imposes a typing-dependent condition on it, and a cross-serial counting block makes the target non-context-free.

Throughout, $\mathtt{1}, \mathtt{2}, \mathtt{3}, \mathtt{4}, \$, \mathtt{l}, \mathtt{r}$ are letters, and for a word w we write $\#_{\mathtt{ab}}(w)$ for the number of occurrences of the factor $\mathtt{ab}$ in w. Put $C_\times := \{\mathtt{1}^m\mathtt{2}^n\mathtt{3}^m\mathtt{4}^n : m, n \geq 1\}$ and let $\varphi_\times$ map $\mathtt{1}, \mathtt{3}, \mathtt{2}, \mathtt{4}$ to $(1,0), (-1,0), (0,1), (0,-1) \in \mathbb{Z}^2$ and every other letter to $(0,0)$.

Lemma 5.6 (Flag separation)**.** *Let* t *be a finite typing of* Σ^+*, let* $u, v \in \Sigma^+$ *with* $t(u) = t(v)$*, and let* $z \in \Sigma^*$*. If* L *contains* $\mathtt{r}uz$*,* $\mathtt{r}vz$ *and* $\mathtt{l}uz$ *but not* $\mathtt{l}vz$*, then* $L \notin \mathcal{C}^{\mathrm{mcf}}_{f,t}$ *for every* $f \geq 1$*.*

Proof. The arity-one contexts $E = \mathtt{r}\Box_1 z$ and $F = \mathtt{l}\Box_1 z$ belong to $\mathcal{E}_1$. E is a common accepting context of u and v, while F accepts u but not v, so $\mathcal{D}^{(1)}_L(u) \neq \mathcal{D}^{(1)}_L(v)$. Every (f,t) condition includes arity one. □

A separation from window–group typings. Let $\Sigma_{\mathrm{A}} = \{\mathtt{l}, \mathtt{r}, \mathtt{a}, \mathtt{b}, \mathtt{c}, \$, \mathtt{1}, \mathtt{2}, \mathtt{3}, \mathtt{4}\}$ and

$$L_{\mathrm{A}} := \{\sigma w\,\$\,c : \sigma \in \{\mathtt{l}, \mathtt{r}\},\ w \in \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}^+,\ c \in C_\times,\ (\sigma = \mathtt{r} \text{ or } \#_{\mathtt{ab}}(w) \text{ is even})\}.$$

Let $s_{\mathrm{A}} := \lambda_{1,\nu}|_{\Sigma^+_{\mathrm{A}}}$, where $\nu : \Sigma^2_{\mathrm{A}} \to \mathbb{Z}_2$ sends $\mathtt{ab}$ to 1 and every other factor to 0 (Definition B.1; this is $h_{\mathtt{ab}}$ of Proposition B.6 on the larger alphabet). Thus s_{A} records the first and last letters and, for words of length at least two, the parity of $\#_{\mathtt{ab}}$.

Proposition 5.7 (A target separating local-group and window–group typings)**.** (i) $L_{\mathrm{A}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}$.

(ii) $L_{\mathrm{A}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{A}}}$ *for every* $f \geq 2$, *and* $s_{\mathrm{A}}(\Sigma_{\mathrm{A}}^{+}) \in \mathbf{LG}$.

(iii) *For every finite typing* t *of* Σ_{A}^{+} *that is window–group-determined, that is,* $t \preceq (s_{k,\ell}, g'|_{\Sigma_{\mathrm{A}}^{+}})$ *for some* k, ℓ *and some morphism* g' *into a finite group, and in particular for every* t *whose positive image lies in* $\mathbf{LI} \vee \mathbf{Grp}$, *we have* $L_{\mathrm{A}} \notin \mathcal{C}^{\mathrm{mcf}}_{f,t}$ *for all* $f \geq 1$.

Consequently, among the window-type results, L_{A} *is covered by Theorem 4.18 but by none of Theorems 4.8, 4.14 and 4.15 or Corollary 4.17, for any choice of typing (but see Remark 5.9); and* $s_{\mathrm{A}}(\Sigma_{\mathrm{A}}^{+}) \notin \mathbf{LI} \vee \mathbf{Grp}$.

Proof. (i) Let W be a rank-one, fan-out-one good grammar for the regular language $\{\sigma w : \sigma \in \{\mathtt{l}, \mathtt{r}\},\ w \in \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}^{+},\ \sigma = \mathtt{r} \text{ or } \#_{\mathtt{ab}}(w) \text{ even}\}$ (a right-linear grammar tracking the last letter and the parity). Add fan-out-two nonterminals C, D and a start symbol S with

$$C \to (\xi_1^1 \$ \mathtt{1},\ \mathtt{3})(W) \mid (\xi_1^1 \mathtt{1},\ \xi_1^2 \mathtt{3})(C),$$
$$D \to (\mathtt{2},\ \mathtt{4}) \mid (\xi_1^1 \mathtt{2},\ \xi_1^2 \mathtt{4})(D),$$
$$S \to (\xi_1^1 \xi_2^1 \xi_1^2 \xi_2^2)(C, D).$$

Then $L_C = \{(\sigma w \$ \mathtt{1}^m, \mathtt{3}^m)\}$ with $\sigma w \in L(W)$, $L_D = \{(\mathtt{2}^n, \mathtt{4}^n)\}$, and S generates L_{A}. The start template is nonmerging because adjacent variables belong to different children. Intersecting with the regular set $\mathtt{ra}\$\mathtt{1}^{+}\mathtt{2}^{+}\mathtt{3}^{+}\mathtt{4}^{+}$ and erasing $\mathtt{ra}\$$ yields $\{\mathtt{1}^m\mathtt{2}^n\mathtt{3}^m\mathtt{4}^n\}$, which is not context-free.

(ii) Write $L_{\mathrm{A}} = Q_{\mathrm{A}} \cap \Pi_{\mathrm{A}} \cap \varphi_{\times}^{-1}(0)$ with $Q_{\mathrm{A}} = \{\mathtt{l}, \mathtt{r}\}\{\mathtt{a}, \mathtt{b}, \mathtt{c}\}^{+}\$\,\mathtt{1}^{+}\mathtt{2}^{+}\mathtt{3}^{+}\mathtt{4}^{+}$ and Π_{A} the set of nonempty words whose first letter is $\mathtt{r}$, or whose first letter is $\mathtt{l}$ and whose number of $\mathtt{ab}$ factors is even. On Q_{A} no factor $\mathtt{ab}$ meets σ, \$ or the counting block, so the parity is that of w.

- Q_{A} is strictly 2-local, with permitted pairs $\vdash\mathtt{l}$, $\vdash\mathtt{r}$, σx and xy for $\sigma \in \{\mathtt{l}, \mathtt{r}\}$ and $x, y \in \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$, $x\$$, $\$\mathtt{1}$, $\mathtt{11}$, $\mathtt{12}$, $\mathtt{22}$, $\mathtt{23}$, $\mathtt{33}$, $\mathtt{34}$, $\mathtt{44}$ and $\mathtt{4}\dashv$. By Lemma 5.2 it is $(f, s_{1,1})$-tuple-substitutable, hence (f, s_{A})-tuple-substitutable since $s_{1,1} \preceq s_{\mathrm{A}}$.
- $\Pi_{\mathrm{A}} = \widehat{s}_{\mathrm{A}}^{-1}(\mathsf{Acc})$ for the set Acc of types of long words whose first letter is $\mathtt{r}$, or whose first letter is $\mathtt{l}$ and whose group coordinate is 0, together with the two short types $\mathtt{r}$ and $\mathtt{l}$ (the latter has zero $\mathtt{ab}$-count). By the observation following Definition 2.8 it is (f, s_{A})-tuple-substitutable.
- $\varphi_{\times}^{-1}(0)$ is handled by Proposition 2.9(ii) with $\mathsf{H} = 0$, $\gamma = 0$.

Proposition 2.9(i) gives the intersection; with (i), $L_{\mathrm{A}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{A}}}$ for $f \geq 2$. The positive image lies in $\mathbf{LG}$ by Theorem B.5, (iv)⇒(i).

(iii) Fix k, ℓ and $g' : \Sigma_{\mathrm{A}}^{*} \to A'$. Let $\hat{k} = \max\{k, \ell, 1\}$, let $q \geq 2$ be an even multiple of the exponent of A', and put

$$u = \mathtt{a}^{\hat{k}}(\mathtt{abc})^q\mathtt{a}^{\hat{k}}, \qquad v = \mathtt{a}^{\hat{k}}(\mathtt{bac})^q\mathtt{a}^{\hat{k}}.$$

As in Proposition B.6(ii), both are long with prefix $\mathtt{a}^k$ and suffix $\mathtt{a}^\ell$, and $g'(u) = g'(\mathtt{a})^{2\hat{k}} = g'(v)$; hence $(\widehat{s}_{k,\ell}, g')(u) = (\widehat{s}_{k,\ell}, g')(v)$, and also $t(u) = t(v)$ for every coarser t. Now $\#_{\mathtt{ab}}(u) = q$ is even and $\#_{\mathtt{ab}}(v) = 1$. With $z = \$\mathtt{1234}$, the words $\mathtt{r}uz$ and $\mathtt{r}vz$ and $\mathtt{l}uz$ lie in L_{A} and $\mathtt{l}vz$ does not. Lemma 5.6 applies. A typing with positive image in $\mathbf{LI} \vee \mathbf{Grp}$ is window–group-determined by Lemma 4.16.

For the consequences: if $s_{\mathrm{A}}(\Sigma_{\mathrm{A}}^{+})$ were in $\mathbf{LI} \vee \mathbf{Grp}$, then by (iii) applied to $t = s_{\mathrm{A}}$ we would contradict (ii). The window and window–group theorems require a window–group-determined typing, and by (iii) no such typing admits L_{A}. □

A target that requires a nonabelian group. Let $\Sigma_{\mathrm{B}} = \{\mathtt{l}, \mathtt{r}, \mathtt{a}, \mathtt{b}, \$, 1, 2, 3, 4\}$, let $\alpha = (12)$ and $\beta = (23)$ in S_3, and let $g_{\mathrm{B}} : \Sigma_{\mathrm{B}}^* \to S_3$ send $\mathtt{a} \mapsto \alpha$, $\mathtt{b} \mapsto \beta$ and every other letter to 1. Put

$$L_{\mathrm{B}} := \{\sigma w \,\$\, c : \sigma \in \{\mathtt{l}, \mathtt{r}\},\ w \in \{\mathtt{a}, \mathtt{b}\}^+,\ c \in C_\times,\ (\sigma = \mathtt{r} \text{ or } g_{\mathrm{B}}(w) = 1)\},$$

and let $s_{\mathrm{B}} := (s_{1,1}, g_{\mathrm{B}}|_{\Sigma_{\mathrm{B}}^+})$ restricted to its positive image. We write $\mathbf{Ab}$ for the pseudovariety of finite abelian groups; $\mathbf{LI} \vee \mathbf{Ab}$ consists of the divisors of products $T \times A$ with $T \in \mathbf{LI}$ and A finite abelian, since both pseudovarieties are closed under finite products.

Proposition 5.8 (A target separating nonabelian and window–abelian typings)**.** (i) $L_{\mathrm{B}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}$.

(ii) $L_{\mathrm{B}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{B}}}$ *for every* $f \geq 2$*; the typing* s_{B} *is window–group-determined with group* S_3*.*

(iii) *For every finite typing* t *of* Σ_{B}^+ *with* $t \preceq (s_{k,\ell}, g'|_{\Sigma_{\mathrm{B}}^+})$ *for some* k, ℓ *and some morphism* g' *into a finite* abelian *group, and in particular for every* t *whose positive image lies in* $\mathbf{LI} \vee \mathbf{Ab}$*,* $L_{\mathrm{B}} \notin \mathcal{C}^{\mathrm{mcf}}_{f,t}$ *for all* $f \geq 1$*.*

Consequently L_{B} *is covered by Theorem 4.14 with the nonabelian constant* $c_{S_3,f} = M_{S_3,f} - 1$*, but by no instance of Theorem 4.8 or 4.15 (but see Remark 5.9).*

Proof. (i) As for L_{A}, with W a right-linear grammar for $\{\sigma w : \sigma \in \{\mathtt{l}, \mathtt{r}\}, w \in \{\mathtt{a}, \mathtt{b}\}^+, \sigma = \mathtt{r} \text{ or } g_{\mathrm{B}}(w) = 1\}$, which tracks the last letter and the current value in S_3.

(ii) As for L_{A}: $L_{\mathrm{B}} = Q_{\mathrm{B}} \cap \Pi_{\mathrm{B}} \cap \varphi_\times^{-1}(0)$, where $Q_{\mathrm{B}} = \{\mathtt{l}, \mathtt{r}\}\{\mathtt{a}, \mathtt{b}\}^+\$\,\mathtt{1}^+\mathtt{2}^+\mathtt{3}^+\mathtt{4}^+$ is strictly 2-local and Π_{B} is the set of nonempty words beginning with $\mathtt{r}$, or beginning with $\mathtt{l}$ and having g_{B}-value 1. Since g_{B} is trivial on all letters other than $\mathtt{a}, \mathtt{b}$, on Q_{B} the g_{B}-value of the whole word equals $g_{\mathrm{B}}(w)$, and $\Pi_{\mathrm{B}} = \widehat{s_{\mathrm{B}}}^{-1}(\mathsf{Acc})$ for a suitable Acc.

(iii) Let $g' : \Sigma_{\mathrm{B}}^* \to A'$ with A' abelian, let $\hat{k}$ be even with $\hat{k} \geq \max\{k, \ell, 1\}$, and put

$$u = \mathtt{a}^{\hat{k}}(\mathtt{ab})^3\mathtt{a}^{\hat{k}}, \qquad v = \mathtt{a}^{\hat{k}}(\mathtt{ab})^2\mathtt{ba}\,\mathtt{a}^{\hat{k}}.$$

Both are long with prefix $\mathtt{a}^k$ and suffix $\mathtt{a}^\ell$, and they have the same Parikh vector, so $g'(u) = g'(v)$ by commutativity. In S_3, $\alpha^2 = \beta^2 = 1$, $\alpha\beta$ has order 3 and $\beta\alpha = (\alpha\beta)^{-1}$. Since $\hat{k}$ is even,

$$g_{\mathrm{B}}(u) = (\alpha\beta)^3 = 1, \qquad g_{\mathrm{B}}(v) = (\alpha\beta)^2(\beta\alpha) = \alpha\beta \neq 1.$$

With $z = \$\mathtt{1234}$, Lemma 5.6 applies. For the clause on $\mathbf{LI} \vee \mathbf{Ab}$, the proof of Lemma 4.16 goes through verbatim with an abelian group factor and produces an abelian g'. □

Remark 5.9 (What the two examples do and do not show). Propositions 5.7 and 5.8 separate *typing classes*: no window–group-determined (respectively, window–abelian-determined) typing admits L_{A} (respectively, L_{B}). They do *not* show that the local-group or nonabelian shortening is needed to obtain polynomial characteristic data for these targets. Indeed, let σ_{Q} be the positive restriction of the syntactic morphism of the regular language Q_{A} and put $s' = (s_{\mathrm{A}}, \sigma_{\mathrm{Q}})$, restricted to its positive image. Then $Q_{\mathrm{A}} \cap \Pi_{\mathrm{A}}$ is recognized by $\widehat{s'}$, so $L_{\mathrm{A}} = \widehat{s'}^{-1}(\mathsf{Acc}') \cap \varphi_\times^{-1}(0)$ for some Acc', and $L_{\mathrm{A}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s'}$ because $s_{\mathrm{A}} \preceq s'$. With the grammar of Proposition 5.7(i), Theorem 4.28 gives polynomial characteristic data for $\mathcal{B}^{\leq 2}_{f,s'}$, with no algebraic hypothesis on s'; the same applies to L_{B}. Moreover the separation happens at arity one (Lemma 5.6): the typing-sensitive material lies in a single fan-out-one constituent, and the crossing block $C_\times$ only makes the targets non-context-free. The examples therefore isolate the algebraic requirement on the typing, not its interaction with interleaved components, which is what Example A.6 illustrates inside the proof. Neither example asserts a lower bound on characteristic data.

5.3 An Intrinsically Rank-Two Boundary-Typed Target

In this subsection, $s_{1,1}$ denotes the positive-word restriction $\widehat{s}_{1,1}|_{\Sigma_{\mathrm{intr}}^+}$ of the prefix–suffix typing from Definition 2.4, with alphabet Σ_{intr} defined below.

Let

$$\Sigma_{\mathrm{intr}} := \{\mathtt{a},\mathtt{b},\mathtt{l},\mathtt{r},\mathtt{p},\mathtt{q},\mathtt{c},\mathtt{d},\#,\$,\mathtt{u},\mathtt{v},\mathtt{w}\}, \qquad P_{\mathrm{intr}} := \{\mathtt{lpc},\mathtt{lpd},\mathtt{lqc},\mathtt{rpc},\mathtt{rpd}\},$$

$$\mathrm{MIX}_2^+ := \{x \in \{\mathtt{a},\mathtt{b}\}^+ : |x|_{\mathtt{a}} = |x|_{\mathtt{b}}\}, \qquad T_3^+ := \{\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}, \qquad L_{\mathrm{intr}} := P_{\mathrm{intr}} \# \mathrm{MIX}_2^+ \,\$\, T_3^+.$$

The MIX_2^+ factor carries the rule-rank obstruction, which is not created by finite typing: $\mathrm{MIX}_2^+ = \{\mathtt{a},\mathtt{b}\}^+ \cap \varphi_{\mathrm{mix}}^{-1}(0)$ for $\varphi_{\mathrm{mix}}(\mathtt{a}) = 1$, $\varphi_{\mathrm{mix}}(\mathtt{b}) = -1$ in $\mathbb{Z}$, so Proposition 2.9(ii) already places it in $\mathsf{YSub}(f)$ for every f. The T_3^+ factor forces minimum fan-out two, and the wrapper P_{intr} separates the target from $\mathsf{YSub}(f)$.

Lemma 5.10 (Rank-one MCFGs have finite-index EDT0L representations)**.** *For every fixed finite fan-out bound f', each language in $\mathsf{MCFL}(f',1)$ has a finite-index EDT0L representation whose index is at most f'.*

Proof. Reduce the rank-one grammar H to productive reachable symbols. For each nonterminal B and component j introduce one EDT0L symbol $[B,j]$. Start with $[S,1]$. For each rank-one rule $A \to \rho(B)$ take a deterministic table that replaces every $[A,j]$ simultaneously by the jth component of ρ, with ξ_1^k represented by $[B,k]$. Every other symbol is fixed. Rank-zero rules have the corresponding terminal-only tables. Because the rule has at most one child and is linear, at every reachable nonterminal-bearing sentential form the active markers belong to a single nonterminal and appear at most once per component. Thus no more than f' such markers occur at a time. Every MCFG derivation chooses a corresponding sequence of tables. Conversely, a successful table sequence acts on the unique active nonterminal; unobserved child components can be supplied with productive derivations in H. It therefore corresponds to a valid MCFG derivation with exactly the same terminal word. Hence the obtained EDT0L system generates $L(H)$ and has finite index at most f'. □

Theorem 5.11 (A non-context-free intrinsic-rank-two typed witness)**.** *The language L_{intr} satisfies $L_{\mathrm{intr}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}$, and, for every $f \geq 2$, $L_{\mathrm{intr}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{1,1}} \setminus \mathsf{YSub}(f)$. Moreover, for every finite $f' \geq 1$, $L_{\mathrm{intr}} \notin \mathsf{MCFL}(f',1)$. Consequently the minimum fan-out of L_{intr} is exactly two, and its minimum rule rank among all finite-fan-out MCFG presentations is exactly two. In particular, Theorem 4.8 applies to a non-context-free language that cannot be covered by Theorem 4.6 at any finite fan-out.*

Proof. Grammar. Let B have the rules

$$B \to \mathtt{a}B\mathtt{b}B \mid \mathtt{b}B\mathtt{a}B \mid \mathtt{a}B\mathtt{b} \mid \mathtt{ab}B \mid \mathtt{ab} \mid \mathtt{b}B\mathtt{a} \mid \mathtt{ba}B \mid \mathtt{ba};$$

the first-return induction on $|x|_{\mathtt{a}} - |x|_{\mathtt{b}}$ gives $L_B = \mathrm{MIX}_2^+$. Let the fan-out-two nonterminal A have rules $A \to (\mathtt{u},\mathtt{w}) \mid (\mathtt{u}\xi_1^1\mathtt{v},\mathtt{w}\xi_1^2)(A)$, so $L_A = \{(\mathtt{u}^n\mathtt{v}^{n-1},\mathtt{w}^n) : n \geq 1\}$, and for every $\alpha \in P_{\mathrm{intr}}$ add $S \to \alpha\#\xi_1^1\$\xi_2^1\mathtt{v}\xi_2^2(B,A)$. The grammar generates L_{intr}, is reduced, λ-free, linear, nondeleting and nonpermuting, and is nonmerging because the two components of A are separated by the literal $\mathtt{v}$. It is good, of fan-out two and rule rank two.

Non-context-freeness. With $R := \mathtt{lpc}\#\mathtt{ab}\$\,\mathtt{u}^+\mathtt{v}^+\mathtt{w}^+$, $L_{\mathrm{intr}} \cap R = \{\mathtt{lpc}\#\mathtt{ab}\$\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}$, which a homomorphism erasing the fixed prefix maps onto the non-context-free $\{\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}$.

By closure of CFL under intersection with regular languages and homomorphism, L_{intr} is not context-free; fan-out-one MCFGs are exactly CFGs here, so the minimum fan-out is two.

Typed membership. Let $Q_{\mathrm{intr}} := P_{\mathrm{intr}} \# \{\mathtt{a},\mathtt{b}\}^+ \$ \mathtt{u}^+\mathtt{v}^+\mathtt{w}^+$. It is strictly 2-local, with the allowed adjacent pairs

$$\vdash \mathtt{l},\ \vdash \mathtt{r},\ \mathtt{lp},\ \mathtt{lq},\ \mathtt{rp},\ \mathtt{pc},\ \mathtt{pd},\ \mathtt{qc},\ \mathtt{c\#},\ \mathtt{d\#},\ \mathtt{\#a},\ \mathtt{\#b},\ \mathtt{aa},\ \mathtt{ab},\ \mathtt{ba},\ \mathtt{bb},$$
$$\mathtt{a\$},\ \mathtt{b\$},\ \mathtt{\$u},\ \mathtt{uu},\ \mathtt{uv},\ \mathtt{vv},\ \mathtt{vw},\ \mathtt{ww},\ \mathtt{w} \dashv .$$

Let $\Phi_{\mathrm{intr}} : \Sigma_{\mathrm{intr}}^* \to \mathbb{Z}^3$ send $\mathtt{a},\mathtt{b},\mathtt{u},\mathtt{v},\mathtt{w}$ to $(1,0,0)$, $(-1,0,0)$, $(0,1,0)$, $(0,-1,1)$, $(0,0,-1)$ and all other letters to 0. Its three coordinates impose $|x|_{\mathtt{a}} = |x|_{\mathtt{b}}$, $|x|_{\mathtt{u}} = |x|_{\mathtt{v}}$ and $|x|_{\mathtt{v}} = |x|_{\mathtt{w}}$, so $L_{\mathrm{intr}} = Q_{\mathrm{intr}} \cap \Phi_{\mathrm{intr}}^{-1}(0)$. Lemma 5.2 and Proposition 2.9(ii) make this intersection $(f, s_{1,1})$-tuple-substitutable for every f; with the grammar above, $L_{\mathrm{intr}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{1,1}}$ for every $f \geq 2$.

Separation. For $x = \mathtt{p}$, $y = \mathtt{q}$, $E = \mathtt{l}\Box_1\mathtt{c\#ab\$uvw}$ and $F = \mathtt{r}\Box_1\mathtt{d\#ab\$uvw}$, we have $E[x], E[y], F[x] \in L_{\mathrm{intr}}$ but $F[y] \notin L_{\mathrm{intr}}$, so the arity-one substitution implication fails and $L_{\mathrm{intr}} \notin \mathsf{YSub}(f)$ for every $f \geq 1$; the boundary typing separates the offending factors, $\widehat{s}_{1,1}(\mathtt{p}) \neq \widehat{s}_{1,1}(\mathtt{q})$.

Intrinsic rule rank. Lemma 5.10 shows that every finite-fan-out rank-one MCFG language is EDT0L. This agrees with Gallot's non-branching characterization [21, Definition 38 and Theorems 28–29]. Independently, Bishop et al. show that the word problem of the infinite cyclic group is not EDT0L [22]. Over $\{\mathtt{a},\mathtt{b}\}$ that word problem is precisely $\mathrm{MIX}_2 = \{x : |x|_{\mathtt{a}} = |x|_{\mathtt{b}}\}$. Let π preserve $\mathtt{a},\mathtt{b}$ and erase the other letters. If L_{intr} had a finite-fan-out rank-one presentation, its homomorphic image $\pi(L_{\mathrm{intr}}) = \mathrm{MIX}_2^+$ would be EDT0L, since the table construction commutes with terminal homomorphisms. Adding the singleton $\{\varepsilon\}$ preserves EDT0L, giving MIX_2, a contradiction. Hence $L_{\mathrm{intr}} \notin \mathsf{MCFL}(f', 1)$ for every finite f', and the displayed grammar proves the claimed intrinsic rule rank. □

The two lower bounds arise from separate factors rather than from an irreducible interaction between the resources, and the non-branching lower bound itself is prior work. What is specific here is that a strictly local wrapper and the $(1,1)$ boundary typing place the target inside the typed class while the substitution implication keeps it outside Yoshinaka's trivial-typing baseline.

6 Why the Fan-Out-One Lower Bound Does Not Transfer

The companion's exponentially separating family is a fan-out-one lower bound, but it cannot be reused unchanged at higher fan-out. The obstruction is a two-hole phenomenon, rather than a bound on the lengths of higher-arity characteristic samples.

Put $\Gamma_{\mathrm{br}} = \{\mathtt{a}, \lfloor, \rfloor, \lceil, \rceil\}$. We use the finite bracketed doubling family L_n, L_n^- and its exceptional word w_n^*, reproduced with its grammar and properties in Appendix G from the companion [25, Section 6]. No property of the typing s_{U} is used in the next result.

Proposition 6.1 (No fixed finite typing lifts the doubling family)**.** *For every finite-semigroup typing $s : \Gamma_{\mathrm{br}}^+ \to \mathsf{S}$ and every $n \geq 2|\mathsf{S}| + 3$, neither L_n nor L_n^- is $(2,s)$-tuple-substitutable. Consequently neither belongs to $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ for any $f \geq 2$.*

Proof. Put $q = |\mathsf{S}|$ and $a = s(\lfloor)$. Among $a, a^2, \ldots, a^{q+1}$ choose equal powers $a^i = a^j$ with $1 \leq i < j \leq q+1$, and put $\pi = j - i$. Then $a^{t+\pi} = a^t$ for every $t \geq i$; the assumption on n gives $n - j - 1 \geq i$ and $j \leq n - 2$. Hence the nonempty tuples

$$\vec{x} = (\lfloor^i, \lfloor^{n-i-1}), \qquad \vec{y} = (\lfloor^j, \lfloor^{n-j-1})$$

have equal componentwise s-types.

For L_n take $w = w_n^*$. For L_n^- take the word obtained from w_n^* by replacing the rightmost bottom binary production $E_1 \to \lfloor E_0 E_0 \rfloor$ by $E_1 \to \lceil E_0 \rceil$. Both choices lie in the respective language and begin with $\lfloor^n$; write $w = \lfloor^n v$. The admissible context $E = \Box_1 \lfloor \Box_2 v$ satisfies $E[\vec{x}] = E[\vec{y}] = w$.

In the same derivation, the node at depth i on the leftmost spine is labelled E_{n-i} and uses the binary production. Replace it by $E_{n-i} \to \lceil E_{n-i-1} \rceil$, keeping the left child's derivation. The result is $w' = \lfloor^i \lceil \lfloor^{n-i-1} v' \in L_n^- \subseteq L_n$, so $F = \Box_1 \lceil \Box_2 v'$ accepts $\vec{x}$ in both languages.

We show $F[\vec{y}] = \lfloor^j \lceil \lfloor^{n-j-1} v' \notin L_n$. In the well-matched word w', let the $\lceil$ after $\lfloor^i$ be matched by a $\rceil$, and write $v' = v_1 \rceil v_2$ accordingly. Then $\lfloor^{n-i-1} v_1$ is well matched, so reducing v_1 alone leaves exactly $\rfloor^{n-i-1}$. In $F[\vec{y}]$, after $\lfloor^j \lceil \lfloor^{n-j-1}$ the reader of v_1 meets these $n-i-1$ unmatched closers but finds only $n-j-1 < n-i-1$ open floors above the $\lceil$. The $(n-j)$th of them therefore meets $\lceil$, so $F[\vec{y}]$ is not well matched and lies in neither L_n nor L_n^-. Thus $\vec{x}, \vec{y}$ have equal types and the common accepting context E, but F separates them, contradicting $(2, s)$-tuple substitutability. □

The obstruction concerns membership in the *typed higher-arity substitutability class*, not MCFG generability: each L_n and L_n^- is finite and hence is generated by some finite grammar at every fan-out bound.

Remark 6.2 (Scope of the obstruction). The proof uses only that a two-hole context can slide its separator along a long periodic factor: if a fixed word z occurs at more than $|\mathsf{S}|^2$ interior positions of a factor m of some $v_0 m v_2 \in L$, two of the resulting splits of m have equal types and share the context $v_0 \Box_1 z \Box_2 v_2$, so $(2, s)$-substitutability forces their distributions to agree. Arity one never makes such comparisons. The statement is uniform over all fixed finite typings, with n large relative to $|\mathsf{S}|$. It eliminates this particular lower-bound construction at higher arity, but neither proves polynomial data bounds for all finite typings nor excludes other higher-fan-out lower-bound families.

7 Conclusion

This paper converts the qualitative positive-data learning principle for typed MCFGs into presentation-sensitive upper bounds on the *total encoded length* of characteristic samples. For reduced good grammars, window and window–finite-group typings yield $O(\|G\|^2 \bar{\theta}_G)$, including noncommutative groups and interleaved tuple components. Positive-image local-group typings yield $O(\|G\|^3 \bar{\theta}_G)$ by coupling classical sliding-factor coordinates with the MCFG ordered-block decomposition and shortening. For arbitrary standard input grammars, component-pattern thickness controls the transfer of these bounds through normalization. Rank-one presentations admit a polynomial bound for every fixed typing. All these upper bounds are proved independently of the companion CFG lower-bound construction. The internal feature-licensing example in Proposition 5.3 is a non-context-free window–group typed target for which the untyped distributional substitution rule is unsound even with the boundary-window condition alone.

The polynomial bounds above turn shortening of owner-wise ordered components into quantitative characteristic data. The remaining question asks whether analogous shortening holds for every fixed finite typing at fan-out at least two; Section 6 shows that the known CFG lower-bound family does not transfer directly.

Open Problem 7.1 (The algebraic boundary at higher fan-out). Fix $f \geq 2$. Does every fixed finite-semigroup typing admit characteristic samples whose total encoded length is polynomial in the size and ordinary rule thickness of a reduced good rank-bounded presentation, with the alphabet, fan-out, rank cap and typing fixed? If not, can failure already be witnessed by a surjective typing onto the two-element semilattice U_1? More generally, must a pseudovariety admitting such bounds

for each of its individual finite typings be contained in **LG**, as at fan-out one (Corollary 4.20)? A U_1 witness would establish this latter necessity, but the two questions are not asserted to be equivalent.

The sliding-separator constraint (Remark 6.2) and the explicit failure of the companion family to satisfy higher-arity substitutability (Proposition 6.1) explain why the fan-out-one lower-bound construction does not settle this problem.

The key unresolved issue is whether such higher-arity distributional rigidity can be turned into short typed representatives for every fixed finite semigroup, or whether a fixed-typing counterexample exists.

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Data availability

No empirical dataset is associated with this theoretical paper.

A Simultaneous Block Replacement in Finite Groups

Semigroup convention. The finite group A and the morphisms into it are auxiliary to the shortening proofs. The semantic typing is always the positive-image restriction of the window–group (or local-group) product, with the fresh identity used only to evaluate empty template factors; by Lemma E.1 this changes neither nonempty tuple comparisons nor the selected witnesses.

We now prove the purely group-theoretic statement used in Theorem 4.11 and Lemma B.7. The commutative case is a zero-sum argument; the general case needs an order-homogenization argument because, for $f \geq 2$, blocks of different indices can interleave differently on different lines. The proof has three logically separate parts: the classification of homogeneous point orders (Lemma A.4), the group-valued interval invariant (Lemma A.5), and the polynomial-size extraction of homogeneous indices (Lemma A.7). Neither the order classification nor the selection of a replacement interval assumes that the group is abelian.

Definition A.1 (Block-replacement family)**.** Let A be a finite group and $f \geq 1$. A *block-replacement family* of width f over A with index set $[m]$ consists of $d \leq f$ *lines*; on line c, a left-to-right sequence

of blocks $\beta_{c,1}, \dots, \beta_{c,q_c}$ and constants $a_{c,0}, \dots, a_{c,q_c} \in A$; for every block β an *owner* $o(\beta) \in [m]$ and two values $x_\beta, x'_\beta \in A$; every index owns at most f blocks. For $J \subseteq [m]$ put

$$\mathrm{val}_c(J) := a_{c,0}\, y_{c,1}\, a_{c,1} \cdots y_{c,q_c}\, a_{c,q_c}, \qquad y_{c,j} := \begin{cases} x'_{\beta_{c,j}}, & o(\beta_{c,j}) \in J, \\ x_{\beta_{c,j}}, & o(\beta_{c,j}) \notin J. \end{cases}$$

Lemma A.2 (Block replacement)**.** *In a block-replacement family of width f over A there is a nonempty $J \subseteq [m]$ with $\mathrm{val}_c(J) = \mathrm{val}_c(\emptyset)$ for every line c, provided that either* (a) *A is abelian and $m > \mathsf{d}(A^f)$, or* (b) *A is arbitrary and $m \geq M_{A,f}$, where d and $M_{A,f}$ are as in Section 4.4.*

Part (a) is proved below under commutativity, and part (b) under the arbitrary-group order homogenization argument.

Commutative groups: zero sums

Proof of Lemma A.2(a). Write A additively. By commutativity $\mathrm{val}_c(J) - \mathrm{val}_c(\emptyset) = \sum_{i \in J} \delta_{i,c}$, where $\delta_{i,c}$ is the sum of $x'_\beta - x_\beta$ over the blocks β of line c owned by i. Put $\delta_i = (\delta_{i,c})_c \in A^d$. A sequence over A^d of length $m > \mathsf{d}(A^f) \geq \mathsf{d}(A^d)$ has a nonempty zero-sum subsequence $(\delta_i)_{i \in J}$; the inequality holds because A^d embeds in A^f and a zero-sum-free sequence in a subgroup is zero-sum free in the group. □

Remark A.3 (Size of the zero-sum constant). The prefix-sum pigeonhole argument gives $\mathsf{d}(A^f) \leq |A|^f - 1$; the zero-sum bound is usually much stronger. Write $A \cong C_{n_1} \oplus \cdots \oplus C_{n_s}$ and $n = \exp(A) \geq 2$. Then

$$f \sum_{i=1}^{s} (n_i - 1) \;\leq\; \mathsf{d}(A^f) \;\leq\; (n-1) + n \log \frac{|A|^f}{n},$$

where the lower bound is attained by a basis sequence and the upper bound is due to van Emde Boas and Kruyswijk [34, Proposition 5.1.8 and Theorem 5.5.5]; for a p-group the lower bound is an equality [34, Theorem 5.5.9] (Olson). Hence $\mathsf{d}(A^f) = O(\exp(A)\, f \log |A|)$ is linear in f, whereas $|A|^f - 1$ is exponential; for $A = C_2$ the antichain bound drops from $2^f - 1$ to f. The value $\mathsf{d}(A^d)$ is also the limit of the argument as stated, since a zero-sum-free sequence of that length exists; we do not claim that a grammar realizes such a configuration.

A.1 Arbitrary Finite Groups: Order Homogenization

Lemma A.4 (A homogeneous block-order lemma)**.** *Fix positive integers e, n with $n \geq 3$. For each $t \in [e]$ and $i \in [n]$, let (t, i) be a distinct point on a line, with a total left-to-right order. Suppose that (i) the order of (t, i) and (u, i) is independent of i; and (ii) for each ordered pair (t, u), including $t = u$, the order of (t, i) and (u, j) for $i < j$ is independent of the choice of $i < j$. Then the types at the points have a partition into classes, each occupying a consecutive region of the line. Every class has one direction (increasing or decreasing in i); inside its region the points occur in n successive* chunks*, one per rank in that direction. Each chunk consists of exactly one point of each type in the class (in a fixed within-chunk order).*

Proof. By (ii) with $t = u$, each type's points are monotone. For distinct types t, u write X for $(t, i) < (u, j)$ and Y for $(u, i) < (t, j)$, $i < j$; both are uniform by (ii). Suppose first that both types increase. $X \wedge Y$ puts all points of rank i before all points of rank $j > i$. Under $X \wedge \neg Y$ all off-diagonal comparisons put t left of u, and so do equal ranks: $(u, i) < (t, i)$ would give $(t, 1) < (t, 2) < (u, 1) < (t, 1)$; thus the types are separated. The case $\neg X \wedge Y$ is symmetric, and

$\neg X \wedge \neg Y$ gives $(t,1) < (t,2) < (u,1) < (u,2) < (t,1)$, a contradiction. If t increases and u decreases, $X \wedge Y$ gives $(t,2) < (u,3) < (u,1) < (t,2)$ and $\neg X \wedge \neg Y$ gives $(t,3) < (u,2) < (t,1) < (t,3)$; under $X \wedge \neg Y$ all off-diagonal comparisons put t left of u, and a contrary equal-rank comparison would give $(t,1) < (u,2) < (u,1) < (t,1)$, contrary to (i), so the types are separated; $\neg X \wedge Y$ is symmetric. Reversing indices covers two decreasing types. Thus any two types are either *separated*, or have the same direction and are *interleaved by rank*.

Interleaving is transitive: if t interleaves with u and u with v, all three have one direction, say increasing; if t, v were separated with t to the left we would have $(v,1) < (u,2) < (t,3) < (v,1)$, and with v to the left $(t,1) < (u,2) < (v,3) < (t,1)$. Hence interleaving, with equality, partitions the types into classes. A type outside a class is separated from each type inside it and lies on the same side of all of them, since otherwise two interleaved types would be separated. So class regions are consecutive, the rank comparisons give successive chunks, and the within-chunk order is independent of rank by (i). □

Lemma A.5 (Simultaneous finite-group interval shortening)**.** *Let A be any finite group, and consider a block-replacement family of width f over A with $n \geq \max\{3, |A|^f\}$ indices in which all indices have the same number of blocks, the same lines for them and the same within-index order, and in which every cross-index comparison between the t-th block of index i and the u-th block of index j on a common line has the same truth value for all $i < j$. Then a nonempty interval J of indices satisfies $\mathrm{val}_c(J) = \mathrm{val}_c(\emptyset)$ for every line c.*

Proof. Apply Lemma A.4 on each line to the block types going to that line. Across all lines there are $q \leq f$ class regions. For such a region $\mathcal{Q}$, enumerate its chunks $1, \ldots, n$ from left to right. Assign any constant between two successive chunks to the chunk on its left; keep the initial and terminal constants outside the chunks. Since those two outer constants never change, equality of the ordered product between them is sufficient. In the following, P_p and Q_p refer to this interior product (so both begin at 1_A). Let P_p be the original value of its prefix ending after chunk p and Q_p the value of that prefix when chunks $1, \ldots, p$ are replaced; set $P_0 = Q_0 = 1_A$ and $\kappa_p^{\mathcal{Q}} = P_p Q_p^{-1}$. For $0 \leq p < p' \leq n$, the *ordered* value after replacing exactly chunks $p+1, \ldots, p'$ is

$$\underbrace{P_p}_{\text{old prefix}} \underbrace{(Q_p^{-1} Q_{p'})}_{\text{new middle}} \underbrace{(P_{p'}^{-1} P_n)}_{\text{old suffix}}.$$

The middle expression is correct even with constants between chunks: $Q_{p'} = Q_p$ (new middle) by the definition of prefix products. Therefore the displayed product equals P_n exactly when

$$P_p Q_p^{-1} Q_{p'} P_{p'}^{-1} = 1_A, \quad \text{equivalently} \quad \kappa_p^{\mathcal{Q}} = P_p Q_p^{-1} = P_{p'} Q_{p'}^{-1} = \kappa_{p'}^{\mathcal{Q}}.$$

No factors have been commuted. Note that this requires equality of *two boundary states*; it does not require an individual block's original and replacement values to agree.

If $\mathcal{Q}$ increases with the index order, replacing owners $(a, b]$ replaces chunks $(a, b]$ in the region. If the order decreases, the same owners occupy chunks $(n-b, n-a]$ (counted from the left). This reversal is why the two orientations must use distinct state indices. For $j = 0, \ldots, n$ define a state in A^q by taking $\kappa_j^{\mathcal{Q}}$ from each increasing and $\kappa_{n-j}^{\mathcal{Q}}$ from each decreasing class. There are $n+1 > |A|^q$ states, so two coincide, at $a < b$. Replacing the indices $a+1, \ldots, b$ then preserves every region value; everything else on every line is unchanged. □

Example A.6 (Two rows, three owners, reversed order, and noncommuting values). This finite example illustrates the *boundary-state identity*, not an invocation of the numerical pigeonhole threshold (which can be much larger than three). Work in the nonabelian group $A = S_3$ and put

$\alpha = (12)$, $\beta = (23)$, so that $\alpha^2 = \beta^2 = 1$ but $\alpha\beta \neq \beta\alpha$. There are three owners $i = 1, 2, 3$ with three blocks each: two blocks U_i, V_i on row 1, and one block W_i on row 2. Reading left to right, with the fixed group-valued material *shown explicitly*, the rows are

$$\begin{aligned} &\text{row 1:} \quad (U_1\,\beta\,V_1)\,\alpha\,(U_2\,\beta\,V_2)\,\alpha\,(U_3\,\beta\,V_3),\\ &\text{row 2:} \quad W_3\,\beta\,W_2\,\beta\,W_1. \end{aligned}$$

Thus row 1 has two owner types occurring in increasing rank chunks, while row 2 has one type in *decreasing* rank chunks. Originally take $U_i = \beta$, $V_i = 1$, $W_i = 1$ for all i; both complete rows have value 1. Now replace *exactly* owners $J = \{2, 3\}$ using the table

owner	U_i'	V_i'	W_i'	replaced?
1	β	1	1	no
2	1	1	$\beta\alpha\beta$	yes
3	$\alpha\beta\alpha\beta$	1	α	yes

(the unchanged entries in row 1 for owner 1 are listed for comparison). The row 1 chunk values were $(1, 1, 1)$ and become $(1, \beta, \alpha\beta\alpha)$; its complete new value is

$$1 \cdot \alpha \cdot \beta \cdot \alpha \cdot (\alpha\beta\alpha) = 1.$$

In the reverse-order row 2 the new value is

$$\alpha \cdot \beta \cdot (\beta\alpha\beta) \cdot \beta \cdot 1 = 1.$$

These are *ordered* calculations in S_3; neither replaces $\alpha\beta$ by $\beta\alpha$. More precisely, for the interval of owner indices $(a, b] = (1, 3]$, the increasing region of row 1 has $\kappa_1 = \kappa_3 = 1$, whereas the decreasing region of row 2 has $\kappa_{3-b} = \kappa_0 = \kappa_2 = \kappa_{3-a} = 1$. The same owner interval therefore matches the two *different* boundary-index orientations. Note that no individual replacement owner need preserve its original row contribution; only both *full row products* are fixed. In the general proof there can be up to f such regions on multiple rows, and their states are compared simultaneously.

Lemma A.7 (Polynomial order homogenization)**.** *Let $N \geq 3$, and define $m_0 = N$ and $m_{s+1} = (m_s - 1)^2 + 1$, so that $m_k \leq N^{2^k}$. Let T be a finite set of point types, each assigned to one of several lines. For every index $i \in [n]$ and $t \in T$ let $p_t(i)$ be a point on the line of t, the points on each line being distinct. Let Π be the set of ordered pairs (t, u), including $t = u$, of types on a common line, and put $k = |\Pi|$. If $n \geq m_k$, there is a set $I \subseteq [n]$ of N indices such that, for every $(t, u) \in \Pi$, the truth value of $p_t(i) < p_u(j)$ is the same for all $i < j$ in I.*

Proof. Treat the pairs one at a time, the pairs (t, t) first. Each step passes from a current index set of size at least $(m-1)^2 + 1$ to a subset of size at least m; homogeneity already obtained, and monotonicity of each type, persist in subsets. For (t, t), homogeneity means that p_t is monotone, and the Erdős–Szekeres theorem gives a monotone subsequence of length m.

For $t \neq u$, both p_t and p_u are already monotone. Relabel the current indices $1, \ldots, m'$ in order, with $m' \geq (m-1)^2 + 1$, and put $X(i, j) = [p_t(i) < p_u(j)]$ for $i < j$. Because $p_u(j)$ is monotone in j, every row $X(i, \cdot)$ is monotone; because $p_t(i)$ is monotone in i, every column $X(\cdot, j)$ is monotone. The respective directions are fixed over all rows and columns by the directions of the two point types. Replacing X by $1 - X$ if necessary, assume each row goes from 0 to 1.

If the columns go from 1 to 0, put $J(i) = \min\{j > i : X(i, j) = 1\}$ (or ∞); J is nondecreasing. Set $i_1 = 1$ and $i_{s+1} = J(i_s)$ while this is at most m'. For $a < b$, $i_b \geq J(i_a)$ and so $X(i_a, i_b) = 1$; if

the chain has m terms we are done. Otherwise its at most $m-1$ terms cut $[m']$ into at most $m-1$ intervals $[i_s, i_{s+1})$ and $[i_L, m']$, one of which has more than $m'/(m-1) > m-1$ indices; inside it every $i < j$ satisfies $j < J(i_s) \le J(i)$, so $X(i,j) = 0$ (and $J \equiv \infty$ in the last interval).

If the columns go from 0 to 1, the set $\{X = 1\}$ is closed upward in both coordinates. Let κ be the least index with $X(\kappa, \kappa+1) = 1$, or $\kappa = m'$. For $\kappa \le i < j$, $X(i,j) \ge X(\kappa, \kappa+1) = 1$; for $i < j \le \kappa$, $X(i,j) \le X(j-1, j) = 0$. One of $[1, \kappa]$ and $[\kappa, m']$ has at least $(m'+1)/2 \ge m$ indices, because $(m-1)^2 + 1 \ge 2m - 1$.

After all k pairs at least $m_0 = N$ indices remain, and $m_{s+1} \le m_s^2$ gives $m_k \le N^{2^k}$. □

Proof of Lemma A.2(b). If an index owns no blocks, choosing that index alone preserves every line, so assume each index owns at least one block. The *unary pattern* of an index records the number of its blocks, the line of each, and their order within any shared line; there are at most $f \cdot f^f \cdot f! = U_f$ patterns. Hence at least $N_{A,f}^{2^{f^2}}$ indices share one pattern; the blocks of all other indices are never replaced and are absorbed into the constants. Number the blocks of the common pattern as types $1, \dots, f'$ ($f' \le f$) and use their positions as points on their lines. There are at most f^2 ordered pairs of types on a common line, including $t = u$, and $m_k \le N^{2^k} \le N^{2^{f^2}}$ for $k \le f^2$, so Lemma A.7 with $N = N_{A,f}$ yields $N_{A,f}$ indices with constant cross-index comparisons for all $i < j$. These satisfy the hypotheses of Lemma A.5, which supplies the required nonempty interval. □

For fixed f, the constant $M_{A,f} = U_f \max\{3, |A|^f\}^{2^{f^2}}$ is polynomial in $|A|$; its exponent depends doubly exponentially on f.

B Local-Group Coordinates and MCFG Block Shortening

Semigroup convention. Throughout this subsection **LG** is tested on the *positive image* $S = \widehat{s}(\Sigma^+)$ of the typing, not on the canonical completion $S^\sharp$. Auxiliary group-valued maps and the displayed monoid morphisms are implementations whose restrictions to nonempty words define the semantic bias (Lemma E.1).

LG is the pseudovariety of finite semigroups S in which every local monoid eSe, with e idempotent, is a group; the classical relation $\mathbf{LG} = \mathbf{Grp} * \mathbf{D}$ places it within the sliding-window and delay-theorem theory [38, 39]. The equivalent exclusion of the two-element semilattice $U_1 = \{1, \zeta\}$ ($\zeta^2 = \zeta$) is given by [37, Proposition 2.3]. This semilattice is the positive image of the typing behind the companion CFG lower-bound example [25, Proposition 6.4]. The sliding-factor coordinates below are a self-contained implementation of this established algebra, tailored to the precise boundary and group values used here; they are not claimed as a new algebraic characterization. The contribution to MCFG learning is the factorization of group values into fixed separators and variable blocks in Lemma B.3, followed by simultaneous replacement of those blocks.

Throughout, Σ is nonempty, $k \ge 1$, and A is a finite group. For a word w with $|w| \ge k+1$ let $\sigma_k(w)$ be the sequence of its factors of length $k+1$, ordered by their starting positions.

Definition B.1 (Local-group typing)**.** Let $\nu : \Sigma^{k+1} \to A$ be any map, the *sliding-factor labelling*. For $|w| \ge 2k$ put

$$\gamma_{k,\nu}(w) := \nu(F_1)\nu(F_2)\cdots\nu(F_{|w|-k}), \qquad \sigma_k(w) = (F_1, \dots, F_{|w|-k}),$$

the ordered product in A, and put $\gamma_{k,\nu}(w) := 1_A$ for $|w| < 2k$. The *local-group typing* is $\lambda_{k,\nu}(w) := \big(\widehat{s}_{k,k}(w), \gamma_{k,\nu}(w)\big)$. A finite typing $\widehat{s}$ is *local-group-determined* if $\widehat{s} \preceq \lambda_{k,\nu}$ for some k, A and ν.

For $\ell = k$, words of length $< 2k$ are *short* and recorded exactly, while a *long* w is recorded by $\langle \mathrm{pref}_k(w), \mathrm{suf}_k(w)\rangle$ (Definition 2.4). For $s, p \in \Sigma^k$ write $J_\nu(s,p) := \prod_{i=1}^{k} \nu((sp)[i..i+k])$, the ordered product over the k factors of length $k+1$ of sp that start in s.

Lemma B.2 (Local-group typings are morphisms)**.** *The kernel of $\lambda_{k,\nu}$ is a congruence of finite index on Σ^*. Hence $\lambda_{k,\nu}$ is a homomorphism onto a finite monoid $M_{k,\nu}$. For long u, v,*

$$\lambda_{k,\nu}(uv) = \big(\langle \mathrm{pref}_k\, u, \mathrm{suf}_k\, v\rangle,\ \gamma_{k,\nu}(u)\, J_\nu(\mathrm{suf}_k\, u, \mathrm{pref}_k\, v)\, \gamma_{k,\nu}(v)\big).$$

Proof. The window component is a morphism by Definition 2.4. For long u, v, the factors of length $k+1$ of uv are, in order, those of u, the k factors starting in the last k positions of u (inside $\mathrm{suf}_k(u)\,\mathrm{pref}_k(v)$), and those of v. If u is short and v long, the factors starting in u lie inside $u\,\mathrm{pref}_k(v)$ and the rest are those of v, so $\gamma_{k,\nu}(uv)$ is determined by u, $\mathrm{pref}_k(v)$ and $\gamma_{k,\nu}(v)$; the case of long u and short v is symmetric, and if both are short uv is determined exactly. Thus $\lambda_{k,\nu}(uv)$ is a function of $\lambda_{k,\nu}(u)$ and $\lambda_{k,\nu}(v)$, and the image has at most $|\Sigma|^{<2k} + |\Sigma|^{2k}|A|$ elements. □

Lemma B.3 (Block decomposition)**.** *Let $w = x_0\beta_1 x_1 \cdots \beta_q x_q$ be a factorization of a word into designated blocks β_j and fixed words x_j. Let $w' = x_0\beta'_1 x_1 \cdots \beta'_q x_q$, where $\widehat{s}_{k,k}(\beta'_j) = \widehat{s}_{k,k}(\beta_j)$ for every j. Then $\widehat{s}_{k,k}(w') = \widehat{s}_{k,k}(w)$. If w is long, there are $c_0, \ldots, c_q \in A$, depending only on the x_j and the window types of the β_j, such that*

$$\gamma_{k,\nu}(w) = c_0\, \gamma_{k,\nu}(\beta_1)\, c_1 \cdots \gamma_{k,\nu}(\beta_q)\, c_q, \qquad \gamma_{k,\nu}(w') = c_0\, \gamma_{k,\nu}(\beta'_1)\, c_1 \cdots \gamma_{k,\nu}(\beta'_q)\, c_q.$$

Proof. A short block is recorded exactly, so $\beta'_j = \beta_j$, and we absorb it into the fixed words. Every remaining block is long and its replacement is long with the same first and last k letters; hence w' is long exactly when w is, with the same first and last k letters, which gives the window claim. A factor F of length $k+1$ of w either lies inside a long block β_j or meets each long block it touches only in a prefix or a suffix of length at most k (meeting both ends would force $|\beta_j| \le k$). Factors of the second kind are spelled by the fixed words and the windows of the blocks, and occur in the same order in w'. Ordering all factors by starting position, those inside β_j form one consecutive run contributing $\gamma_{k,\nu}(\beta_j)$; let c_j be the ordered product of the remaining factors between consecutive runs. □

Lemma B.4 (Explicit sliding-factor coordinates for block shortening)**.** *Let S be a finite semigroup with minimal ideal K and suppose every idempotent of S lies in K. Put $n = |S|$ and fix any morphism $h : \Sigma^+ \to S$. In a Rees representation $\mathsf{K} = \mathcal{M}(H; I, \Lambda; P)$ there is a map $\nu : \Sigma^{n+1} \to H$ such that, for every word w with $|w| \ge 2n$,*

$$h(w) = \big(\mathrm{row}(h(\mathrm{pref}_n\, w)),\ \mathrm{grp}(h(\mathrm{pref}_n\, w))\, \gamma_{n,\nu}(w),\ \mathrm{col}(h(\mathrm{suf}_n\, w))\big). \qquad (\star)$$

Here row, grp *and* col *send $(i, a, \lambda) \in \mathsf{K}$ to $i \in I$, $a \in H$ and $\lambda \in \Lambda$, and $\gamma_{n,\nu}(w)$ is the* ordered *product of the $|w| - n$ sliding-factor labels from Definition B.1. Consequently $h(w)$ is determined by the first and last n letters and this ordered product, with no commutativity hypothesis on H.*

Proof. (1) Uniform entry into K. The Rees quotient S/K has no idempotent except zero. Every element therefore has a power equal to zero. If a product of $m = |S/\mathsf{K}|$ factors were nonzero, its m nonempty prefix products would all be nonzero, so two, say $p_i = p_j$ with $i < j$, would coincide. With $x = s_{i+1} \cdots s_j$ we would have $p_i = p_i x^q$ for every q; taking q with $x^q = 0$ is a contradiction. Thus every product of $n \ge |S/\mathsf{K}|$ letters belongs to K.

(2) Outer coordinates and the right action. Write $(i,a,\lambda)(j,b,\mu) = (i, ap_{\lambda j}b, \mu)$. Fix $i_0 \in I$ and set $e_\lambda = (i_0, p^{-1}_{\lambda i_0}, \lambda) \in \mathsf{K}$. For all $(i,a,\lambda) \in \mathsf{K}$, multiplication on the right by e_λ is the identity. For $s \in S$ write $e_\lambda s = (j,c,\mu) \in \mathsf{K}$ and put $\rho(\lambda, s) = p_{\lambda j}c$. Associativity then gives the exact identity

$$(i,a,\lambda)s = (i, a\rho(\lambda,s), \mu(\lambda,s)).$$

In particular the I-coordinate is unchanged by right multiplication, even if $s \notin \mathsf{K}$. Dually, left multiplication preserves the Λ-coordinate of a product already in K: choose an idempotent $f_i \in \mathsf{K}$ with $f_i(i,a,\lambda) = (i,a,\lambda)$; then $s(i,a,\lambda) = (sf_i)(i,a,\lambda)$, and the Rees rule forces the same Λ-coordinate. The I-coordinate of a long word is therefore fixed by its first n letters, and its Λ-coordinate by its last n letters.

(3) The sliding-factor group coordinate. Write $w = a_1 \cdots a_m$, $m \geq 2n$, and set $h(a_1 \cdots a_n) = (i, a, \lambda_n)$. For $j \geq n$ let λ_j denote the Λ-coordinate of $h(a_1 \cdots a_j)$. Since left multiplication preserves the Λ-coordinate on K, this equals the Λ-coordinate of $h(a_{j-n+1} \cdots a_j)$; both products have length at least n. Define for each $F \in \Sigma^{n+1}$

$$\nu(F) = \rho\big(\mathrm{col}(h(F[1..n])),\, h(F[n+1])\big).$$

Successive applications of the right-action identity, in their actual left-to-right order, yield

$$h(w) = \Big(i,\ a \prod_{j=n}^{m-1} \nu(a_{j-n+1} \cdots a_{j+1}),\ \lambda_m\Big).$$

The displayed product is precisely $\gamma_{n,\nu}(w)$, and $\lambda_m = \mathrm{col}(h(\mathrm{suf}_n\, w))$. This proves $(\star)$. The case $m = 2n$ is included: it has exactly n sliding factors and the same boundary-coordinate formula. □

Theorem B.5 (Classical local-group criteria in sliding-factor coordinates)**.** *For a finite typing $s : \Sigma^+ \to \mathsf{S}$ put $S := \widehat{s}(\Sigma^+)$. The following are equivalent.*

(i) *$S \in \mathbf{LG}$; that is, eSe is a group for every idempotent e of S.*

(ii) *S has no subsemigroup isomorphic to U_1.*

(iii) *Every idempotent of S lies in the minimal ideal of S.*

(iv) *$\widehat{s}$ is local-group-determined. One may take $k = |S|$.*

Proof. The equivalence of (i)–(iii) is classical. More precisely, [37, Proposition 2.3] proves that (i) and (ii) are equivalent to $S^m = \mathsf{K}$ for some m, where K is the minimal ideal. By [37, Lemma 2.2], for $n = |S|$ one has $S^n = SE(S)S$. Thus $S^m = \mathsf{K}$ for some m if and only if $E(S) \subseteq \mathsf{K}$: the forward implication uses $e = e^m$ for every idempotent e, while the converse follows from $S^n = SE(S)S \subseteq \mathsf{K}$ and the fact that S^n is a nonempty ideal containing K.

(iii)⇒(iv). Apply Lemma B.4 to $h = \widehat{s}|_{\Sigma^+}$ with $n = |S|$. Equation $(\star)$ gives the value of $\widehat{s}(w)$ for every $|w| \geq 2n$ from its n-letter prefix, n-letter suffix, and the ordered sliding-factor product $\gamma_{n,\nu}(w)$. The window congruence $\widehat{s}_{n,n}$ records all shorter words exactly and separates ε by the fresh identity. Hence $\widehat{s} \preceq \lambda_{n,\nu}$.

(iv)⇒(i). Since $\mathbf{LG}$ is closed under homomorphic images, it suffices to show that the positive image $T := \lambda_{k,\nu}(\Sigma^+)$ belongs to $\mathbf{LG}$. No short element is idempotent: squaring a short word increases its length, which the window component records exactly up to the long threshold. Let $e = (\langle p, s\rangle, \gamma_e)$ be idempotent. By Lemma B.2, $\gamma_e J_\nu(s,p)\gamma_e = \gamma_e$; moreover, every exe ($x \in T$) has window $\langle p, s\rangle$. Thus eTe lies in the group $H_e := \{(\langle p, s\rangle, \alpha) : \alpha \in A\}$, whose multiplication is $\alpha * \beta = \alpha J_\nu(s,p)\beta$ and which is isomorphic to A via $\alpha \mapsto \alpha J_\nu(s,p)$. The finite subsemigroup eTe is therefore a group. □

Proposition B.6 (Window–group typings are local-group typings, strictly)**.** (i) *If $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$ for a homomorphism g into a finite group, then $\widehat{s} \preceq \lambda_{\hat{k},\nu}$ with $\hat{k} = \max\{k, \ell, 1\}$ and $\nu(F) := g(\text{last letter of } F)$. Hence $\mathbf{LI} \vee \mathbf{Grp} \subseteq \mathbf{LG}$.*

(ii) *Let $\Sigma = \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$. Let $h_{\mathtt{ab}} := \lambda_{1,\nu}$ with $\nu(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and $\nu(F) = 0$ otherwise; it records the parity of the number of occurrences of $\mathtt{ab}$, together with the window. Then $h_{\mathtt{ab}}$ is not window–group-determined. Hence $h_{\mathtt{ab}}(\Sigma^+) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{Grp})$.*

Proof. (i) For $|w| \geq 2\hat{k}$, $g(w) = g(\mathrm{pref}_{\hat{k}}\, w)\, \gamma_{\hat{k},\nu}(w)$, and shorter words are recorded exactly. For the inclusion, let $S \in \mathbf{LI} \vee \mathbf{Grp}$ and take $\Sigma = S$ with the evaluation typing $s : \Sigma^+ \twoheadrightarrow S$. Lemma 4.16 makes $\widehat{s}$ window–group determined, hence local-group-determined by the first part, so $S \in \mathbf{LG}$ by Theorem B.5.

(ii) Suppose $h_{\mathtt{ab}} \preceq (\widehat{s}_{k,\ell}, g')$ with $g' : \Sigma^* \to A'$. Let $\hat{k} = \max\{k, \ell, 1\}$ and let q be an even multiple of the exponent of A'. Put $u = \mathtt{a}^{\hat{k}}(\mathtt{abc})^q \mathtt{a}^{\hat{k}}$ and $v = \mathtt{a}^{\hat{k}}(\mathtt{bac})^q \mathtt{a}^{\hat{k}}$. Both are long, with prefix $\mathtt{a}^k$ and suffix $\mathtt{a}^\ell$, and $g'(u) = g'(\mathtt{a})^{2\hat{k}} = g'(v)$. But u contains q occurrences of $\mathtt{ab}$, while v contains exactly one, at $\mathtt{a}^{\hat{k}}\mathtt{b}$, so $h_{\mathtt{ab}}(u) \neq h_{\mathtt{ab}}(v)$. Membership in $\mathbf{LG}$ follows from Theorem B.5; if the positive image were in $\mathbf{LI} \vee \mathbf{Grp}$, Lemma 4.16 would make $h_{\mathtt{ab}}$ window–group-determined. □

Lemma B.7 (Short tuples with prescribed local-group type)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$, and let $\lambda = \lambda_{k,\nu}$. Let $b_{\mathrm{w}} := f(2k+1)n_{\mathrm{w}}(r_{\max}+1)\bar{\theta}_G$, where $n_{\mathrm{w}} \leq |V||M_{k,k}|^f$ is the number of non-start typed nonterminals of the trimmed $\widehat{s}_{k,k}$-refinement. Put $n_\lambda := |V||M_{k,\nu}|^f$, and let $M_{A,f}$ be the constant of Section 4.4. For every $B \in V$ and every realizable $\mathbf{t} \in M_{k,\nu}^{\mathrm{fan}(B)}$ there is $\vec{u} \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$ and*

$$\|\vec{u}\|_1 \leq \beta_\lambda := (r_{\max} + 1)\, b_{\mathrm{w}} \max\{1, (M_{A,f} - 1)\, n_\lambda\}.$$

If A is abelian, $M_{A,f} - 1$ may be replaced by $\mathtt{d}(A^f)$.

Proof. Let T be minimal as in Lemma 4.9 with root λ-vector $\mathbf{t}$ and threshold $b = b_{\mathrm{w}} \geq \bar{\theta}_G$. For a heavy node v labelled C, with yield $\vec{x}$ and window vector $\boldsymbol{\omega} = \widehat{s}_{k,k}(\vec{x})$, the typed nonterminal $C_{\boldsymbol{\omega}}$ is retained in the trimmed $\widehat{s}_{k,k}$-refinement, because T can be inserted into a successful derivation (G is reduced). Lemma 4.7 (trivial κ) gives a replacement tuple of C with the same window vector and length at most b_{w}. For pairwise incomparable heavy nodes $v_1, \ldots, v_m$ and $J \subseteq [m]$, let T_J replace the subtrees at v_i, $i \in J$. The components of the v_i are disjoint contiguous blocks of the root components. By Lemma B.3, every T_J has the root window vector of T; a short root component is unchanged, and on each long root component $\gamma_{k,\nu}$ is an ordered product $c_0 y_1 c_1 \cdots y_q c_q$ of block values between constants that do not depend on J. The long root components thus form a block-replacement family over A (Definition A.1). If $m \geq M_{A,f}$, or $m > \mathtt{d}(A^f)$ for abelian A, Lemma A.2 gives a nonempty J with the root λ-vector of T, while T_J is strictly shorter. This contradicts minimality, and Lemma 4.9 with $N = M_{k,\nu}$ gives the bound. □

By Proposition B.6(i), this theorem also applies to window and window–group typings, but its general cubic bound is weaker than the quadratic bounds proved for those subclasses in Theorems 4.8 and 4.14.

C Pattern-Thickness Transfer for General MCFG Presentations

Here are the support-fibre details behind Proposition 4.23. Write $F_{A,I} := \{\vec{u} \in L_A(H_{\mathrm{red}}) : \mathrm{supp}(\vec{u}) = I\}$. Every realized fibre has a shortest tuple, of length at most $t_{\mathrm{pat}}(H_{\mathrm{red}})$.

For clarity, the source-language invariant is preserved *through all four stages*: each intermediate decorated nonterminal is the image of a union of realizable original fibres $F_{A,I}$ under one composition of projection, permutation and consecutive-component merging; where an intermediate support is prescribed, retain only the original fibres whose image has that support. The map is length-nonincreasing, and every such fibre has an actual derivation in the trimmed input. The stagewise bookkeeping is:

Stage	Tuple-language operation	Support/length information
(a) Nondeletion	$\pi_J(L_A)$, union over source fibres	Omitted children have derivations; projection does not increase length
(b) Nonpermutation	$\sigma(\pi_J(L_A))$	Support is permuted; length is unchanged
(c) Empty-pattern removal	Restrict to a realized support I, then π_I	Select preimage source fibres with the required support; length does not increase
(d) Nonmerging	$\mu(\sigma\pi_J(L_A))$, with the support restriction inherited from (c)	Merged support is a function of previous support; length is unchanged

In this table, L_A refers to the original input nonterminal and the operations are understood as applying to the appropriate union of $F_{A,I}$, not to an unrestricted L_A after stage (c). The original rank and fan-out bounds survive because the four constructions introduce only finitely many component decorations per rule for fixed f, r.

At stage (a) of Lemma F.1, a new nonterminal has language $\pi_J(L_A(H_{\mathrm{red}}))$, a union of the images $\pi_J(F_{A,I})$. Stage (b) permutes components and thus permutes support patterns without changing total length. At stage (c), restricting to a realized output pattern selects the images of those original support fibres whose projected/permuted support is exactly that pattern. Because each such original fibre is nonempty, selecting its shortest tuple yields an output tuple of total length at most $t_{\mathrm{pat}}(H_{\mathrm{red}})$. Stage (d) merges consecutive components; concatenation cannot increase total length, and its output support is determined by the input support, so the same argument applies after merging. The stagewise equalities are equalities of tuple languages, not merely inclusions; hence each retained nonterminal has an actual bounded-length terminal derivation. Fresh nonterminal names are only the finite decorators of these source nonterminals. Final trimming removes symbols without changing the tuple languages of the surviving ones. This proves $t_G \le t_{\mathrm{pat}}(H_{\mathrm{red}})$.

The normal form has $\|G\| = O_{f,r}(\|H\|)$ by Proposition 3.9; consequently $\bar{\theta}_G \le 2\max\{1, \|G\|\}\max\{1, t_G\} = O_{f,r}(\|H\|\bar{t}_{\mathrm{pat}}(H_{\mathrm{red}}))$ by Lemma 4.1(iii). This argument does not assert polynomial control by the *ordinary* untyped rule thickness of H: Proposition F.3 shows why elimination of empty components can be expensive.

Example C.1 (An explicit four-stage fan-out-two transfer). Here is a finite example in which *deletion, permutation, an empty component, and a merge* all occur. It illustrates the exact source-fibre invariant used above, rather than replacing the proof of Lemma F.1. Let the terminal alphabet contain $a, b, c, d, z, \#$ and take fan-out-one start symbol S, fan-out-two symbols A, B, C, and the rank-≤ 2 grammar

$$\begin{aligned} B &\to (a, \varepsilon) \mid (a, b), \\ C &\to (c, d), \\ A &\to (\xi_1^2\xi_1^1, \xi_2^1)(B, C) \mid (\varepsilon, z), \\ S &\to (\xi_1^1 \# \xi_1^2)(A) \mid (\varepsilon). \end{aligned}$$

The first A-rule *deletes* C's second component d, uses B's components in *reverse* order, and puts the adjacent pieces B^2B^1 together. The terminal tuple (ε, z) realizes the support pattern $\{2\}$, whereas the other A-derivations realize $\{1,2\}$. Direct evaluation gives

X	realized tuple fibres (by support)	minimum lengths per fibre
B	$F_{B,\{1\}} = \{(a,\varepsilon)\}, \quad F_{B,\{1,2\}} = \{(a,b)\}$	$1, 2$
C	$F_{C,\{1,2\}} = \{(c,d)\}$	2
A	$F_{A,\{2\}} = \{(\varepsilon,z)\}, \quad F_{A,\{1,2\}} = \{(a,c),(ba,c)\}$	$1, 2$
S	$F_{S,\emptyset} = \{(\varepsilon)\}, \quad F_{S,\{1\}} = \{(\#z),(a\#c),(ba\#c)\}$	$0, 2$

The last column records the minimum length *within each realized support fibre*, rather than the lengths of all its tuples. For example, $\|(a,c)\|_1 = 2$ and $\|(ba,c)\|_1 = 3$, so the minimum for $F_{A,\{1,2\}}$ is 2. Thus $t_{\mathrm{pat}}(H) = 2$ for this H and $L(H) = \{\varepsilon, \#z, a\#c, ba\#c\}$.

Stage (a): projecting C to its used component gives $C^{\{1\}} \to (c)$ with language $\pi_{\{1\}}(L_C) = \{(c)\}$; the deleted d is restored by the omitted component in the original C-derivation. All derived A-tuples are unchanged.

Stage (b): the decorated reverse-order copy of B has language $\{(\varepsilon,a),(b,a)\}$. Reading this copy in its own component order makes the first A-rule nonpermuting. This is precisely a permutation image of L_B, and neither support nor length is silently changed (the support is permuted).

Stage (c): the two nonempty support fibres of this reordered B become $\{(a)\}$ from $\{2\}$ and $\{(b,a)\}$ from $\{1,2\}$. The A-copy of support $\{2\}$ generates (z), whereas the $\{1,2\}$-copy generates (a,c) and (ba,c). The positive start language is $\{\#z, a\#c, ba\#c\}$; its separate ε-rule is removed. In particular, the all-empty output is not accidentally reconstructed by a good-rule template.

Stage (d): the contiguous pair (b,a) in the reordered B-copy is merged to (ba); the singleton fibre remains (a). The corresponding A-copy still generates exactly (a,c) or (ba,c) and S retains exactly the three positive words. After collapsing duplicate names, one possible *final good grammar* for the nonempty core is the following fully explicit one:

$$
\begin{array}{lcl@{\qquad}lcl}
D_0 & \to & (a), & D_1 & \to & (ba), \\
C_0 & \to & (c), & A_2 & \to & (z), \\
A_{12} & \to & (\xi_1^1, \xi_2^1)(D_0, C_0) \mid (\xi_1^1, \xi_2^1)(D_1, C_0), & & & \\
S_+ & \to & (\#\,\xi_1^1)(A_2) \mid (\xi_1^1\,\#\,\xi_1^2)(A_{12}). & & &
\end{array}
$$

Here D_0 and D_1 are exactly the two transformed B fibres: $L_{D_0} = \{(a)\}$ and $L_{D_1} = \{(ba)\}$; the omitted d from C contributes no root output. Each displayed component is nonempty, no child component is deleted, permuted, or separately merged at a rule, and the fan-out and rank remain 2. For each output copy, a representative is the image of a tuple in one of the displayed *original* fibres under a composition $\mu\sigma\pi_J$, possibly after conditioning on a realized support pattern. Each map is length-nonincreasing, and the shortest chosen original-fibre representative has length at most two. Thus every surviving decorated nonterminal has a terminal yield of length at most $2 = t_{\mathrm{pat}}(H)$, as asserted by the general transfer. In this example the difference between preserving the *whole source language* and preserving a *support fibre* is visible at stage (c).

D Auxiliary Verification of Qualitative Reconstruction

D.1 Qualitative antecedent in Yoshinaka (2015)

Remark D.1 (Finite-signature antecedent). Yoshinaka's finite-signature learning principle [15, Theorem 1, Example 3 and Section 8] provides the qualitative MCFG-learning comparison for fixed

rule rank. One can refine the noninitial tuple sorts by their fixed componentwise s-types; the initial sort and its promotion operations remain distinct. The present paper does not use a signature-embedding assertion as a premise: soundness, exact reconstruction, polynomial construction and Gold convergence for the concrete Const/Link/Comp grammar are verified in the following results.

D.2 Soundness

Lemma D.2 (Composition preserves equivalence)**.** *Let $L \subseteq \Sigma^*$, let ρ be a good template of rank r, and let $\vec{x}_j, \vec{u}_j \in (\Sigma^+)^{d_j}$ satisfy $\vec{u}_j \equiv_{L,s} \vec{x}_j$ for every j. Then $\rho(\vec{u}_1, \dots, \vec{u}_r) \equiv_{L,s} \rho(\vec{x}_1, \dots, \vec{x}_r)$.*

Proof. Replace the children one at a time, keeping all other children fixed and nonempty. Given an accepting outer context E for the result of any intermediate replacement, Lemma 3.1 induces an admissible context E_j of the child being replaced. Since $\vec{u}_j \equiv_{L,s} \vec{x}_j$, membership in L is unchanged under that replacement, in either direction. The same argument applies at each step even if earlier children have already been replaced. Finally, the componentwise s-types of the output tuples agree because the template is fixed and $\widehat{s}$ is multiplicative on concatenation, including the fresh identity for empty terminal gaps. Therefore the output distributions coincide. □

Proposition D.3 (Soundness of every hypothesis)**.** *Let L be (f, s)-tuple-substitutable and $K \subseteq L$ finite. For every $r_{\max} \geq 1$, if $[\vec{x}] \Rightarrow^* \vec{u}$ in $\widehat{G}_{\leq r_{\max}}(K)$, then $\vec{u} \equiv_{L,s} \vec{x}$. Consequently $L(\mathcal{B}_{f,s}^{\leq r_{\max}}(K)) \subseteq L$ and $L(\mathcal{B}_{f,s}(K)) \subseteq L$.*

Proof. Proceed by induction on the height of a derivation rooted at $[\vec{x}]$. A Const rule derives exactly $\vec{x}$. For Link, $[\vec{x}] \to [\vec{y}]$ is witnessed by a common admissible context of same-type tuples in $K \subseteq L$, hence tuple substitutability gives $\vec{x} \equiv_{L,s} \vec{y}$; compose this equivalence with the inductive hypothesis for $[\vec{y}]$. For Comp, each child derivation produces a tuple equivalent to its witnessed child, and Lemma D.2 transfers those equivalences through the witnessed template. Its original output is exactly the observed parent $\vec{x}$, establishing the invariant. Every positive Start rule points to an observed sentence (w), whose derivational context is the identity; the invariant implies $w \in L$. The empty Start rule is included only for $\varepsilon \in K \subseteq L$. This argument holds at every finite rank cap; for nonempty K the uncapped construction has cap $\|K\|_+$, and the empty case is immediate. □

D.3 Effective Reconstruction and Conservative Learning

Theorem D.4 (Polynomial construction under a rule-rank cap)**.** *Fix $f, r_{\max} \geq 1$ and s, and implement type tests by the completion $\widehat{s}$. The encoding of $\widehat{G}_{\leq r_{\max}}(K)$ is computable in time $(1 + \|K\|_+)^{O(f r_{\max})}$; its fan-out is at most f and its rank at most $r_{\max}$.*

Proof. Put $n_K = \max\{1, \|K\|_+\}$. An occurrence is a sample word with at most $2f$ cut positions, so there are $n_K^{O(f)}$ occurrences, typed by fixed-monoid products; grouping occurrences by context and type lists the Link rules. A composition witness of rank $r \leq r_{\max}$ is determined by a parent occurrence and at most $2f r_{\max}$ further cut positions inside it, with a labelling whose number depends only on $f, r_{\max}$; goodness is checked directly. This gives $n_K^{O(f r_{\max})}$ candidate Comp rules; the remaining rules have rank at most one, and deduplication leaves an encoding of the same size. □

Proof of Corollary 3.12. For fixed rule rank, qualitative identification follows from the direct soundness and exact-reconstruction results (Proposition D.3 and Theorem 3.10) with the conservative policy of [14, Algorithm 1]; the finite-signature antecedent is Yoshinaka [15, Theorem 1]. For unbounded rank, Theorem 3.10 provides a finite characteristic sample, while Proposition D.3 gives

soundness. Membership is decidable [30]. After a text has presented the characteristic sample, the next rebuild (if any) is correct and permanent; if no rebuild follows, sample consistency and conservativeness imply that the current hypothesis already generates every text word and hence L. For fixed rank, the per-datum bound follows from Theorem D.4 and [14, Proposition 4]. □

Update complexity versus convergence stage. The polynomial-time guarantee of Theorem D.4 and Corollary 3.12 concerns hypothesis reconstruction and membership testing at each finite stage, measured against the data seen so far. TxtEx convergence in Corollary 3.12 does not bound the first stage of stabilization or the number of mind changes polynomially: a text may postpone the first presentation of a particular positive example arbitrarily long. Short characteristic data by itself gives no presentation-independent convergence-stage bound.

Every CFG has a rank-two presentation, so $\mathcal{A}_{1,s}^{\leq 2}$ covers $\mathcal{C}_{1,s}^{\mathrm{mcf}}$; this does not imply polynomial characteristic-data length at rank two (Section 4).

Proof of Proposition 3.8. This is the rule-witness argument of [15, Lemma 3], applied to the concrete Link/Const/Comp presentation. For each typed rule put $\vec{z} = \rho(\iota(Y_1), \ldots, \iota(Y_q))$. The two witnesses supply Link $[\iota(X)] \to [\vec{z}]$, followed by Const if $q = 0$ or Comp of rank $q \leq r_G$ otherwise. The latter also belongs to the uncapped grammar, since $q \leq \|\vec{z}\|_1 \leq \|K\|_+$. Induction on a source derivation thus simulates $X \Rightarrow_{\widetilde{G}_0}^{*} \vec{u}$ from $[\iota(X)]$; $\chi(X) = \square_1$ at a successful start link supplies Start. Apply Proposition 3.5(iii). □

Proof of Theorem 3.10. If $L \cap \Sigma^+ = \emptyset$ then $K = L \in \{\emptyset, \{\varepsilon\}\}$ and the claim is immediate from Definition 3.3. Otherwise $\mathcal{F}_L \subseteq L$ is finite by the remark after Definition 3.6. Proposition 3.8 with $\iota = \omega$ (admissible by Proposition 3.5(i)) gives $L \cap \Sigma^+ \subseteq L(\mathcal{B}(K))$, and Proposition D.3 gives $L(\mathcal{B}(K)) \subseteq L$, for both operators. The hypothesis contains ε iff $\varepsilon \in K$, iff $\varepsilon \in L$, because $\mathcal{F}_L \subseteq K \subseteq L$. Rank-preserving normalization gives the last sentence. □

E Auxiliary Proofs for Typed Witness Construction

The statements used for quantitative witness bounds remain in Section 3.2; the routine verification details are recorded here.

Proof of Proposition 3.5. At every rule application, the monoid morphism property gives

$$\widehat{s}^{(\mathrm{fan}(A))}(\rho(\vec{u}_1, \ldots, \vec{u}_r)) = \mathrm{out}_\rho(\widehat{s}^{(d_1)}(\vec{u}_1), \ldots, \widehat{s}^{(d_r)}(\vec{u}_r)).$$

Induction on derivation height gives (i) and the unique lifting (ii). Erasing indices gives $L(\widehat{G^s}) \subseteq L(G)$, (ii) the converse, and trimming keeps every rule of every successful derivation. □

Lemma E.1 (Monoid implementations)**.** *Let $h_0 : \Sigma^* \to M_0$ be a finite-monoid morphism, $S_0 := h_0(\Sigma^+)$ and $s_0 := h_0|_{\Sigma^+}$.*

(i) *For $1 \leq d \leq f$ and $\vec{x}, \vec{y} \in (\Sigma^+)^d$, the equalities $h_0^{(d)}(\vec{x}) = h_0^{(d)}(\vec{y})$, $s_0^{(d)}(\vec{x}) = s_0^{(d)}(\vec{y})$ and $\widehat{s_0}^{(d)}(\vec{x}) = \widehat{s_0}^{(d)}(\vec{y})$ are equivalent. Hence $\mathcal{C}_{f,h_0}^{\mathrm{mcf}} = \mathcal{C}_{f,s_0}^{\mathrm{mcf}}$, and the reconstruction operators built with h_0 and with $\widehat{s_0}$ return the same grammar on every sample.*

(ii) *For every good MCFG G, the trimmed refinements of G under h_0 and under $\widehat{s_0}$ coincide (the untrimmed ones need not); hence the same canonical tuples, contexts and witness words may be chosen for both.*

Proof of Lemma E.1. (i) The three maps agree on Σ^+ as maps into S_0, and the class and the Link guard compare types of nonempty components only. (ii) S_0 is a subsemigroup of M_0 and of $S_0^\sharp$ with the same multiplication. By λ-freeness and Proposition 3.5(i), every typed nonterminal occurring in a successful derivation under either morphism is some $A_{\mathbf{t}}$ with $\mathbf{t} \in S_0^{\mathrm{fan}(A)}$, and the start links used are $\widetilde{S} \to S_{(\eta)}$ with $\eta \in S_0$. For child indices in S_0, every component α_i of a good template is a nonempty word whose letters and variables are evaluated in S_0 identically under both morphisms, so both give the same out_ρ. Thus the two full refinements contain the same rules among S_0-indexed symbols, which carry all successful derivations. □

Lemma E.2 (Canonical contexts for typed nonterminals)**.** *Let G be good. Every typed nonterminal X of arity d has a context $\chi(X) \in \mathcal{E}_d$ with $\chi(X)[\vec{u}] \in L(G)$ for all $\vec{u} \in L_X$; if $\widetilde{S} \to X$ is in $\widetilde{G}_0$, one may take $\chi(X) = \square_1$.*

Proof of Lemma E.2. Take a successful typed derivation containing X, and follow the path from the root to an occurrence of X, starting with $\square_1$. At each rule on the path apply Lemma 3.1 with the (nonempty) yields of the sibling subderivations fixed; this gives $\chi(X) \in \mathcal{E}_d$. The rules on the path depend only on the label X, so replacing the subtree of X by any X-derivation yields a successful derivation of $\chi(X)[\vec{u}]$. If $\widetilde{S} \to X$ is a rule, use the path of length zero. □

Lemma E.3 (Typed nonterminals are distributionally homogeneous)**.** *Let G be good with $L := L(G) \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. For every typed nonterminal X of arity d, L_X lies in a single $\equiv^d_{L,s}$-class, denoted $[X]$.*

Proof of Lemma E.3. For $\vec{u}, \vec{v} \in L_X$, Proposition 3.5(i) gives equal types and $\chi(X)$ accepts both; substitutability gives $\mathcal{D}_L(\vec{u}) = \mathcal{D}_L(\vec{v})$. □

F Details of the Good Normalization

The normalization uses the four stagewise constructions in Lemma F.1; their semantic and size invariants are verified below.

We record the four conversions used in Proposition 3.9 once, together with the tuple-language identity that each satisfies. Both the fixed-parameter size bound of that proposition and the thickness transfer of Proposition F.2 are read off from these identities. Semantic correctness of the nondeleting and nonmerging conversions is standard [14, 30, 32]; Kanazawa's existence statement [32, Section 2.1] does not by itself give the size or thickness bounds, which are therefore derived here. For $\vec{w} = (w_1, \ldots, w_d)$ and nonempty $I \subseteq [d]$, $\pi_I(\vec{w})$ retains the components indexed by I in increasing order, $\sigma\vec{w}$ permutes components, and $\mu(\vec{w})$ concatenates consecutive blocks of components; thus $\|\pi_I(\vec{w})\|_1 \le \|\vec{w}\|_1$ and $\|\sigma\vec{w}\|_1 = \|\mu(\vec{w})\|_1 = \|\vec{w}\|_1$.

Lemma F.1 (Stagewise tuple languages)**.** *Fix f and $r_{\max}$, and let G_0 be a reduced f-MCFG of rule rank at most $r_{\max}$. Each of the following conversions preserves linearity, does not increase fan-out or rule rank, and multiplies the number of nonterminal and rule copies by a factor depending only on f and $r_{\max}$, without enlarging any template.*

(a) Nondeletion *(the construction (f3) in the proof of [30, Lemma 2.2]). Introduce A^I for nonempty $I \subseteq [\mathrm{fan}(A)]$. For $R : A \to \rho(B_1, \ldots, B_q)$ let J_j be the set of components of B_j used by the I-indexed outputs of ρ; keep these outputs, rename their variables, use $B_j^{J_j}$ when $J_j \neq \emptyset$ and omit B_j otherwise. The resulting rules are nondeleting, and $L_{A^I} = \pi_I(L_A(G_0))$.*

(b) Nonpermutation. *Decorate a fan-out-d nonterminal by $\sigma \in \mathfrak{S}_d$; the order in which a permuted parent template uses a child's variables determines the child's permutation. The decorated nonterminal generates σL_A.*

(c) Empty components. *Let $\mathsf{Pat}(A)$ be the set of nonempty-component patterns of tuples in L_A, computed by finite saturation. For realizable child patterns, replace variables of absent components by ε, delete empty parent components and omit all-empty children; introduce A^I for each nonempty $I \in \mathsf{Pat}(A)$. Then L_{A^I} is the set of π_I-projections of the tuples of L_A with pattern I, the result is λ-free, and $S^{\{1\}}$ generates the nonempty part of the language. If every derived component is already nonempty, this stage is the identity.*

(d) Nonmerging *(the construction in the proof of [14, Section 2.3, Lemma 1]). Decorate a nonterminal by a composition μ of its fan-out; in a merged parent template, maximal adjacent runs of consecutive variables of a child determine that child's merge. The merge-decorated nonterminal generates the* image $L_{A^\mu} = \mu(L_A)$.

Stages (b)–(d) preserve the conditions established by earlier stages.

Proof. (a) For arbitrary child tuples, $\pi_I(\rho(\vec{v}_1, \dots, \vec{v}_q)) = \rho^I\big((\pi_{J_j}\vec{v}_j)_{J_j \neq \emptyset}\big)$, so induction on derivations gives $\subseteq$. For $\supseteq$, lift every retained child tuple by the induction hypothesis and supply a terminal derivation of each omitted child; this uses that G_0 is reduced, hence every omitted child is productive. Without prior trimming only $\pi_I(L_A) \subseteq L_{A^I}$ would be automatic. There are at most 2^f copies of each nonterminal, and I determines the J_j, so at most 2^f copies of each rule; omitting children cannot increase rank.

(b) Linearity and nondeletion make the induced child permutations well defined; induction in both directions identifies the language with σL_A. There are at most $f!$ copies, no child is duplicated, and total length, λ-freeness, fan-out and rank are preserved.

(c) Retained variables still occur once and in order, so nondeletion, nonpermutation, fan-out and rank are preserved. Induction on derivations gives the projection identity; omitted all-empty children are restored by terminal derivations realizing their all-empty pattern. There are at most 2^f patterns per nonterminal and $2^{f r_{\max}}$ choices per rule, and projection only shortens templates.

(d) Nonpermutation makes the maximal runs compatible with each child's component order, and nondeletion makes them cover all its components. The merged template satisfies $\rho^\mu(\mu_1(\vec{v}_1), \dots, \mu_q(\vec{v}_q)) = \mu(\rho(\vec{v}_1, \dots, \vec{v}_q))$, and induction gives the image equality. No inverse-image claim is made, because the merge map need not be injective. Concatenation preserves total length, λ-freeness, nonpermutation, fan-out and rank; for fixed dimension and rank the conversion has linear size [14, Section 2.3, after Example 2]. □

Proof of Proposition 3.9. Apply (a)–(d) to the trimmed input and trim again. Each stage multiplies the encoding by a factor depending only on $f, r_{\max}$ and trimming can only decrease it, so $\|G_{\text{good}}\| = O_{f,r_{\max}}(\|G_{\text{in}}\|)$, and $L(G_{\text{good}}) = L(G_{\text{in}}) \setminus \{\varepsilon\}$ by (c). □

Thickness transfer for λ-free presentations

Proposition F.2 (Thickness under good-form conversion)**.** *Fix f and $r_{\max}$. Let G_{in} be a reduced λ-free f-MCFG of maximum rule rank at most $r_{\max}$, possibly with deleting rules, and suppose $L(G_{\text{in}}) \cap \Sigma^+ \neq \emptyset$. The good normalization of Proposition 3.9 yields an equivalent reduced good grammar G with the same fan-out and rule-rank bounds such that*

$$\|G\| = O_{f,r_{\max}}(\|G_{\text{in}}\|), \qquad t_G \leq t_{G_{\text{in}}}, \qquad \theta_G = O_{f,r_{\max}}(\|G_{\text{in}}\| \max\{1, t_{G_{\text{in}}}\}).$$

Proof. Because G_{in} is λ-free, every component of every derived tuple is nonempty, by induction on derivations; this persists through (a) and (b), so stage (c) is the identity. By Lemma F.1, every nonterminal X of the final grammar is associated with an original nonterminal A such that

$$L_X(G) = \mu(\sigma\pi_I(L_A(G_{\mathrm{in}}))), \qquad \min_{\vec{u}\in L_X(G)} \|\vec{u}\|_1 \le \min_{\vec{w}\in L_A(G_{\mathrm{in}})} \|\vec{w}\|_1.$$

Trimming leaves the tuple languages of surviving nonterminals unchanged, so $t_G \le t_{G_{\mathrm{in}}}$, and the size bound was shown above. Finally Lemma 4.1(iii) gives $\theta_G \le 2\|G\| \max\{1, t_G\} = O_{f,r_{\max}}(\|G_{\mathrm{in}}\| \max\{1, t_{G_{\mathrm{in}}}\})$. □

Thus deleting rules do not by themselves obstruct polynomial thickness transfer from a reduced λ-free presentation; genuine empty components are the problematic step.

Proposition F.3 (Empty-component elimination can blow up thickness)**.** *For every $n \ge 1$ there is a reduced fan-out-two, rule-rank-two MCFG E_n of size $O(n)$ with $t_{E_n} \le 2$ and $\theta_{E_n} = O(n)$ such that the empty-component elimination stage of Proposition 3.9 contains a retained nonterminal and a retained rule of thickness at least 2^n.*

Proof. Let E_n have fan-out-one S, fan-out-two $A_0, \dots, A_n$, and

$$A_0 \to (\mathtt{a}, \varepsilon), \qquad A_i \to (\varepsilon, \mathtt{c}) \mid (\xi_1^1\xi_2^1, \xi_1^2\xi_2^2)(A_{i-1}, A_{i-1}) \quad (1 \le i \le n), \qquad S \to (\xi_1^1 \mathtt{d}\, \xi_1^2)(A_n).$$

It is reduced, nondeleting, nonpermuting, and nonmerging. Short side derivations give $t_{E_n} \le 2$, and placing any chosen rule on one branch of the binary spine while completing siblings minimally gives $\theta_{E_n} = O(n)$. If an A_i-tuple has nonempty first and empty second component, the rule $(\varepsilon, \mathtt{c})$ cannot occur in it; inductively the shortest such tuple is $(\mathtt{a}^{2^i}, \varepsilon)$. Thus pattern $\{1\}$ for A_n survives, and the projected start rule $S \to \xi_1^1 \mathtt{d}\, (A_n^{\{1\}})$ succeeds only with a child of length at least 2^n. □

This is the obstruction of Proposition 4.5: finite state refinement can force exponentially long shortest witnesses inside one refined state.

G A Distinguishing-Pair Lower Bound at Fan-Out One

The family below is the bracketed doubling family of the companion paper [25, Section 6]. For self-containment of Corollary 4.20, we reproduce the construction and the distinguishing-pair argument, including a proof of substitutability. This lower-bound construction is attributed to the companion and is not claimed as new here.

Lemma G.1 (Distinguishing pair)**.** *Let $\mathcal{O}$ be a set-driven reconstruction operator and let $L' \subsetneq L$ be languages that both admit characteristic samples for $\mathcal{O}$. Then every characteristic sample for L meets $L \setminus L'$.*

Proof. Let F be characteristic for L and suppose $F \subseteq L'$. Let F' be characteristic for L' and put $K := F \cup F'$. Then $F \subseteq K \subseteq L$ gives $L(\mathcal{O}(K)) = L$, while $F' \subseteq K \subseteq L'$ gives $L(\mathcal{O}(K)) = L'$, contradicting $L' \ne L$. □

Let $\Gamma_{\mathrm{br}} = \{\mathtt{a}, \lfloor, \rfloor, \lceil, \rceil\}$ and let $U_1 = \{1, \zeta\}$ be the two-element semilattice ($1\zeta = \zeta 1 = \zeta^2 = \zeta$). The typing $s_{\mathrm{U}} : \Gamma_{\mathrm{br}}^+ \twoheadrightarrow U_1$ sends $\mathtt{a}, \lfloor, \rfloor$ to 1 and $\lceil, \rceil$ to ζ; thus $s_{\mathrm{U}}(w) = 1$ exactly for nonempty words without ceiling brackets. For $n \ge 1$ let G_n be the fan-out-one MCFG

$$S \to E_n, \qquad E_0 \to \mathtt{a}, \qquad E_i \to \lfloor E_{i-1}E_{i-1} \rfloor \;\mid\; \lceil E_{i-1} \rceil \quad (1 \le i \le n),$$

with templates written in CFG notation, and put $L_n := L(G_n)$. Let $w_0^* = \mathtt{a}$, $w_i^* = \lfloor w_{i-1}^* w_{i-1}^* \rfloor$, and $L_n^- := L_n \setminus \{w_n^*\}$.

Lemma G.2 (Substitutability of the bracketed doubling family)**.** *For every $n \geq 1$, the language L_n defined above is Clark–Eyraud substitutable: for all nonempty $x, y \in \Gamma_{\mathrm{br}}^{+}$,*

$$D_{L_n}(x) \cap D_{L_n}(y) \neq \varnothing \quad \Longrightarrow \quad D_{L_n}(x) = D_{L_n}(y),$$

where $D_{L_n}(z) = \{(p, q) : pzq \in L_n\}$.

Proof. Put $L_0 = \{\mathtt{a}\}$ and $L_i = \lceil L_{i-1} \rceil \cup \lfloor L_{i-1} L_{i-1} \rfloor$ for $i \geq 1$. Use a deterministic top-down parser whose stacks, with top on the left, are words over $E_0, \ldots, E_n, \rceil, \rfloor$. For every stack suffix ψ its only transitions are

$$E_0\psi \xrightarrow{\mathtt{a}} \psi, \quad E_i\psi \xrightarrow{\lceil} E_{i-1}\rceil\psi, \quad E_i\psi \xrightarrow{\lfloor} E_{i-1}E_{i-1}\rfloor\psi$$
$$\rceil\psi \xrightarrow{\rceil} \psi, \quad \rfloor\psi \xrightarrow{\rfloor} \psi \qquad\qquad (1 \leq i \leq n).$$

No transition leaves an empty stack. Write $\sigma \xrightarrow{w} \theta$ for the unique run, if defined, reading w from σ and ending at θ. Three properties will be used.

(P1) Letterwise reversibility. For each fixed letter the transition is a partial injective function on stacks. Opening brackets reveal the original index from the expanded top and preserve the untouched suffix; the other letters pop a uniquely determined top marker. Consequently every word acts as a partial injection. In particular, if $\sigma \xrightarrow{w} \theta$ and $\sigma' \xrightarrow{w} \theta$, then $\sigma = \sigma'$.

(P2) Strict descent. Give E_i weight 4^i and each closing marker weight 1. Every transition strictly decreases total stack weight; for the binary expansion this follows from $4^i > 2 \cdot 4^{i-1} + 1$. Thus $\sigma \xrightarrow{w} \sigma$ implies $w = \varepsilon$.

(P3) Suffix locality and first exposure. Transitions inspect only the top stack symbol, so $\sigma \xrightarrow{w} \theta$ implies $\sigma\psi \xrightarrow{w} \theta\psi$. Conversely, in any run $\sigma_1\psi \xrightarrow{w} \theta$ with $\sigma_1 \neq \varepsilon$, either the run first reaches the fixed suffix ψ and splits as $w = uv$, with $\sigma_1 \xrightarrow{u} \varepsilon$ and $\psi \xrightarrow{v} \theta$, in which case $\mathrm{wt}(\theta) \leq \mathrm{wt}(\psi)$; or it never reaches ψ, so $\theta = \theta_1\psi$ for some nonempty θ_1 and $\mathrm{wt}(\theta) > \mathrm{wt}(\psi)$. In the latter case, *if* the bottom symbol of σ_1 is a closing marker, that marker is still the bottom symbol of θ_1. We use this assertion only for prefixes ending in $\rceil$ or $\rfloor$ (not for arbitrary σ_1). The weight comparison therefore distinguishes the two cases.

We first show by induction on i that, for every suffix ψ,

$$E_i\psi \xrightarrow{w} \psi \quad \Longleftrightarrow \quad w \in L_i. \tag{1}$$

Indeed, a successful run cannot leave a nonempty stack prefix over ψ; (P3) and (P2) show that it must empty E_i exactly and then stop. For $i = 0$ its only input is $\mathtt{a}$. For $i \geq 1$ the first symbol is $\lceil$ or $\lfloor$. In the first case the parser must consume one L_{i-1}-word, then $\rceil$; in the second it must consume two L_{i-1}-words, then $\rfloor$. The converse runs exist by (P3), giving (1). By (P1), (1) also implies the useful inverse statement

$$\sigma \xrightarrow{w} \theta, \quad w \in L_i \quad \Longrightarrow \quad \sigma = E_i\theta. \tag{2}$$

If $u, v \in L_i$, then $u \equiv_{L_n} v$ (ordinary syntactic congruence). Indeed, if $puq \in L_n$, the successful parsing run factors as $E_n \xrightarrow{p} \sigma \xrightarrow{u} \theta \xrightarrow{q} \varepsilon$. By (2), $\sigma = E_i\theta$, and (1) lets us replace u by v. The converse follows symmetrically.

Next suppose $\sigma \xrightarrow{x} \theta$ and $\sigma \xrightarrow{y} \theta$. We claim that

$$x \equiv_{L_n} y. \tag{3}$$

Induct on $|x| + |y|$. If either word is empty, (P2) forces both words to be empty. If $x = cx'$ and $y = cy'$ begin with the same letter, determinism gives a common successor stack, and the induction

hypothesis applies to x', y'. If their first letters differ, the top of σ must be E_i with $i \geq 1$, and the two initial letters must be $\lceil, \lfloor$. Write $\sigma = E_i\psi$. After these respective opening letters the stacks are $E_{i-1}\rceil\psi$ and $E_{i-1}E_{i-1}\rfloor\psi$. Apply (P3) to the remaining runs with suffix ψ. They lie in the same case because their common final stack θ has a fixed weight. They cannot both be in the nonexposure case: each original stack prefix ends in its displayed closing marker, so the nonempty remainder above ψ would end in $\rceil$ in the first run and $\rfloor$ in the second, contradicting equality of θ. Therefore both runs first reach ψ. We may split $x = uz$, $y = vz'$ with $u, v \in L_i$ by (1), and $\psi \xrightarrow{z} \theta$, $\psi \xrightarrow{z'} \theta$. Induction gives $z \equiv_{L_n} z'$, while the preceding paragraph gives $u \equiv_{L_n} v$. Since syntactic congruence is a monoid congruence, $uz \equiv_{L_n} vz'$. This proves (3).

Finally suppose $(p, q) \in D_{L_n}(x) \cap D_{L_n}(y)$. The two successful runs reading pxq and pyq agree on the stack σ reached after p by determinism, and on the stack θ reached immediately before q by the injectivity in (P1). Thus $\sigma \xrightarrow{x} \theta$ and $\sigma \xrightarrow{y} \theta$; (3) gives $x \equiv_{L_n} y$, i.e. $D_{L_n}(x) = D_{L_n}(y)$. □

Proposition G.3 (Properties of the family)**.** *For every $n \geq 1$:*

(i) *G_n is reduced and good, of rank two, with $\|G_n\| = O(n)$ and $\theta_{G_n} \leq 4n$.*

(ii) *w_n^* is the only word of L_n without ceiling brackets, and $|w_n^*| = 3 \cdot 2^n - 2$; hence $L_n^- = L_n \cap \widehat{s}_{\mathrm{U}}^{-1}(\{\zeta\})$.*

(iii) *$L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_{\mathrm{U}}}$ and $L_n^- \in \mathcal{C}^{\mathrm{mcf}}_{1,s_{\mathrm{U}}}$.*

Proof. (i) Every nonterminal is reachable and productive. Each template is λ-free and uses each child variable exactly once; the two children of $E_i \to \lfloor E_{i-1}E_{i-1}\rfloor$ are distinct child occurrences, so the rule is nonmerging, and nonpermutation is vacuous at fan-out one. There are $2n + 2$ rules of constant size. For thickness, use every rule other than the chosen one in its $\lceil\cdot\rceil$ form. A rule $\lceil E_{i-1}\rceil$, $E_0 \to \mathtt{a}$ or $S \to E_n$ then occurs in $\lceil^n \mathtt{a}\rceil^n$, of length $2n + 1$. The rule $E_i \to \lfloor E_{i-1}E_{i-1}\rfloor$ occurs in $\lceil^{n-i}\lfloor u\, u\rfloor\rceil^{n-i}$ with $u = \lceil^{i-1}\mathtt{a}\rceil^{i-1}$, of length $2(n-i) + 2 + 2(2i-1) = 2n + 2i \leq 4n$.

(ii) A derivation avoiding the rules $E_i \to \lceil E_{i-1}\rceil$ is the complete binary derivation of height n, whose yield is w_n^*. From $|w_0^*| = 1$ and $|w_i^*| = 2|w_{i-1}^*| + 2$ we get $|w_i^*| = 3 \cdot 2^i - 2$. Every other word contains $\lceil$, so its $\widehat{s}_{\mathrm{U}}$-value is ζ, while $\widehat{s}_{\mathrm{U}}(w_n^*) = 1$.

(iii) L_n is Clark–Eyraud substitutable by Lemma G.2; in the present notation it is $(1, s_{\mathrm{triv}})$-tuple-substitutable, hence $(1, s_{\mathrm{U}})$-tuple-substitutable because $s_{\mathrm{triv}} \preceq s_{\mathrm{U}}$. The language $\widehat{s}_{\mathrm{U}}^{-1}(\{\zeta\})$ is $(1, s_{\mathrm{U}})$-tuple-substitutable by the observation following Definition 2.8, so L_n^- is $(1, s_{\mathrm{U}})$-tuple-substitutable by (ii) and Proposition 2.9(i). Both languages are finite, hence context-free. □

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the author used ChatGPT (OpenAI) and Claude (Anthropic) to assist with organization of the manuscript, expression and translation of text, cross-checks of references and notation, and editing of LaTeX source. After using these tools, the author reviewed and edited the material as needed and takes full responsibility for the content of the publication.

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